The Mexican Health and Aging Study:
Cognitive Function Measures
Scoring and Classification Across Waves
2018-2021
Version 2

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Introduction

The Mexican Health and Aging Study (MHAS, or *Estudio Nacional de Salud y Envejecimiento en México*, ENASEM in Spanish) is a national longitudinal study of adults over the age of 50. The survey was designed to capture a broad range of health, functional, and socioeconomic data from this demographic. The MHAS baseline survey was completed in 2001 targeting a nationally representative sample of individuals born in 1951 or earlier to represent adults aged 50 years and older. In 2012, the sample was refreshed adding a new cohort of those born between 1952 and 1962. Every two waves, the study adds a refresher sample to yield a representative sample of these 50 years and older.

As of 2021, six waves of data collection have been completed, allowing for an in-depth understanding of aging over time in the Mexican context. The MHAS protocols and survey instruments are highly comparable to the U.S. Health and Retirement Study (HRS), allowing for cross-country comparisons on a wide range of aging-related issues.

For more information about the MHAS, we recommend that you read the cohort profile by Wong, Michaels-Obregón, and Palloni[1] and visit the study website www.MHASweb.org (www.ENASEM.org for the Spanish-language website).

Overall Description

The main goals of the current MHAS document are:

- To describe application and scoring of the different cognitive function tasks included in Waves 5 and 6 (2018 and 2021) and the comparability across the six waves of the study.
- To provide the updated normative standards and z-scores for each task as well as the composite score specifically constructed for Waves 5 and 6 of the study.
- To distribute the STATA program codes used for the following constructed variables:
    - individual task score, corresponding z-scores and classification as normal or impaired (see Table 4 for all variables names).
    - composite score for direct interviews and corresponding z-scores (see Table 4 for all variables names).
    - score for proxy interviews and classification as normal or impaired (see Table 6 for all variables names).
    - classification of subjects as normal, CIND, dementia (`cognitive_status_xx` and `cognitive_status_v01_xx`).

[1] Wong R, Michaels-Obregon A, Palloni A. Cohort Profile: The Mexican Health and Aging Study (MHAS). Int J Epidemiol. 2017 Apr 1;46(2):e2. doi: 10.1093/ije/dyu263. PMID: 25626437; PMCID: PMC5837398.

This descriptive document accompanies the databases that contain the 2018 and 2021 constructed variables for the MHAS users. For more information on the cognitive function tasks, application and scoring instruction, and the process to classify subjects and normal, CIND, dementia, please refer to the 2001-2015 document “The Mexican Health and Aging Study: Cognitive Function Measures Scoring and Classification Across Waves 2001-2015” available here.

MHAS 2018 and 2021 Cognitive Function Measures

a. Direct Interview

Since the MHAS baseline in 2001, measuring cognitive function has been part of the study. Cognitive function has been assessed using the screening portion[2] of the Cross-Cultural Cognitive Examination (CCCE) for direct interviews and the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) for proxy interviews.

The CCCE was created as a brief and sensitive tool for diagnosing dementia in community settings. It includes an initial screening section designed for use by individuals without medical training in surveys. The second section of the CCCE, intended to enhance the tool's specificity, is to be administered by trained medical professionals. However, this second section is not part of the MHAS study. All tasks in the CCCE are recognized as indicators of cognitive function, and the influence of literacy and education level is considered negligible (Glosser et al. 1993).

The IQCODE is a short screening tool designed to detect cognitive changes in older populations based on reports from informants. The Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) was used for proxy interviews (Glosser et al. 1993; Jorm 1994).

Differences in 2018-2021 compared to the previous waves

The study aims to maintain comparability across waves to enable cross-wave comparisons and allow for the analysis of cognitive function trajectories. Most of the battery has been kept consistent across waves, though some changes have been made to incorporate new cognitive tasks.

This is especially important for longitudinal analysis using the MHAS data, as it enables users to identify items that are consistent across all waves, ensuring reliable comparisons. By including these common items throughout the study, researchers can track cognitive changes using the same set of tasks, which is essential for assessing individual trajectories over time. This consistency helps reduce the impact of variations in the cognitive exercises across waves.

[2] The CCCE has been modified over the MHAS waves to include a more comprehensive cognitive evaluation.

Table 1 below lists the items included in each wave from 2001 to 2021.

In 2001, five tasks were included as part of the direct interview. In 2003, an additional task was included to measure orientation (marked in green in the list below). In 2012, semantic verbal fluency and backward counting tasks were added (marked in purple in the list below). In 2015, an additional task – serial 7 – was added (marked in red in the list below). In 2018, the backwards counting task was removed. Finally, in 2021, the serial 3 task was used instead of the serial 7 task (marked in blue in the list below).

The different tasks to measure cognitive function are briefly described below.

1. Constructional Praxis (visuo-constructional): measured by presenting two geometrical figures and asking respondents to copy the figures. Respondents were given 90 seconds to complete both figures.
2. Verbal Fluency: measured by asking respondents to list all the animals they can think of in the next 60 seconds.
3. Serial 7s: measured by asking the respondent to sequentially subtract 7 starting from 100, until they complete five successive subtractions.
4. Verbal learning: measured by asking respondents to listen to a list of eight words and to repeat them. Three consecutive trials are administered and the number of recalled words in each trial is recorded.
5. Visual scanning: measured by asking respondents to circle all figures that are identical to a specific stimulus shown previously within an array of different stimuli. Respondents are given 60 seconds for this task.
6. Backward Counting: measured by asking respondents to count backwards from 20 to 0 as fast as possible. Respondents are given 60 seconds (max) to complete this task.
7. Constructional Praxis Recall: measured by asking respondents to remember the figures they copied and to draw them from memory on a blank piece of paper; three minutes are allowed to complete this task.
8. Delayed Verbal recall: measured by asking respondents to repeat as many of the words they remember from the list provided in the verbal learning task; verbal recall was administered after the visual scanning task to allow a time delay.
9. Day, Month and Year: This task asks respondents to indicate the day, month and year of the interview.
10. Serial 3s: measured by asking the respondent to sequentially subtract 3 starting from 20, until they complete five successive subtractions.

Table 1. MHAS 2001-2021 Cognitive Measures - Variable Names by Task

Cognitive Domain	Task	MHAS Variables					
		2001	2003	2012	2015	2018	2021
Executive Function	Constructional Praxis	e8_e9_01	e6_e7_03	e8_12	e8_15	e8a_c_18	e8a_c_21
	Verbal Fluency			e9b_12	e9b_15	e9a_c_18	e9a_c_21
	Serial 7s				e15a_15 e15b_15 e15c_15 e15d_15 e15e_15	e15a_18 e15b_18 e15c_18 e15d_18 e15e_18	e15a_21 e15b_21 e15c_21 e15d_21 e15e_21
Immediate Memory	Verbal Learning	**List A**: e11_1i_01 e11_2i_01 e11_3i_01	**List A**: e9_a1_9_03 e9_a2_9_03 e9_a3_9_03	**List A**: e7a_1_12 e7a_2_12 e7a_3_12	**List A**: e7a_1_15 e7a_2_15 e7a_3_15	**List A:** e7a1_tot_18 e7a2_tot_18 e7a3_tot_18	**List A:** e7a1_tot_21 e7a2_tot_21 e7a3_tot_21
		List B: e11_4i_01 e11_5i_01 e11_6i_01	**List B**: e9_b1_9_03 e9_b2_9_03 e9_b3_9_03	**List B**: e7b_1_12 e7b_2_12 e7b_3_12	**List B**: e7b_1_15 e7b_2_15 e7b_3_15	**List B:** e7b1_tot_18 e7b2_tot_18 e7b3_tot_18	**List B:** e7b1_tot_21 e7b2_tot_21 e7b3_tot_21
Attention	Visual Scanning	e12_01	e10_03	e10_12	e10_15	e10_c_18	e10_c_21
	Backward Counting (time)			**Trial 1** e12a_12 **Trial 2** e12b_12 **Time** e12c_12	**Trial 1** e12a_15 **Trial 2** e12b_15 **Time** e12c_15		
Delayed Memory	Constructional Praxis Recall	e13_01	e11_03	e13_12	e13_15	e13a_c_18	e13_c_21
	Delayed Verbal Recall	**List A**: e14_1i_01	**List A**: e12a_9_03	**List A**: e14a_12	**List A**: e14a_15	**List A:** e14a_tot_18	**List A:** e14a_tot_21
		List B: e14_2i_01	**List B**: e12b_9_03	**List B**: e14b_12	**List B**: e14b_15	**List B:** e14b_tot_18	**List B:** e14b_tot_21
Orientation	Orientation		**Day** e13a_03 **Month** e13b_03 **Year** e13c_03	**Day** e11a_12 **Month** e11b_12 **Year** e11c_12	**Day** e11a_15 **Month** e11b_15 **Year** e11c_15	**Day** e11a_18 **Month** e11b_18 **Year** e11c_18	**Day** e11a_21 **Month** e11b_21 **Year** e11c_21

Constructed variables and composite scores

The differences across waves described above are mostly due to the addition or deletion of tasks. However, in some instances the tasks have been modified in a way that it impacts the calculation of individual task scores as well as composites. For example, the constructional praxis tasks were changed in 2012 impacting the new scoring for constructional praxis and constructional praxis recall from 0 to 6 points instead of 0 to 2. It is important to note that despite this change, it is still possible to compare the score between waves. More information about this change and how to compare the scores across waves can be found in the 2001-2015 document "The Mexican Health and Aging Study: Cognitive Function Measures Scoring and Classification Across Waves 2001-2015" available here.

Table 2 lists the names of the constructed variables included in the datafiles, the possible score for each task, and how these variables are added to calculate the total composite score which varies across waves. The datafile includes multiple composite scores to facilitate the comparison across waves as follows (the names in parenthesis refer to the variable name, with the suffix `_xx` referring to the year of the survey, whereas 01 refers to 2001, and so forth):

1) **Total Composite Score for each wave** (`ccce_xx`)

Constructed using all the items available at each wave. As illustrated in Table 2 below the number of tasks and total scores varies across waves. This score is commonly used only when analyzing a cross-section.

2) **Total Composite Score for each wave, comparable to 2001** (`ccce_v01_xx`)

Constructed for each wave using only the five items included in 2001, that is: constructional praxis, constructional praxis recall, verbal learning (immediate word recall), delayed verbal recall (delayed verbal recall), and visual scanning (visual route)[3].

These composite scores facilitate longitudinal analysis.

[3] We added in parenthesis alternative names that are uses interchangeably for these tasks.

Table 3. Names of Constructed Variables and Scores by Task, and Total Composite Score for each Wave

Task	Constructed Variables Names						Notes on Calculation	Score					
	2001	2003	2012	2015	2018	2021		2001	2003	2012	2015	2018	2021
Constructional Praxis	construction_xx						Total score	0-2		0-6			
	construction_v01_xx						Total score (comparable)			0-2			
Verbal Fluency			verbal_fluency_xx				Categorical variable: 1=0-8 animals 2=9-18 animals 3=19-24 animals 4=25-60 animals			1-4[a]			
Serial 7s				serial7_xx			Total 5 subtractions scores: 0=Incorrect 1=Correct				0-5		
Serial 3s						serial3_xx	Total 5 subtractions scores: 0=Incorrect 1=Correct				0-5		
Verbal Learning	iwr_xx						Average 3 Trials (including both lists)	0-8					
Visual Scanning	visual_scan_xx						Total score	0-60					
Backward Counting (time)			bwc_xx				Categorical variable (time): 0=Incorrect 1=31-60 sec 2=21-30 sec 3=11-20 sec 4=3-10 sec			0-4[b]			
Constructional Praxis Recall	construction_m_xx						Total score (see Table 2)	0-2		0-6	0-6		
Delayed Verbal Recall	dwr_xx						Total of both lists	0-8					
Orientation		orientation_xx					Sum of 3 items 0=Incorrect 1=Correct		0-3				
Total Composite Score	**ccce_xx**						**Sum of all tasks [c]**	**0-80**	**0-83**	**0-99**	**0-104**	**0-99**	**0-99**
Number of tasks missing	ccce_miss_xx						Counts of tasks missing	0-5	0-6	0-8	0-9	0-8	0-8
Total Composite Score Comparable to 2001		**ccce_v01_xx**					**Sum of 5 tasks included since 2001 [c]**		**0-80**				
Number of tasks missing		ccce_miss_v01_xx					Counts of tasks missing		0-5				

Notes: "xx" refers to the year of the corresponding wave, where "01" indicates 2001;
[a] The verbal fluency categorical variable was constructed following previous work by Mejia et al. 2015;
[b] The backward counting categorical variable was constructed using in combination the correct/incorrect variable from trial 1 and 2, and the time variable, following previous work by Mejia-Arango et al. 2015.
[c] Composite scores are constructed adding the tasks scores if at least one task is non-missing.

Constructed norms and z-scores

While the study aims to maintain the comparability across waves, changes in the scores due to tasks that are included, as well as the new cohorts added to the survey sample, introduce a complexity to longitudinal analysis. Norms and z-scores eliminate this issue and allow researchers to compare standardized scores across waves and different cohorts or demographic groups. The norms are used to classify the participants into Normal function, Mild, or Severe cognitive impairment for each task.

The main motivation to periodically review the norms previously used in the MHAS resides in the addition of new samples from different birth years every two waves. Because the participants are aging, this addition is usually of younger old adults, and is done to maintain the representation of the sample as 'adults aged 50 and older living in Mexico.' As expected, younger cohorts are more educated and will need new norms that reflect a new distribution of cognition scores.

In 2022, for the first time we published normative standardized data (z-scores) by age and education for each task in the cognitive function battery and for total cognition score. A previous version of the norms and z-scores was also described in the manuscript "Normative and Standardized data for Cognition measures in the MHAS" (Mejia-Arango, Wong and Michaels-Obregon 2015). While such manuscript referred to only three waves of the data (2001, 2003, 2012), the 2022 document provided a revised version of the z-scores for 2001, 2003 and 2012 and added the 2015 data.

The current document provides an update to the previously published document, to include the 2018 and 2021 data. New norms were constructed with the entire 2018 sample and using the same methodology as in the previous waves, as described in the manuscript Mejia-Arango et al. 2015 (see Figure 1 below).

As explained above, in 2021 the Serial 3 was used instead of Serial 7. To address this change, a new set of norms was constructed for Serial 3 and for the Composite Score (CCCE). Norms were calculated following the same procedures as in 2018. Figure 2 illustrates how we selected a sample for the construction of these norms.

Table 4 below, includes the norms for the Composite Score for both 2018 and 2021. As described above, to allow for comparison across waves, we provide norms for the 'Composite Score for each wave, comparable to 2001' that uses the 5 measures included in the study since 2001. A second set of norms is provided for the 'Composite Score for each wave' constructed using all measures available in each wave.

In addition to the constructed variables, we also provide the norms and z-scores for individual tasks (Table 3) and for the total composite score (all tasks included) for each wave in Table 4. Note that norms are the same for 2018 and 2021 for all individual tasks except for Serial 7.

Figure 1. Flowchart for selection of MHAS 2018 Normative sample and Classification of Cognitive Status

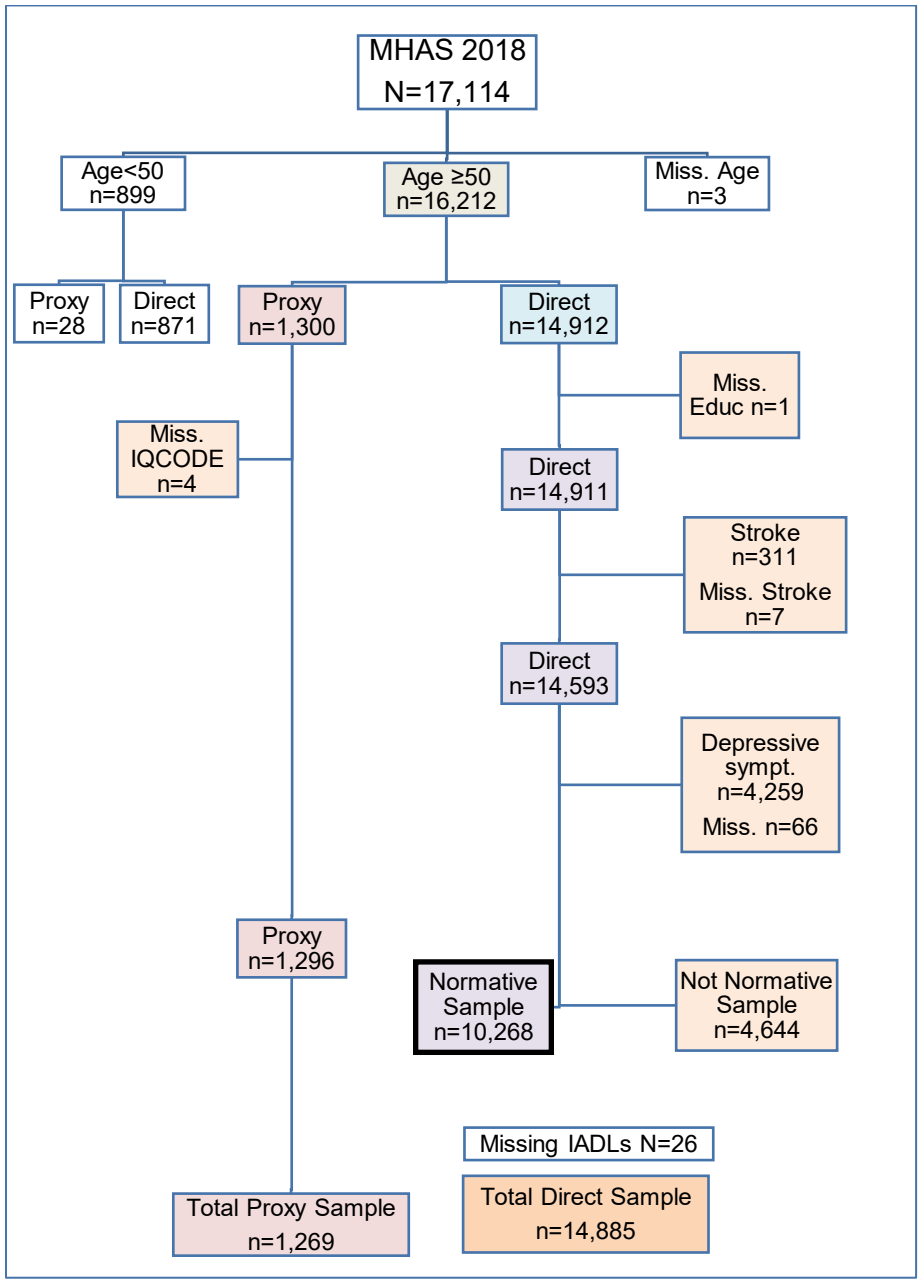


Figure 2. Flowchart for selection of MHAS 2021 Normative sample and Classification of Cognitive Status

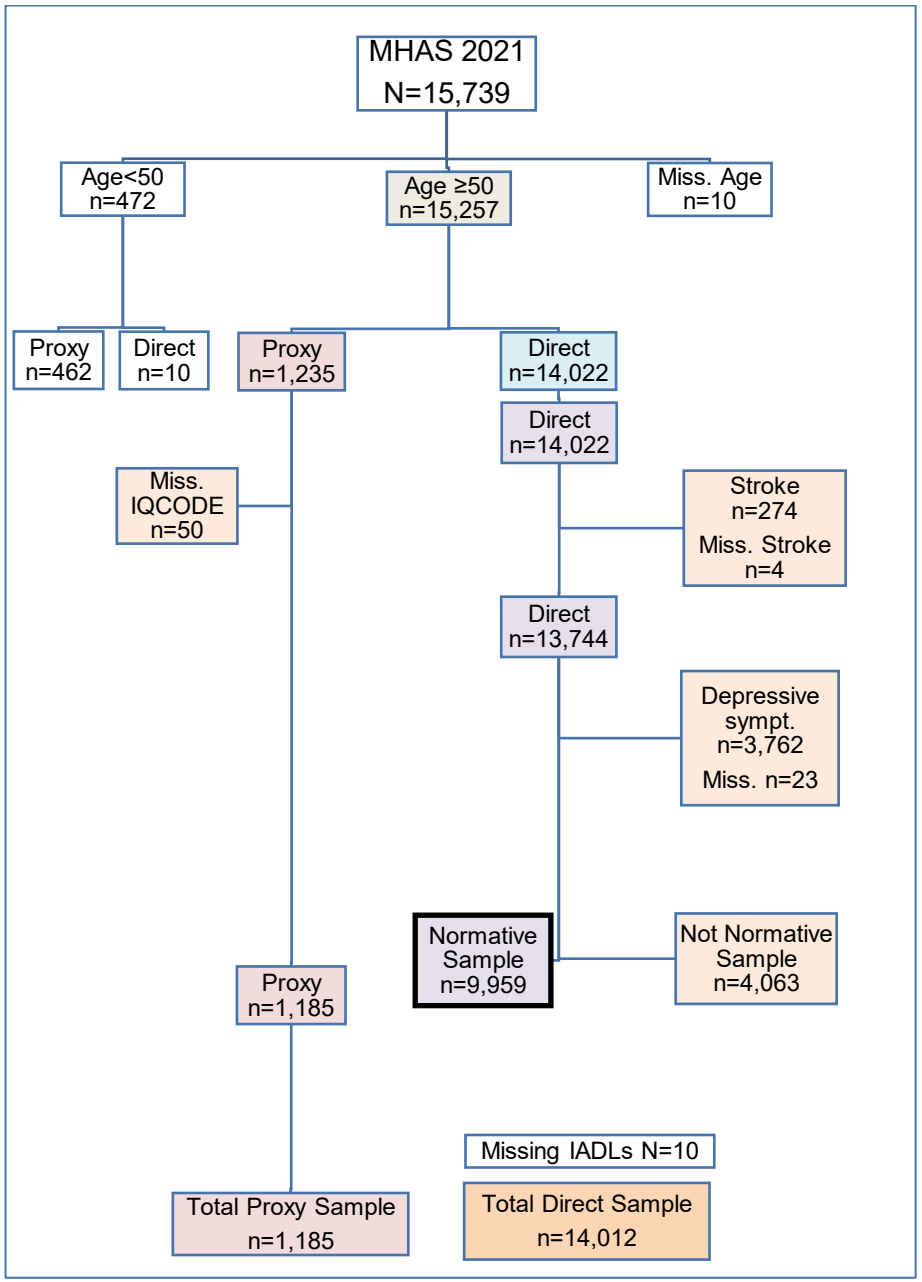


Table 3. MHAS 2018 and 2021 Norms and Z-Scores for each Task, by Age Group and Education Level

		ORIENTATION (Score 0-3)			VISUAL SCANNING (Score 0-60)			VERBAL LEARNING (Score 0-8)			VERBAL RECALL (Score 0-8)			CONSTRUCTIONAL PRAXIS						CONSTRUCTIONAL PRAXIS RECALL						VERBAL FLUENCY (Score 0~50)			SERIAL 7 2018 ONLY (Score 0-5)			SERIAL 3 2021 ONLY (Score 0-5)		
														2001 & 2003 (Score 0-2)			2012-2021 (Score 0-6)			2001 & 2003 (Score 0-2)			2012-2021 (Score 0-6)										
Z-Score		Education (years)																														
		0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+
Age =<59 years old																																		
Normal	3.0		2 - 3	1 - 3	51 - 60			8	8																	24 - 40	27 - 40							
	2.0	3			42 - 50	55 - 60		7	7	8	8	8	8													20 - 23	23 - 26	27 - 40						
	1.5		1		38 - 41	50 - 54				7	7	7														19	22	26	5					
	1.0	2			28 - 37	38 - 49	50 - 60	6	6		5 - 6	5 - 6	7				6			2			6			15 - 18	18 - 21	22 - 25	4	5	5	5	5	5
	0.0	1	0		17 - 27	26 - 37	38 - 49	4 - 5	5	5 - 6	3 - 4	4	5 - 6	2	2	2	5	6	6	1	2	2	4 - 5	5 - 6	5 - 6	12 - 14	14 - 17	17 - 21	2 - 3	3 - 4	4	3 - 4	4	4
	-1.0	0			8 - 16	14 - 25	27 - 37	3	3 - 4		2	3	4	1			4	5			1		3	4	4	8 - 11	10 - 13	13 - 16	1	2	3	2	3	
Mild	-1.5				3 - 7	10 - 13	23 - 26			4	1	2	3				3						2	3		7	9	11 - 12		1	2			
Severe	-2.0			0	0 - 2	0 - 9	13 - 22	2	2	3	0	1	2				2	4	5	0		1	1	2	3	4 - 6	5 - 8	7 - 10	0	0	1	1	2	3
	-3.0						0 - 12	0 - 1	0 - 1	0 - 2		0	0 - 1	0	0 - 1	0 - 1	0 - 1	0 - 3	0 - 4		0	0	0	0 - 1	0 - 2	0 - 3	0 - 4	0 - 6			0	0	0 - 1	0 - 2
Age 60-69 years old																																		
Normal	3.0	3	2 - 3	1 - 3	47 - 60	60		7 - 8	8																	24 - 40	27 - 40	31 - 40						
	2.0				38 - 46	50 - 59	59 - 60	6	7	8	8	7 - 8	8													20 - 23	24 - 26	27 - 30	5					
	1.5	2			34 - 37	46 - 49					7															19	22 - 23	25 - 26		5				
	1.0		1		23 - 33	34 - 45	47 - 58	5	6	7	5 - 6	6	7				6			2						15 - 18	18 - 21	21 - 24	3 - 4	4	5	5	5	5
	0.0	1	0	0	13 - 22	23 - 33	35 - 46	4	5	5 - 6	4 - 3	4 - 5	5 - 6	2	2	2	4 - 5	6	6	1	2	2	4 - 6	5 - 6	5 - 6	11 - 14	14 - 17	16 - 20	2	3	4	3 - 4	4	4
	-1.0	0			2 - 12	11 - 22	23 - 34	3	4		2	3	4	1	1		3	5			1		2 - 3	4	4	8 - 10	10 - 13	12 - 15	1	2	3	2	3	
Mild	-1.5				0 - 1	8 - 10	19 - 22	2		4	1	2	3						5			1		3		7	8 - 9	11	0	1	2		2	
Severe	-2.0					0 - 7	8 - 18		3	3	0	0 - 1	2				2	4		0			0 - 1	2	3	4 - 6	5 - 7	7 - 10		0	1	0 - 1		3
	-3.0						0 - 7	0 - 1	0 - 2	0 - 2			0 - 1	0	0	0 - 1	0 - 1	0 - 3	0 - 4		0	0		0 - 1	0 - 2	0 - 3	0 - 4	0 - 6			0		0 - 1	0 - 2
Age 70-79 years old																																		
Normal	3.0		2 - 3	2 - 3	38 - 60	53 - 60		7 - 8	8		8															23 - 40	25 - 40	30 - 40						
	2.0	3		1	30 - 37	44 - 52	59 - 60	6	7	8	7	8	8													20 - 22	22 - 24	26 - 29	5					
	1.5				27 - 29	40 - 43	55 - 58		6	7	6	7														19	21	25	4	5				
	1.0	2	1		19 - 26	29 - 39	43 - 54	5	5	6	5	5 - 6	6 - 7	2			6			2			5 - 6			15 - 18	16 - 20	20 - 24	3	4	5	5		
	0.0	1	0	0	10 - 18	18 - 28	31 - 42	3 - 4	4	5	3 - 4	3 - 4	4 - 5		2	2	4 - 5	5 - 6	6	1	2	2	3 - 4	4 - 6	4 - 6	11 - 14	12 - 15	16 - 19	2	2 - 3	3 - 4	3 - 4	3 - 5	4 - 5
	-1.0	0			2 - 9	7 - 17	19 - 30	2	3	4	1 - 2	2	3	1			3		5		1	1	2	3		7 - 10	8 - 11	11 - 15	1	1	2	2		
Mild	-1.5				0 - 1	3 - 6	15 - 18	1	2		0	1	2					4		0			1		3	6	7	10	0			1	2	3
Severe	-2.0					0 - 2	5 - 14	0		3		0	0 - 1	0	1		2	3	4		0		0	1 - 2	2	3 - 5	4 - 6	6 - 9		0	1	0	1	
	-3.0						0 - 4		0 - 1	0 - 2					0	0 - 1	0 - 1	0 - 2	0 - 3			0		0	0 - 1	0 - 2	0 - 3	1 - 5			0		0	0 - 2
Age 80 or more years old																																		
Normal	3.0		3	3	28 - 60	42 - 60		7 - 8	8	8	8	8														21 - 40	21 - 40	27 - 40						
	2.0			2	26 - 27	34 - 41	49 - 60	6	7	7	6 - 7	7	7 - 8													18 - 20	18 - 20	23 - 26	5					
	1.5	3	2		24 - 25	31 - 33	44 - 48		6		5	6								2			6			17	17	21 - 22	4	5				
	1.0	2		1	15 - 23	21 - 30	31 - 43	5	5	6	4	5	5 - 6	2			6				2	2	5	6	6	13 - 16	13 - 16	18 - 20	3	4	5	4 - 5	5	
	0.0	1		0	7 - 14	12 - 20	19 - 30	3 - 4	4	4 - 5	3	2 - 4	3 - 4	1	2	2	4 - 5	5 - 6	5 - 6	1	1	1	2 - 4	3 - 5	4 - 5	9 - 12	9 - 12	17	2	2 - 3	3 - 4	3	3 - 4	4 - 5
	-1.0	0	0 - 1		0 - 6	3 - 11	6 - 18	2	3	3	1 - 2	1	1 - 2		1		2 - 3	4		0			1	2	3	5 - 8	5 - 8	8 - 16	0 - 1	1	2	2	2	
Mild	-1.5					0 - 2	2 - 5	1	2		0	0	0					3	4						2	4	4	7		0	1	1		3
Severe	-2.0						0 - 1	0		2				0		1		2			0	0	0	0 - 1	1	1 - 3	1 - 3	3 - 6			0	0	1	2
	-3.0								0 - 1	0 - 1					0	0	0 - 1	0 - 1	0 - 3						0	0	0	0 - 2				0	0 - 1

Table 4. Composite Score Norms and Z-Scores, by Age Group and Education Level

		2018 & 2021 5 measures (CCCE Score Range 0-80)			2018 8 measures (CCCE Score Range 0-100)			2021 8 measures (CCCE Score Range 0-100)		
Z-Score		Education (years) 0	1 to 6	7+	0	1 to 6	7+	0	1 to 6	7+
Age =<59 years old										
Normal	3.0	66 - 80			81 - 100			79 - 100	93 - 100	
	2.0	55 - 65	69 - 80		71 - 80	85 - 100	99 - 100	66 - 78	81 - 92	94 - 100
	1.5	51 - 54	65 - 68	77 - 80	66 - 70	81 - 84	95 - 98	61 - 65	75 - 80	89 - 93
	1.0	40 - 50	52 - 64	65 - 76	57 - 65	67 - 80	81 - 94	46 - 60	58 - 74	72 - 88
	0.0	28 - 39	39 - 51	53 - 64	37 - 56	53 - 66	67 - 80	37 - 45	50 - 57	64 - 71
	-1.0	16 - 27	26 - 38	40 - 52	27 - 36	41 - 52	54 - 66	28 - 36	40 - 49	54 - 63
Mild	-1.5	12 - 15	22 - 25	36 - 39	23 - 26	35 - 40	50 - 53	23 - 27	34 - 39	49 - 53
Severe	-2.0	0 - 11	10 - 21	25 - 35	14 - 22	23 - 34	38 - 49	13 - 22	22 - 33	36 - 48
	-3.0		0 - 9	0 - 24	0 - 13	0 - 22	0 - 37	0 - 12	0 - 21	0 - 35
Age 60-69 years old										
Normal	3.0	63 - 80	75 - 80		76 - 100	92 - 100		74 - 100	91 - 100	
	2.0	51 - 62	64 - 74	80	64 - 75	81 - 91	98 - 100	64 - 73	78 - 90	93 - 100
	1.5	47 - 50	60 - 63	75 - 79	60 - 63	76 - 80	93 - 97	56 - 63	72 - 77	87 - 92
	1.0	35 - 46	48 - 59	62 - 74	47 - 59	62 - 75	79 - 92	49 - 55	62 - 71	76 - 86
	0.0	23 - 34	35 - 47	49 - 61	34 - 46	48 - 61	64 - 78	33 - 48	46 - 61	61 - 75
	-1.0	11 - 22	23 - 34	36 - 48	21 - 33	35 - 47	50 - 63	23 - 32	36 - 45	51 - 60
Mild	-1.5	7 - 10	18 - 22	32 - 35	17 - 20	30 - 34	45 - 49	18 - 22	31 - 35	44 - 50
Severe	-2.0	0 - 6	7 - 17	21 - 31	0 - 16	19 - 29	33 - 44	8 - 17	18 - 30	31 - 43
	-3.0		0 - 6	0 - 20		0 - 18	0 - 32	0 - 7	0 - 17	0 - 30
Age 70-79 years old										
Normal	3.0	51 - 80	68 - 80		65 - 100	85 - 100		67 - 100	81 - 100	93 - 100
	2.0	42 - 50	57 - 67	74 - 80	56 - 64	73 - 84	92 - 100	54 - 66	68 - 80	86 - 92
	1.5	39 - 41	53 - 56	70 - 73	52 - 55	68 - 72	87 - 91	49 - 53	62 - 67	78 - 85
	1.0	29 - 38	41 - 52	57 - 69	41 - 51	55 - 67	73 - 86	40 - 48	53 - 61	69 - 77
	0.0	19 - 28	29 - 40	44 - 56	29 - 40	41 - 54	58 - 72	27 - 39	38 - 52	52 - 68
	-1.0	9 - 18	17 - 28	31 - 43	18 - 28	28 - 40	44 - 57	18 - 26	28 - 37	41 - 51
Mild	-1.5	5 - 8	13 - 16	27 - 30	15 - 17	23 - 27	39 - 43	13 - 17	23 - 27	35 - 40
Severe	-2.0	0 - 4	0 - 12	15 - 26	0 - 14	12 - 22	26 - 38	0 - 12	10 - 22	21 - 34
	-3.0			0 - 14		0 - 11	0 - 25		0 - 9	0 - 20
Age 80 or more years old										
Normal	3.0	45 - 80	56 - 80	75 - 80	53 - 100	71 - 100	80 - 100	61 - 100	70 - 100	84 - 100
	2.0	37 - 44	46 - 55	62 - 74	46 - 52	60 - 70	71 - 79	50 - 60	58 - 69	75 - 83
	1.5	33 - 36	43 - 45	58 - 61	44 - 45	56 - 59	68 - 70	44 - 49	52 - 57	68 - 74
	1.0	24 - 32	32 - 42	44 - 57	34 - 43	44 - 55	59 - 67	28 - 43	43 - 51	57 - 67
	0.0	14 - 23	26 - 31	36 - 43	23 - 33	32 - 43	49 - 58	21 - 27	28 - 42	40 - 56
	-1.0	5 - 13	10 - 25	15 - 35	13 - 22	20 - 31	42 - 48	12 - 20	18 - 27	29 - 39
Mild	-1.5	0 - 4	7 - 9	10 - 14	10 - 12	16 - 19	26 - 41	7 - 11	13 - 17	24 - 28
Severe	-2.0		0 - 6	0 - 9	0 - 9	0 - 15	0 - 25	0 - 6	5 - 12	13 - 23
	-3.0								0 - 4	0 - 12

b. Proxy Interviews

The MHAS uses a proxy-cognitive section (Section PC) for participants unable to complete the Cognitive Function in Direct Interviews (Section E). This section includes questions on daily functioning based on the IQCODE (Jorm, 1994) and relies on the proxy respondent's observations to assess the participant's cognitive status compared to two years prior. The MHAS utilizes a brief 16-item version of the IQCODE, focusing on observed changes in behavior and functionality, as cognitive decline often manifests gradually across various activities. While some questions may seem repetitive, they provide different indicators of cognitive deterioration.

The questions cover general cognitive areas such as memory, judgment, organization, and daily activities, as well as more specific aspects like family memory, spatial orientation, and the ability to use new devices. Other cognitive function measures were included in this section, including global ratings on behavioral problems. These assess the frequency with which the proxy informant rates the presence of different psychiatric symptoms: aggressive and self-harm behavior, sleeping disorders, wandering behavior and paranoid symptoms.

More information about the proxy interview cognitive section is provided in the 2001-2015 document "The Mexican Health and Aging Study: Cognitive Function Measures Scoring and Classification Across Waves 2001-2015" available here.

Different from the Cognitive function battery used during direct interviews (Section E), the proxy-cognitive section (Section PC) has not changed across waves.

There is a total of 16 items that are used to calculate a composite score as the average score of all these items, resulting in a range of 1 to 5. To calculate the score for each item, three variables need to be considered to determine the presence of each symptom: 1 "Much Improved", 2" A bit improved", 3 "Not much changed", 4 "A bit worse", 5 "Much worse". In Table 6, we provide notes on the calculation of the scores using as an example the first item "Remembering things about family & friends" which uses questions pc5, pc6, pc7. The resulting constructed variable is 'iqcode1'. The same algorithm is applied for the other items.

To classify participants based on the proxy interview, we adopted a cut-off point of 3.4 in the IQCODE Composite Score, as proposed by (Cherbuin and Jorm 2017) for community settings with a sensitivity of 84% and specificity of 80%. The cognitive function categories are "Normal" (below 3.4) or "Impaired" (3.4 and above).

References

Cherbuin, N.and A.F. Jorm. 2017. "The IQCODE: Using Informant Reports to Assess Cognitive Change in the Clinic and in Older Individuals Living in the Community." Pp. 275-295 in *Cognitive Screening Instruments*.

Glosser, G., N. Wolfe, M. Albert, L. Lavine, J. Steele, D. Calne, and B. Schoenberg. 1993. "Cross-cultural cognitive examination: validation of a dementia screening instrument for neuroepidemiological research." *Journal of the American Geriatrics Society* 41(9):931-939.

Jorm, A. 1994. "A short form of the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE): development and cross-validation." *Psychological medicine* 24(01):145-153.

Mejia-Arango, S., R. Wong, and A. Michaels-Obregon. 2015. "Normative and standardized data for cognitive measures in the Mexican Health and Aging Study." *Salud Publica Mex* 57 Suppl 1:S90-96.

Appendix I. STATA Codes for the Creation of the 2018 Cognitive Function Variables (Direct and Proxy Interviews)

The following STATA codes can be used to construct the score variable for each task included in the 2018 wave.

```
**********************************************************
**** MHAS 2018 COGNITIVE EXERCISES - CCCE (SECTION E) ****
**********************************************************

**** CREATING VARIABLES BY TASK AND DOMAIN ****

* NOTE: Special missing values were assigned as follows:
* .p indicates proxy interview, .d don't know, .skip, .r refusal and .v indicates visual limination, and .l physical limitation

*** INMEDIATE MEMORY ***
*** Repeat 8 words (score 0-8) ***
recode e7a1_tot_18 e7a2_tot_18 e7a3_tot_18 e7b1_tot_18 e7b2_tot_18 e7b3_tot_18 (9=0)
egen iwr_18=rowmean (e7a1_tot_18 e7a2_tot_18 e7a3_tot_18 e7b1_tot_18 e7b2_tot_18 e7b3_tot_18) if [inrange(e7a1_tot_18,0,8) | inrange(e7a2_tot_18,0,8) | inrange(e7a3_tot_18,0,8) | inrange(e7b1_tot_18,0,8) | inrange(e7b2_tot_18,0,8) | inrange(e7b3_tot_18,0,8)]  & inlist(tipent_18,1,2)
replace iwr_18 = .r if resul_ec_18 == 4 | e6_18 == 8
replace iwr_18 = .i if resul_ec_18 == 3
replace iwr_18 = .p if inlist(tipent_18,3,4)
label variable iwr_18 "MHAS 2018 Immediate Memory - Repeat 8 words (0-8)"

*** ATTENTION ***
*** Visual Scan (score 0-60) ***
gen visual_scan_18=.
replace visual_scan_18 = .s if e10_c_18 == .s
replace visual_scan_18 = .l if e10_c_18 == 80
replace visual_scan_18 = .d if e10_c_18 == 99
replace visual_scan_18 = .r if resul_ec_18 == 4 | e10_c_18 == 88
```

```
replace visual_scan_18 = .i if resul_ec_18 == 3
replace visual_scan_18 = .p if inlist(tipent_18,3,4)
replace visual_scan_18 = e10_c_18 if inrange(e10_c_18, 0,60)
replace visual_scan_18 = 0 if e10a_18 == 2
label variable visual_scan_18 "MHAS 2018 Attention - Visual Scan (0-60)"

*** ORIENTATION ***
*** Day of the month (score 0-1) ***
gen name_day_18=.
replace name_day_18 = .d if e11a_18 == 9
replace name_day_18 = .r if resul_ec_18 == 4 | e11a_18 == 8
replace name_day_18 = .i if resul_ec_18 == 3
replace name_day_18 = .p if inlist(tipent_18,3,4)
replace name_day_18 = 1 if e11a_18 == 1
replace name_day_18 = 0 if e11a_18 == 2

*** Month (score 0-1) ***
gen name_month_18=.
replace name_month_18 = .d if e11b_18 == 9
replace name_month_18 = .r if resul_ec_18 == 4 | e11b_18 == 8
replace name_month_18 = .i if resul_ec_18 == 3
replace name_month_18 = .p if inlist(tipent_18,3,4)
replace name_month_18 = 1 if e11b_18 == 1
replace name_month_18 = 0 if e11b_18 == 2

*** Year (score 0-1) ***
gen name_year_18=.
replace name_year_18 = .d if e11c_18 == 9
replace name_year_18 = .r if resul_ec_18 == 4 | e11c_18 == 8
replace name_year_18 = .i if resul_ec_18 == 3
replace name_year_18 = .p if inlist(tipent_18,3,4)
replace name_year_18 = 1 if e11c_18 == 1
replace name_year_18 = 0 if e11c_18 == 2

*** Orientation (score 0-3)
egen orientation_18= rowtotal(name_day_18 name_month_18 name_year_18) if inlist(tipent_18,1,2)
replace orientation_18 = .d if e11a_18 == 9 & e11b_18 == 9 & e11c_18 == 9
replace orientation_18 = .r if resul_ec_18 == 4 | e11a_18 == 8 | e11b_18 == 8 | e11c_18 == 8
```

```
replace orientation_18 = .i if resul_ec_18 == 3
replace orientation_18 = .p if inlist(tipent_18,3,4)
label variable orientation_18 "MHAS 2018 Orientation (0-3)"

*** EXECUTIVE FUNCTION ***
*** Verbal fluency (score 0-4) ***
** Number of repeated animals
gen verbal_fluency_rep_18= e9b_c_18 if inrange(e9b_c_18,0,60)
replace verbal_fluency_rep_18 = .s if e9a_18 == 2
replace verbal_fluency_rep_18 = .d if e9b_c_18 == 88
replace verbal_fluency_rep_18 = .r if resul_ec_18 == 4 | e9b_c_18==99
replace verbal_fluency_rep_18 = .i if resul_ec_18 == 3
replace verbal_fluency_rep_18 = .p if inlist(tipent_18,3,4)
label variable verbal_fluency_rep_18 "MHAS 2018 Executive function - Verbal Fluency: number of repeted animals (0-50)"

** Number of different animals (88 "No response" recorded as a missing)
gen verbal_fluency_num_18=e9a_c_18
replace verbal_fluency_num_18 = 0 if e9a_18 == 2
replace verbal_fluency_num_18 = .d if e9a_c_18 == 88
replace verbal_fluency_num_18 = .r if resul_ec_18 == 4 | e9a_c_18==99
replace verbal_fluency_num_18 = .i if resul_ec_18 == 3
replace verbal_fluency_num_18 = .p if inlist(tipent_18,3,4)
label variable verbal_fluency_num_18 "MHAS 2018 Executive function - Verbal Fluency: number of animals (0-50)"

** Categorical variable (using S. Mejia 2014 SPM)
gen verbal_fluency_18=.
replace verbal_fluency_18 = 0 if e9a_18 == 2
replace verbal_fluency_18=1 if inrange(verbal_fluency_num_18,0,8)
replace verbal_fluency_18=2 if inrange(verbal_fluency_num_18,9,18)
replace verbal_fluency_18=3 if inrange(verbal_fluency_num_18,19,24)
replace verbal_fluency_18=4 if inrange(verbal_fluency_num_18,25,50)
replace verbal_fluency_18=.r if resul_ec_18 == 4 | e9a_c_18 == 88
replace verbal_fluency_18=.i if resul_ec_18 == 3
replace verbal_fluency_18=.p if inlist(tipent_18,3,4)
label variable verbal_fluency_18 "MHAS 2018 Executive function - Verbal Fluency (0-4)"
```

```
*** Serial 7 (score 0-5) ***
** Correct/Incorrect
gen serial7_1=.
replace serial7_1=1 if inlist(e15a_18,93,100)
replace serial7_1=0 if !inlist(e15a_18,93,100,.) | e15a_18==999
replace serial7_1=.i if e15a_18==.i
replace serial7_1=.p if inlist(tipent_18,3,4)
replace serial7_1=.r if e15a_18==888

gen serial7_2=.
replace serial7_2=1 if e15b_18==e15a_18-7 & e15b_18!=.
replace serial7_2=0 if !inlist(e15b_18,e15a_18-7,.) | e15b_18==999
replace serial7_2=.p if inlist(tipent_18,3,4)
replace serial7_2=.i if e15b_18==.i
replace serial7_2=.r if e15b_18==888

gen serial7_3=.
replace serial7_3=1 if e15c_18==e15b_18-7 & e15c_18!=.
replace serial7_3=0 if !inlist(e15c_18,e15b_18-7,.) | e15c_18==999
replace serial7_3=.p if inlist(tipent_18,3,4)
replace serial7_3=.i if e15c_18==.i
replace serial7_3=.r if e15c_18==888

gen serial7_4=.
replace serial7_4=1 if e15d_18==e15c_18-7 & e15d_18!=.
replace serial7_4=0 if !inlist(e15d_18,e15c_18-7,.) | e15d_18==999
replace serial7_4=.p if inlist(tipent_18,3,4)
replace serial7_4=.i if e15d_18==.i
replace serial7_4=.r if e15d_18==888

gen serial7_5=.
replace serial7_5=1 if e15e_18==e15d_18-7 & e15e_18!=.
replace serial7_5=0 if !inlist(e15e_18,e15d_18-7,.) | e15e_18==999
replace serial7_5=.p if inlist(tipent_18,3,4)
replace serial7_5=.i if e15e_18==.i
replace serial7_5=.r if e15e_18==888
```

```
** Serial 7 score variable
egen serial7_18 = rowtotal(serial7_1 serial7_2 serial7_3 serial7_4 serial7_5) if inlist(serial7_1,0,1) |
inlist(serial7_2,0,1) | inlist(serial7_3,0,1) | inlist(serial7_4,0,1) | inlist(serial7_5,0,1)
replace serial7_18=.r if e15a_18==888 | resul_ec_18 == 4
replace serial7_18=.p if inlist(tipent_18,3,4)
replace serial7_18=.i if resul_ec_18 == 3
label variable serial7_18 "MHAS Executive Function - Series of subtractions 7 from 100 (0-5)"


*** CONSTRUCTIONAL PRAXIS ***
*** Copy one figure (score 0-6) ***
gen construction_18= .
replace construction_18 = .s if e8a_c_18==.s
replace construction_18 = .r if e8a_c_18 == 8 | resul_ec_18 == 4
replace construction_18 = .d if e8a_c_18 == 9
replace construction_18 = .i if resul_ec_18 == 3
replace construction_18 = .p if inlist(tipent_18,3,4)
replace construction_18 = e8a_c_18 if inrange(e8a_c_18,0,6)
replace construction_18 = 0 if e8a_18 == 2
label variable construction_18 "MHAS 2018 Constructional Praxis (0-6)"

*** Copy one figure comparable to 2018 & 2003 (score 0-2) ***
gen construction_v01_18= .
replace construction_v01_18 = .s if e8a_c_18==.s
replace construction_v01_18 = .r if e8a_c_18 == 8 | resul_ec_18 == 4
replace construction_v01_18 = .d if e8a_c_18 == 9
replace construction_v01_18 = .i if resul_ec_18 == 3
replace construction_v01_18 = .p if inlist(tipent_18,3,4)
replace construction_v01_18 = 0 if inrange(e8a_c_18,0,2)
replace construction_v01_18 = 0 if e8a_18 == 2
replace construction_v01_18 = 1 if inrange(e8a_c_18,3,4)
replace construction_v01_18 = 2 if inrange(e8a_c_18,5,6)
label variable construction_v01_18 "MHAS 2018 Constructional Praxis v2001-2003 (0-2)"


*** DELAYED MEMORY ***
*** Repeat 8 words (score 0-8) ***
```

```
egen dwr_18=rowtotal (e14a_tot_18 e14b_tot_18) if inlist(tipent_18,1,2)
replace dwr_18 = 0 if e14a_tot_18 == 9 | e14b_tot_18 ==9
replace dwr_18 = .r if resul_ec_18 == 4 | e6_18 == 8
replace dwr_18 = .i if resul_ec_18 == 3
replace dwr_18 = .p if inlist(tipent_18,3,4)
label variable dwr_18 "MHAS 2018 Delayed Memory - Repeat 8 words (0-8)"

*** Delayed recall/copy of figure (score 0-6) ***
gen construction_m_18= .
replace construction_m_18 = .s if e13a_c_18==.s
replace construction_m_18 = .d if e13a_c_18 == 9
replace construction_m_18 = .r if e13a_c_18 == 8 | resul_ec_18 == 4
replace construction_m_18 = .i if resul_ec_18 == 3
replace construction_m_18 = .p if inlist(tipent_18,3,4)
replace construction_m_18 = e13a_c_18 if inrange(e13a_c_18,0,6)
replace construction_m_18 = 0 if e13a_18==2
label variable construction_m_18 "MHAS 2018 Delayed Memory - Delayed recall of figure (0-6)"

*** Delayed recall/copy of figure comparable to 2001 & 2003 (score 0-2) ***
gen construction_m_v01_18= .
replace construction_m_v01_18 = .s if e13a_c_18==.s
replace construction_m_v01_18 = .d if e13a_c_18 == 9
replace construction_m_v01_18 = .r if e13a_c_18 == 8 | resul_ec_18 == 4
replace construction_m_v01_18 = .i if resul_ec_18 == 3
replace construction_m_v01_18 = .i if e13a_c_18 == .i & resul_ec_18 == 2
replace construction_m_v01_18 = .p if inlist(tipent_18,3,4)
replace construction_m_v01_18 = 0 if inrange(e13a_c_18,0,2)
replace construction_m_v01_18 = 0 if e13a_18==2
replace construction_m_v01_18 = 1 if inrange(e13a_c_18,3,4)
replace construction_m_v01_18 = 2 if inrange(e13a_c_18,5,6)
label variable construction_m_v01_18 "MHAS 2018 Delayed Memory - Delayed recall of figure v2001-2003 (0-
2)"


*** TOTAL CCCE SCORE ***
*** NUMBER of Tasks missing (out of 8 tasks) ***
egen ccce_miss_18=rowmiss(iwr_18 dwr_18 visual_scan_18 orientation_18 verbal_fluency_18
construction_18 construction_m_18 serial7_18) ///
```

```
                                                    if inlist(tipent_18,1,2)
replace ccce_miss_18=.p if inlist(tipent_18,3,4)
label variable ccce_miss_18 "MHAS 2018 Number of CCCE Tasks missing (0-8)"


*** TOTAL MHAS 2018 CCCE SCORE (score 0-99) - Including all 2018 Items ***
egen ccce_18 = rowtotal(iwr_18 dwr_18 visual_scan_18 orientation_18 verbal_fluency_18 construction_18
construction_m_18 serial7_18) if inlist(tipent_18,1,2)
replace ccce_18 = .d  if [iwr_18==.d ///
                                                          | dwr_18==.d ///
                                                          | visual_scan_18==.d ///
                                                          | orientation_18==.d ///
                                                          | verbal_fluency_18==.d ///
                                                          | construction_18==.d ///
                                                          | construction_m_18==.d ///
                                                          | serial7_18==.d] & ccce_18==.
replace ccce_18 = .r   if [iwr_18==.r ///
                                                          | dwr_18==.r ///
                                                          | visual_scan_18==.r ///
                                                          | orientation_18==.r ///
                                                          | verbal_fluency_18==.r ///
                                                          | construction_18==.r ///
                                                          | construction_m_18==.r ///
                                                          | serial7_18==.r] & ccce_18==.
replace ccce_18 = .s if [e3a_18 == 2 | inlist(e5_18,1,2,3) | inlist(e8a_18,2)] & ccce_18==.
replace ccce_18 = .r if resul_ec_18==4 & ccce_18==.
replace ccce_18 = .i if resul_ec_18==3 & ccce_18==.
replace ccce_18 = .p if inlist(tipent_18,3,4)
replace ccce_18 = .m if ccce_miss_18 == 8

label variable ccce_18 "MHAS 2018 CCCE Score (0-99)"


*** NUMBER of Tasks missing (out of 5 tasks) ***
egen ccce_miss_v01_18=rowmiss(iwr_18 dwr_18 visual_scan_18 construction_v01_18
construction_m_v01_18) if inlist(tipent_18,1,2)
replace ccce_miss_v01_18=.p if inlist(tipent_18,3,4)
label variable ccce_miss_v01_18 "MHAS 2018 Number of CCCE Tasks missing - Comparable to 2001 (0-5)"
```

```
*** TOTAL MHAS 2018 CCCE SCORE (score 0-80) - With only the 5 items included since 2001***
egen ccce_v01_18 = rowtotal(iwr_18 dwr_18 visual_scan_18 construction_v01_18 construction_m_v01_18) ///
                                                    if inlist(tipent_18,1,2)
replace ccce_v01_18 = .d  if [iwr_18==.d ///
                                                    | dwr_18==.d ///
                                                    | visual_scan_18==.d ///
                                                    | construction_v01_18==.d ///
                                                    | construction_m_v01_18==.d] & ccce_v01_18==.
replace ccce_v01_18 = .r   if [iwr_18==.r ///
                                                    | dwr_18==.r ///
                                                    | visual_scan_18==.r ///
                                                    | construction_v01_18==.r ///
                                                    | construction_m_v01_18==.r] & ccce_v01_18==.

replace ccce_v01_18 = .s if [e3a_18 == 2 | inlist(e5_18,1,2,3) | inlist(e8a_18,2)] & ccce_v01_18==.
replace ccce_v01_18 = .r if resul_ec_18==4 & ccce_v01_18==.
replace ccce_v01_18 = .i if resul_ec_18==3 & ccce_v01_18==.
replace ccce_v01_18 = .p if inlist(tipent_18,3,4)
replace ccce_v01_18 = .m if ccce_miss_v01_18 == 5

label variable ccce_v01_18 "MHAS 2018 CCCE Score - Comparable to 2001 (0-80)"



***********************************************************
******* MHAS 2018 IQCODE SHORT -Jorm (SECTION PC) ********
***********************************************************

* Remembering things about family & friends
gen iqcode1_18=.
replace iqcode1_18=.r if pc5_18==8 | pc6_18==8 | pc7_18==8
replace iqcode1_18=.d if pc5_18==9 | pc6_18==9 | pc7_18==9
replace iqcode1_18=3 if pc5_18==2
replace iqcode1_18=pc6_18 if inlist(pc6_18,1,2)
replace iqcode1_18=4 if pc7_18==2
replace iqcode1_18=5 if pc7_18==1

* Remembering things that happened recently
```

```
gen iqcode2_18=.
replace iqcode2_18=.r if pc8_18==8 | pc9_18==8 | pc10_18==8
replace iqcode2_18=.d if pc8_18==9 | pc9_18==9 | pc10_18==9
replace iqcode2_18=3 if pc8_18==2
replace iqcode2_18=pc9_18 if inlist(pc9_18,1,2)
replace iqcode2_18=4 if pc10_18==2
replace iqcode2_18=5 if pc10_18==1

* Recalling conversations a few name_day_18s later
gen iqcode3_18=.
replace iqcode3_18=.r if pc11_18==8 | pc12_18==8 | pc13_18==8
replace iqcode3_18=.d if pc11_18==9 | pc12_18==9 | pc13_18==9
replace iqcode3_18=3 if pc11_18==2
replace iqcode3_18=pc12_18 if inlist(pc12_18,1,2)
replace iqcode3_18=4 if pc13_18==2
replace iqcode3_18=5 if pc13_18==1

* Remembering his/her address & phone number
gen iqcode4_18=.
replace iqcode4_18=.r if pc14_18==8 | pc15_18==8 | pc16_18==8
replace iqcode4_18=.d if pc14_18==9 | pc15_18==9 | pc16_18==9
replace iqcode4_18=3 if pc14_18==2
replace iqcode4_18=pc15_18 if inlist(pc15_18,1,2)
replace iqcode4_18=4 if pc16_18==2
replace iqcode4_18=5 if pc16_18==1

* Remembering what name_day_18 and name_month_18 it is
gen iqcode5_18=.
replace iqcode5_18=.r if pc17_18==8 | pc18_18==8 | pc19_18==8
replace iqcode5_18=.d if pc17_18==9 | pc18_18==9 | pc19_18==9
replace iqcode5_18=3 if pc17_18==2
replace iqcode5_18=pc18_18 if inlist(pc18_18,1,2)
replace iqcode5_18=4 if pc19_18==2
replace iqcode5_18=5 if pc19_18==1

* Remembering where things are usually kept
gen iqcode6_18=.
replace iqcode6_18=.r if pc20_18==8 | pc21_18==8 | pc22_18==8
```

```
replace iqcode6_18=.d if pc20_18==9 | pc21_18==9 | pc22_18==9
replace iqcode6_18=3 if pc20_18==2
replace iqcode6_18=pc21_18 if inlist(pc21_18,1,2)
replace iqcode6_18=4 if pc22_18==2
replace iqcode6_18=5 if pc22_18==1

* Remembering where to find things which have been put in a different place from usual
gen iqcode7_18=.
replace iqcode7_18=.r if pc23_18==8 | pc24_18==8 | pc25_18==8
replace iqcode7_18=.d if pc23_18==9 | pc24_18==9 | pc25_18==9
replace iqcode7_18=3 if pc23_18==2
replace iqcode7_18=pc24_18 if inlist(pc24_18,1,2)
replace iqcode7_18=4 if pc25_18==2
replace iqcode7_18=5 if pc25_18==1

* Knowing how to work familiar machines around the house
gen iqcode8a_c_18=.
replace iqcode8a_c_18=.r if pc26_18==8 | pc27_18==8 | pc28_18==8
replace iqcode8a_c_18=.d if pc26_18==9 | pc27_18==9 | pc28_18==9
replace iqcode8a_c_18=3 if pc26_18==2
replace iqcode8a_c_18=pc27_18 if inlist(pc27_18,1,2)
replace iqcode8a_c_18=4 if pc28_18==2
replace iqcode8a_c_18=5 if pc28_18==1

* Learning to use a new gadget or machine around the house
gen iqcode9_18=.
replace iqcode9_18=.r if pc29_18==8 | pc30_18==8 | pc31_18==8
replace iqcode9_18=.d if pc29_18==9 | pc30_18==9 | pc31_18==9
replace iqcode9_18=3 if pc29_18==2
replace iqcode9_18=pc30_18 if inlist(pc30_18,1,2)
replace iqcode9_18=4 if pc31_18==2
replace iqcode9_18=5 if pc31_18==1

* Learning new things in general
gen iqcode10_c_18=.
replace iqcode10_c_18=.r if pc32_18==8 | pc33_18==8 | pc34_18==8
replace iqcode10_c_18=.d if pc32_18==9 | pc33_18==9 | pc34_18==9
replace iqcode10_c_18=3 if pc32_18==2
```

```
replace iqcode10_c_18=pc33_18 if inlist(pc33_18,1,2)
replace iqcode10_c_18=4 if pc34_18==2
replace iqcode10_c_18=5 if pc34_18==1

* Following a story in a book or TV
gen iqcode11_18=.
replace iqcode11_18=.r if pc35_18==8 | pc36_18==8 | pc37_18==8
replace iqcode11_18=.d if pc35_18==9 | pc36_18==9 | pc37_18==9
replace iqcode11_18=3 if pc35_18==2
replace iqcode11_18=pc36_18 if inlist(pc36_18,1,2)
replace iqcode11_18=4 if pc37_18==2
replace iqcode11_18=5 if pc37_18==1

* Making decision on everyname_day_18 matters
gen iqcode12_18=.
replace iqcode12_18=.r if pc38_18==8 | pc39_18==8 | pc40_18==8
replace iqcode12_18=.d if pc38_18==9 | pc39_18==9 | pc40_18==9
replace iqcode12_18=3 if pc38_18==2
replace iqcode12_18=pc39_18 if inlist(pc39_18,1,2)
replace iqcode12_18=4 if pc40_18==2
replace iqcode12_18=5 if pc40_18==1

* Handling money for shopping
gen iqcode13a_c_18=.
replace iqcode13a_c_18=.r if pc41_18==8 | pc42_18==8 | pc43_18==8
replace iqcode13a_c_18=.d if pc41_18==9 | pc42_18==9 | pc43_18==9
replace iqcode13a_c_18=3 if pc41_18==2
replace iqcode13a_c_18=pc42_18 if inlist(pc42_18,1,2)
replace iqcode13a_c_18=4 if pc43_18==2
replace iqcode13a_c_18=5 if pc43_18==1

* Handling financial matters
gen iqcode14_18=.
replace iqcode14_18=.r if pc44_18==8 | pc45_18==8 | pc46_18==8
replace iqcode14_18=.d if pc44_18==9 | pc45_18==9 | pc46_18==9
replace iqcode14_18=3 if pc44_18==2
replace iqcode14_18=pc45_18 if inlist(pc45_18,1,2)
replace iqcode14_18=4 if pc46_18==2
```

```
replace iqcode14_18=5 if pc46_18==1

* Handling other everyname_day_18 arithmetic problems
gen iqcode15_18=.
replace iqcode15_18=.r if pc47_18==8 | pc48_18==8 | pc49_18==8
replace iqcode15_18=.d if pc47_18==9 | pc48_18==9 | pc49_18==9
replace iqcode15_18=3 if pc47_18==2
replace iqcode15_18=pc48_18 if inlist(pc48_18,1,2)
replace iqcode15_18=4 if pc49_18==2
replace iqcode15_18=5 if pc49_18==1

* Using his/her intelligence to understand what's going on and to reason things through
gen iqcode16_18=.
replace iqcode16_18=.r if pc50_18==8 | pc51_18==8 | pc52_18==8
replace iqcode16_18=.d if pc50_18==9 | pc51_18==9 | pc52_18==9
replace iqcode16_18=3 if pc50_18==2
replace iqcode16_18=pc51_18 if inlist(pc51_18,1,2)
replace iqcode16_18=4 if pc52_18==2
replace iqcode16_18=5 if pc52_18==1


***  TOTAL IQCODE SCORE (score 1-5) ***
egen iqcode= rowmean(iqcode1_18 iqcode2_18 iqcode3_18 iqcode4_18 iqcode5_18 iqcode6_18 iqcode7_18
iqcode8a_c_18 iqcode9_18 iqcode10_c_18 iqcode11_18 iqcode12_18 iqcode13a_c_18 iqcode14_18
iqcode15_18 iqcode16_18) if inlist(tipent_18,3,4)

gen iqcode_18= round(iqcode,0.1)
replace iqcode_18 = .d if  [iqcode1_18==.d | ///
                                                    iqcode2_18==.d | ///
                                                    iqcode3_18==.d | ///
                                                    iqcode4_18==.d | ///
                                                    iqcode5_18==.d | ///
                                                    iqcode6_18==.d | ///
                                                    iqcode7_18==.d | ///
                                                    iqcode8a_c_18==.d | ///
                                                    iqcode9_18==.d | ///
                                                    iqcode10_c_18==.d | ///
                                                    iqcode11_18==.d | ///
```

```
                                                    iqcode12_18==.d | ///
                                                    iqcode13a_c_18==.d | ///
                                                    iqcode14_18==.d | ///
                                                    iqcode15_18==.d | ///
                                                    iqcode16_18==.d]

replace iqcode_18 = .r if  [iqcode1_18==.r | ///
                                                    iqcode2_18==.r | ///
                                                    iqcode3_18==.r | ///
                                                    iqcode4_18==.r | ///
                                                    iqcode5_18==.r | ///
                                                    iqcode6_18==.r | ///
                                                    iqcode7_18==.r | ///
                                                    iqcode8a_c_18==.r | ///
                                                    iqcode9_18==.r | ///
                                                    iqcode10_c_18==.r | ///
                                                    iqcode11_18==.r | ///
                                                    iqcode12_18==.r | ///
                                                    iqcode13a_c_18==.r | ///
                                                    iqcode14_18==.r | ///
                                                    iqcode15_18==.r | ///
                                                    iqcode16_18==.r]
replace iqcode_18 = .p if inlist(tipent_18,1,2)
label variable iqcode_18 "MHAS 2018 IQCODE Score (1-5)"


*** COGNITIVE IMPAIRMENT - IQCODE (only proxy interviews) ***
gen cog_imp_iqcode_18 = .
replace cog_imp_iqcode_18=0 if inrange(iqcode_18,1,3.4)
replace cog_imp_iqcode_18=1 if inrange(iqcode_18,3.4,5)
replace cog_imp_iqcode_18 = .d if iqcode_18==.d
replace cog_imp_iqcode_18 = .r if iqcode_18==.r
replace cog_imp_iqcode_18 = .n if inlist(tipent_18,1,2)
replace cog_imp_iqcode_18 = .i if inlist(tipent_18,3,4) & pc1_18==.i
label variable cog_imp_iqcode_18 "MHAS 2018 IQ CODE Cognitive Impairment - Normal, Impaired"
label define imp_iqcode 0 "Normal" 1 "Impaired", replace
label values cog_imp_iqcode_18 imp_iqcode
```

drop name_day_18 name_month_18 name_year_18 serial7_1-serial7_5 iqcode*_18 iqcode

Appendix II. STATA Codes for the Creation of the 2018 Z-Scores and the Classification of Subjects

The following STATA codes can be used to construct the z-scores for each task and total cognition score included in the 2018 wave. In addition, we provide the code for the cognitive function classification (categories Normal, CIND, Dementia).

```
*********************************************************
**** MHAS 2018 COGNITIVE EXERCISES - CCCE (SECTION E) ****
*********************************************************

*** CREATING Z-SCORES & IMPAIREMENT STATUS VARIABLES ***
***         BY TOTAL SCORE, TASK & DOMAIN           ***

*** CCCE Total Score (0-80) - Including only the items in 2001 *** ***
** Z-SCORES **
gen ccce_z_score_v01_18=.
replace             ccce_z_score_v01_18=-3.0      if [
                                    [inrange(ccce_v01_18,0,10)  & inrange(yrschool,1,6)] |
[inrange(ccce_v01_18,0,25)  & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace             ccce_z_score_v01_18=-2.0      if [[inrange(ccce_v01_18,0,12)  & yrschool==0]
| [inrange(ccce_v01_18,10,22) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,25,36) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace             ccce_z_score_v01_18=-1.5      if [[inrange(ccce_v01_18,12,16) & yrschool==0]
| [inrange(ccce_v01_18,22,26) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,36,40) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace          ccce_z_score_v01_18=-1.0         if [[inrange(ccce_v01_18,16,28) & yrschool==0]
| [inrange(ccce_v01_18,26,39) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,40,53) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
```

```
replace      ccce_z_score_v01_18=0.0      if [[inrange(ccce_v01_18,28,40) & yrschool==0]
| [inrange(ccce_v01_18,39,52) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,53,65) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace      ccce_z_score_v01_18=1.0      if [[inrange(ccce_v01_18,40,51) & yrschool==0]
| [inrange(ccce_v01_18,52,65) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,65,77) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace      ccce_z_score_v01_18=1.5      if [[inrange(ccce_v01_18,51,55) & yrschool==0]
| [inrange(ccce_v01_18,65,69) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,77,80) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace      ccce_z_score_v01_18=2.0      if [[inrange(ccce_v01_18,55,66) & yrschool==0]
| [inrange(ccce_v01_18,69,80) & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,17,59)
replace      ccce_z_score_v01_18=3.0      if [[inrange(ccce_v01_18,66,80) & yrschool==0]

                                                                          ] &
inrange(age_18,17,59)

replace      ccce_z_score_v01_18=-3.0     if [
                        [inrange(ccce_v01_18,0,7)   & inrange(yrschool,1,6)] |
[inrange(ccce_v01_18,0,21)  & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=-2.0     if [[inrange(ccce_v01_18,0,7)   & yrschool==0] |
[inrange(ccce_v01_18,7,18)  & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,21,32) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=-1.5     if [[inrange(ccce_v01_18,7,11)  & yrschool==0]
| [inrange(ccce_v01_18,18,23) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,32,36) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=-1.0     if [[inrange(ccce_v01_18,11,23) & yrschool==0]
| [inrange(ccce_v01_18,23,35) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,36,49) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
```

```
replace      ccce_z_score_v01_18=0.0      if [[inrange(ccce_v01_18,23,35) & yrschool==0]
| [inrange(ccce_v01_18,35,48) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,49,62) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=1.0      if [[inrange(ccce_v01_18,35,47) & yrschool==0]
| [inrange(ccce_v01_18,48,60) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,62,75) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=1.5      if [[inrange(ccce_v01_18,47,51) & yrschool==0]
| [inrange(ccce_v01_18,60,64) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,75,80) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=2.0      if [[inrange(ccce_v01_18,51,63) & yrschool==0]
| [inrange(ccce_v01_18,64,75) & inrange(yrschool,1,6)] | [inlist(ccce_v01_18,80)     &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace      ccce_z_score_v01_18=3.0      if [[inrange(ccce_v01_18,63,80) & yrschool==0]
| [inrange(ccce_v01_18,75,80) & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,60,69)

//replace      ccce_z_score_v01_18=-3.0      if [


                ] & inrange(age_18,70,79)
replace        ccce_z_score_v01_18=-2.0     if [[inrange(ccce_v01_18,0,5)   & yrschool==0] |
[inrange(ccce_v01_18,0,7)   & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,0,10)  &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace        ccce_z_score_v01_18=-1.5     if [[inrange(ccce_v01_18,5,9)   & yrschool==0] |
[inrange(ccce_v01_18,7,10)  & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,10,15) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace      ccce_z_score_v01_18=-1.0      if [[inrange(ccce_v01_18,9,19)  & yrschool==0]
| [inrange(ccce_v01_18,10,26) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,15,36) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
```

```
replace      ccce_z_score_v01_18=0.0      if [[inrange(ccce_v01_18,19,29) & yrschool==0]
| [inrange(ccce_v01_18,26,32) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,36,44) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace      ccce_z_score_v01_18=1.0      if [[inrange(ccce_v01_18,29,39) & yrschool==0]
| [inrange(ccce_v01_18,32,43) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,44,58) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace      ccce_z_score_v01_18=1.5      if [[inrange(ccce_v01_18,39,42) & yrschool==0]
| [inrange(ccce_v01_18,43,46) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,58,61) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace      ccce_z_score_v01_18=2.0      if [[inrange(ccce_v01_18,42,51) & yrschool==0]
| [inrange(ccce_v01_18,46,56) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,61,80) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace      ccce_z_score_v01_18=3.0      if [[inrange(ccce_v01_18,51,80) & yrschool==0]
| [inrange(ccce_v01_18,56,80) & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,70,79)

//replace      ccce_z_score_v01_18=-3.0      if [



                    ] & inrange(age_18,80,120)
replace          ccce_z_score_v01_18=-2.0      if [
                            [inrange(ccce_v01_18,0,7)   & inrange(yrschool,1,6)] |
[inrange(ccce_v01_18,0,10)  & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace          ccce_z_score_v01_18=-1.5      if [[inrange(ccce_v01_18,0,5)   & yrschool==0] |
[inrange(ccce_v01_18,7,10)  & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,10,15) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace      ccce_z_score_v01_18=-1.0      if [[inrange(ccce_v01_18,5,14)  & yrschool==0]
| [inrange(ccce_v01_18,10,26) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,15,36) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
```

```
replace ccce_z_score_v01_18=0.0 if [[inrange(ccce_v01_18,14,24) & yrschool==0] | [inrange(ccce_v01_18,26,32) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,36,44) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_v01_18=1.0 if [[inrange(ccce_v01_18,24,33) & yrschool==0] | [inrange(ccce_v01_18,32,43) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,44,58) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_v01_18=1.5 if [[inrange(ccce_v01_18,33,37) & yrschool==0] | [inrange(ccce_v01_18,43,46) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,58,62) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_v01_18=2.0 if [[inrange(ccce_v01_18,37,45) & yrschool==0] | [inrange(ccce_v01_18,46,56) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,62,75) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_v01_18=3.0 if [[inrange(ccce_v01_18,45,80) & yrschool==0] | [inrange(ccce_v01_18,56,80) & inrange(yrschool,1,6)] | [inrange(ccce_v01_18,75,80) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace ccce_z_score_v01_18= ccce_v01_18 if mi(ccce_v01_18) & ccce_z_score_v01_18==.
replace ccce_z_score_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & ccce_z_score_v01_18==.
replace ccce_z_score_v01_18= .p if inlist(tipent_18,3,4)
label variable ccce_z_score_v01_18 "MHAS 2018 CCCE Z-Scores - Comparable to 2001"

** Impairment Status  **
gen cognitive_imp_v01_18 = .
label variable cognitive_imp_v01_18 "MHAS 2018 Cognitive Impairment - Normal, MCI, Impaired Comparable to 2001"
** Age <59 & Years of Education=0
replace cognitive_imp_v01_18=2 if inrange(age_18,17,59) & yrschool==0 & inrange(ccce_v01_18,0,12)
```

```
replace cognitive_imp_v01_18=1 if inrange(age_18,17,59) & yrschool==0 &
inrange(ccce_v01_18,12,16)
replace cognitive_imp_v01_18=0 if inrange(age_18,17,59) & yrschool==0 &
inrange(ccce_v01_18,16,80)

** Age <59 & Years of Education 1-6
replace cognitive_imp_v01_18=2  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,0,22)
replace cognitive_imp_v01_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,22,26)
replace cognitive_imp_v01_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,26,80)

** Age <59 & Years of Education 7+
replace cognitive_imp_v01_18=2  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,0,36)
replace cognitive_imp_v01_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,36,40)
replace cognitive_imp_v01_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,40,80)



** Age 60-69 & Years of Education=0
replace cognitive_imp_v01_18=2 if inrange(age_18,60,69) & yrschool==0 &
inrange(ccce_v01_18,0,7)
replace cognitive_imp_v01_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(ccce_v01_18,7,11)
replace cognitive_imp_v01_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(ccce_v01_18,11,80)

** Age 60-69 & Years of Education 1-6
```

```
replace cognitive_imp_v01_18=2  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,0,18)
replace cognitive_imp_v01_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,18,23)
replace cognitive_imp_v01_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,23,80)

** Age 60-69 & Years of Education 7+
replace cognitive_imp_v01_18=2  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,0,32)
replace cognitive_imp_v01_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,32,36)
replace cognitive_imp_v01_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,36,80)


** Age 70-79 & Years of Education=0
replace cognitive_imp_v01_18=2 if inrange(age_18,70,79) & yrschool==0 &
inrange(ccce_v01_18,0,5)
replace cognitive_imp_v01_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(ccce_v01_18,5,9)
replace cognitive_imp_v01_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(ccce_v01_18,9,80)

** Age 70-79 & Years of Education 1-6
replace cognitive_imp_v01_18=2 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,0,13)
replace cognitive_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,13,17)
replace cognitive_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_v01_18,17,80)
```

** Age 70-79 & Years of Education 7+
replace cognitive_imp_v01_18=2 if inrange(age_18,70,79) & inrange(yrschool,7,25) & inrange(ccce_v01_18,0,27)
replace cognitive_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) & inrange(ccce_v01_18,27,31)
replace cognitive_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) & inrange(ccce_v01_18,31,80)

** Age 80+ & Years of Education=0
//replace cognitive_imp_v01_18=2 if inrange(age_18,80,120) & yrschool==0 & inrange(ccce_v01_18,,)
replace cognitive_imp_v01_18=1 if inrange(age_18,80,120) & yrschool==0 & inrange(ccce_v01_18,0,5)
replace cognitive_imp_v01_18=0 if inrange(age_18,80,120) & yrschool==0 & inrange(ccce_v01_18,5,80)

** Age 80+ & Years of Education 1-6
replace cognitive_imp_v01_18=2 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inrange(ccce_v01_18,0,7)
replace cognitive_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inrange(ccce_v01_18,7,10)
replace cognitive_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inrange(ccce_v01_18,10,80)

** Age 80+ & Years of Education 7+
replace cognitive_imp_v01_18=2 if inrange(age_18,80,120) & inrange(yrschool,7,25) & inrange(ccce_v01_18,0,10)
replace cognitive_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) & inrange(ccce_v01_18,10,15)

```
replace cognitive_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(ccce_v01_18,15,80)

replace cognitive_imp_v01_18= ccce_v01_18 if mi(ccce_v01_18) & cognitive_imp_v01_18==.
replace cognitive_imp_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
cognitive_imp_v01_18==.
replace cognitive_imp_v01_18= .p if inlist(tipent_18,3,4)
label define cognitive_imp_v01_18 0 "0.Normal" 1 "1.MCI" 2 "2.Impaired", replace
label values cognitive_imp_v01_18 cognitive_imp_v01_18



*** CCCE Total Score (0-100) ***
** Z-SCORES **
gen ccce_z_score_18=.
replace          ccce_z_score_18=-3.0   if [[inrange(ccce_18,0,14)    & yrschool==0] |
[inrange(ccce_18,0,23)   & inrange(yrschool,1,6)] | [inrange(ccce_18,0,38)   &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace          ccce_z_score_18=-2.0   if [[inrange(ccce_18,14,23)  & yrschool==0] |
[inrange(ccce_18,23,35)  & inrange(yrschool,1,6)] | [inrange(ccce_18,38,50)  &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace          ccce_z_score_18=-1.5   if [[inrange(ccce_18,23,27)  & yrschool==0] |
[inrange(ccce_18,35,41)  & inrange(yrschool,1,6)] | [inrange(ccce_18,50,54)  &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace        ccce_z_score_18=-1.0      if [[inrange(ccce_18,27,37)  & yrschool==0] |
[inrange(ccce_18,41,53)  & inrange(yrschool,1,6)] | [inrange(ccce_18,54,67)  &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace        ccce_z_score_18=0.0               if [[inrange(ccce_18,37,57)  & yrschool==0] |
[inrange(ccce_18,53,67)  & inrange(yrschool,1,6)] | [inrange(ccce_18,67,81)  &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
```

```
replace ccce_z_score_18=1.0 if [[inrange(ccce_18,57,66) & yrschool==0] |
[inrange(ccce_18,67,81) & inrange(yrschool,1,6)] | [inrange(ccce_18,81,95) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace ccce_z_score_18=1.5 if [[inrange(ccce_18,66,71) & yrschool==0] |
[inrange(ccce_18,81,85) & inrange(yrschool,1,6)] | [inrange(ccce_18,95,99) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace ccce_z_score_18=2.0 if [[inrange(ccce_18,71,81) & yrschool==0] |
[inrange(ccce_18,85,100) & inrange(yrschool,1,6)] | [inrange(ccce_18,99,100) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace ccce_z_score_18=3.0 if [[inrange(ccce_18,81,100) & yrschool==0]

                                                                    ] &
inrange(age_18,17,59)

replace ccce_z_score_18=-3.0 if [
                [inrange(ccce_18,0,19) & inrange(yrschool,1,6)] | [inrange(ccce_18,0,33) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=-2.0 if [[inrange(ccce_18,0,17) & yrschool==0] |
[inrange(ccce_18,19,30) & inrange(yrschool,1,6)] | [inrange(ccce_18,33,45) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=-1.5 if [[inrange(ccce_18,17,21) & yrschool==0] |
[inrange(ccce_18,30,35) & inrange(yrschool,1,6)] | [inrange(ccce_18,45,50) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=-1.0 if [[inrange(ccce_18,21,34) & yrschool==0] |
[inrange(ccce_18,35,48) & inrange(yrschool,1,6)] | [inrange(ccce_18,50,64) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=0.0 if [[inrange(ccce_18,34,47) & yrschool==0] |
[inrange(ccce_18,48,62) & inrange(yrschool,1,6)] | [inrange(ccce_18,64,79) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
```

```
replace ccce_z_score_18=1.0 if [[inrange(ccce_18,47,60) & yrschool==0] | [inrange(ccce_18,62,76) & inrange(yrschool,1,6)] | [inrange(ccce_18,79,93) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=1.5 if [[inrange(ccce_18,60,64) & yrschool==0] | [inrange(ccce_18,76,81) & inrange(yrschool,1,6)] | [inrange(ccce_18,93,98) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=2.0 if [[inrange(ccce_18,64,76) & yrschool==0] | [inrange(ccce_18,81,92) & inrange(yrschool,1,6)] | [inrange(ccce_18,98,100) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace ccce_z_score_18=3.0 if [[inrange(ccce_18,76,100) & yrschool==0] | [inrange(ccce_18,92,100) & inrange(yrschool,1,6)]
                                ] & inrange(age_18,60,69)

replace ccce_z_score_18=-3.0 if [
              [inrange(ccce_18,0,12) & inrange(yrschool,1,6)] | [inrange(ccce_18,0,26) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace ccce_z_score_18=-2.0 if [[inrange(ccce_18,0,15) & yrschool==0] | [inrange(ccce_18,12,23) & inrange(yrschool,1,6)] | [inrange(ccce_18,26,39) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace ccce_z_score_18=-1.5 if [[inrange(ccce_18,15,18) & yrschool==0] | [inrange(ccce_18,23,28) & inrange(yrschool,1,6)] | [inrange(ccce_18,39,44) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace ccce_z_score_18=-1.0 if [[inrange(ccce_18,18,29) & yrschool==0] | [inrange(ccce_18,28,41) & inrange(yrschool,1,6)] | [inrange(ccce_18,44,58) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace ccce_z_score_18=0.0 if [[inrange(ccce_18,29,41) & yrschool==0] | [inrange(ccce_18,41,55) & inrange(yrschool,1,6)] | [inrange(ccce_18,58,73) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace ccce_z_score_18=1.0 if [[inrange(ccce_18,41,52) & yrschool==0] | [inrange(ccce_18,55,68) & inrange(yrschool,1,6)] | [inrange(ccce_18,73,87) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
```

```
replace         ccce_z_score_18=1.5             if [[inrange(ccce_18,52,56)  & yrschool==0] |
[inrange(ccce_18,68,73)  & inrange(yrschool,1,6)] | [inrange(ccce_18,87,92)  &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace         ccce_z_score_18=2.0             if [[inrange(ccce_18,56,65)  & yrschool==0] |
[inrange(ccce_18,73,85)  & inrange(yrschool,1,6)] | [inrange(ccce_18,92,100) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace         ccce_z_score_18=3.0             if [[inrange(ccce_18,65,100) & yrschool==0] |
[inrange(ccce_18,85,100) & inrange(yrschool,1,6)]
                                                ] & inrange(age_18,70,79)

//replace       ccce_z_score_18=-3.0      if [



        ] & inrange(age_18,80,120)
replace             ccce_z_score_18=-2.0   if [[inrange(ccce_18,0,10)   & yrschool==0] |
[inrange(ccce_18,0,16)   & inrange(yrschool,1,6)] | [inrange(ccce_18,0,26)   &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace             ccce_z_score_18=-1.5   if [[inrange(ccce_18,10,13)  & yrschool==0] |
[inrange(ccce_18,16,20)  & inrange(yrschool,1,6)] | [inrange(ccce_18,26,42)  &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         ccce_z_score_18=-1.0      if [[inrange(ccce_18,13,23)  & yrschool==0] |
[inrange(ccce_18,20,32)  & inrange(yrschool,1,6)] | [inrange(ccce_18,42,49)  &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         ccce_z_score_18=0.0             if [[inrange(ccce_18,23,34)  & yrschool==0] |
[inrange(ccce_18,32,44)  & inrange(yrschool,1,6)] | [inrange(ccce_18,49,59)  &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         ccce_z_score_18=1.0             if [[inrange(ccce_18,34,44)  & yrschool==0] |
[inrange(ccce_18,44,56)  & inrange(yrschool,1,6)] | [inrange(ccce_18,59,68)  &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
```

```
replace ccce_z_score_18=1.5 if [[inrange(ccce_18,44,46) & yrschool==0] | [inrange(ccce_18,56,60) & inrange(yrschool,1,6)] | [inrange(ccce_18,68,71) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_18=2.0 if [[inrange(ccce_18,46,53) & yrschool==0] | [inrange(ccce_18,60,71) & inrange(yrschool,1,6)] | [inrange(ccce_18,71,80) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace ccce_z_score_18=3.0 if [[inrange(ccce_18,53,100) & yrschool==0] | [inrange(ccce_18,71,100) & inrange(yrschool,1,6)] | [inrange(ccce_18,80,100) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace ccce_z_score_18= ccce_18 if mi(ccce_18) & ccce_z_score_18==.
replace ccce_z_score_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & ccce_z_score_18==.
replace ccce_z_score_18= .p if inlist(tipent_18,3,4)
label variable ccce_z_score_18 "MHAS 2018 CCCE Z-Scores"

** Impairment Status **
gen cognitive_imp_18 = .
label variable cognitive_imp_18 "MHAS 2018 Cognitive Impairment - Normal, MCI, Impaired"
** Age <59 & Years of Education=0
replace cognitive_imp_18=2 if inrange(age_18,17,59) & yrschool==0 & inrange(ccce_18,0,23)
replace cognitive_imp_18=1 if inrange(age_18,17,59) & yrschool==0 & inrange(ccce_18,23,27)
replace cognitive_imp_18=0 if inrange(age_18,17,59) & yrschool==0 & inrange(ccce_18,27,100)

** Age <59 & Years of Education 1-6
replace cognitive_imp_18=2 if inrange(age_18,17,59) & inrange(yrschool,1,6) & inrange(ccce_18,0,35)
replace cognitive_imp_18=1 if inrange(age_18,17,59) & inrange(yrschool,1,6) & inrange(ccce_18,35,41)
```

```
replace cognitive_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(ccce_18,41,100)

** Age <59 & Years of Education 7+
replace cognitive_imp_18=2  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_18,0,50)
replace cognitive_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_18,50,54)
replace cognitive_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(ccce_18,54,100)


** Age 60-69 & Years of Education=0
replace cognitive_imp_18=2 if inrange(age_18,60,69) & yrschool==0 & inrange(ccce_18,0,17)
replace cognitive_imp_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(ccce_18,17,21)
replace cognitive_imp_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(ccce_18,21,100)

** Age 60-69 & Years of Education 1-6
replace cognitive_imp_18=2  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_18,0,30)
replace cognitive_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_18,30,35)
replace cognitive_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(ccce_18,35,100)

** Age 60-69 & Years of Education 7+
replace cognitive_imp_18=2  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_18,0,39)
```

```
replace cognitive_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_18,39,44)
replace cognitive_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(ccce_18,44,100)


** Age 70-79 & Years of Education=0
replace cognitive_imp_18=2 if inrange(age_18,70,79) & yrschool==0 & inrange(ccce_18,0,15)
replace cognitive_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(ccce_18,15,18)
replace cognitive_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(ccce_18,18,100)

** Age 70-79 & Years of Education 1-6
replace cognitive_imp_18=2 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_18,0,23)
replace cognitive_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_18,23,28)
replace cognitive_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(ccce_18,28,100)

** Age 70-79 & Years of Education 7+
replace cognitive_imp_18=2 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(ccce_18,0,39)
replace cognitive_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(ccce_18,39,44)
replace cognitive_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(ccce_18,44,100)


** Age 80+ & Years of Education=0
```

```
replace cognitive_imp_18=2 if inrange(age_18,80,120) & yrschool==0 &
inrange(ccce_18,0,10)
replace cognitive_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inrange(ccce_18,10,13)
replace cognitive_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(ccce_18,13,100)

** Age 80+ & Years of Education 1-6
replace cognitive_imp_18=2 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(ccce_18,0,16)
replace cognitive_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(ccce_18,16,20)
replace cognitive_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(ccce_18,20,100)

** Age 80+ & Years of Education 7+
replace cognitive_imp_18=2 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(ccce_18,0,26)
replace cognitive_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(ccce_18,26,42)
replace cognitive_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(ccce_18,42,100)

replace cognitive_imp_18= ccce_18 if mi(ccce_18) & cognitive_imp_18==.
replace cognitive_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
cognitive_imp_18==.
replace cognitive_imp_18= .p if inlist(tipent_18,3,4)
label define cognitive_imp_18 0 "0.Normal" 1 "1.MCI" 2 "2.Impaired", replace
label values cognitive_imp_18 cognitive_imp_18
```

```
*** INMEDIATE MEMORY Z-SCORES ***
** Z-SCORES **
gen iwr_z_18=.
replace iwr_z_18=-3.0       if [[inrange(iwr_18,0,2) & yrschool==0] | [inrange(iwr_18,0,2) &
inrange(yrschool,1,6)] | [inrange(iwr_18,0,3) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)
replace iwr_z_18=-2.0       if [[inrange(iwr_18,2,3) & yrschool==0] | [inrange(iwr_18,2,3) &
inrange(yrschool,1,6)] | [inrange(iwr_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)
replace iwr_z_18=-1.5       if [

[inrange(iwr_18,4,5) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace iwr_z_18=-1.0       if [[inrange(iwr_18,3,4) & yrschool==0] | [inrange(iwr_18,3,5) &
inrange(yrschool,1,6)]
        ] & inrange(age_18,17,59)
replace iwr_z_18= 0.0       if [[inrange(iwr_18,4,6) & yrschool==0] | [inrange(iwr_18,5,6) &
inrange(yrschool,1,6)] | [inrange(iwr_18,5,7) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)
replace iwr_z_18= 1.0       if [[inrange(iwr_18,6,7) & yrschool==0] | [inrange(iwr_18,6,7) &
inrange(yrschool,1,6)]
        ] & inrange(age_18,17,59)
replace iwr_z_18= 1.5       if [
                                                        [inrange(iwr_18,7,8) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace iwr_z_18= 2.0       if [[inrange(iwr_18,7,8) & yrschool==0] | [inrange(iwr_18,7,8) &
inrange(yrschool,1,6)] | [ inlist(iwr_18,8)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace iwr_z_18= 3.0       if [[ inlist(iwr_18,8)   & yrschool==0] | [ inlist(iwr_18,8)   &
inrange(yrschool,1,6)]
        ] & inrange(age_18,17,59)
```

```
replace iwr_z_18=-3.0       if [[inrange(iwr_18,0,2) & yrschool==0] | [inrange(iwr_18,0,3) &
inrange(yrschool,1,6)] | [inrange(iwr_18,0,3) & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace iwr_z_18=-2.0       if [
[inrange(iwr_18,3,4) & inrange(yrschool,1,6)] | [inrange(iwr_18,3,4) & inrange(yrschool,7,25)]]
& inrange(age_18,60,69)
replace iwr_z_18=-1.5       if [[inrange(iwr_18,2,3) & yrschool==0] |
                                                    [inrange(iwr_18,4,5) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace iwr_z_18=-1.0       if [[inrange(iwr_18,3,4) & yrschool==0] | [inrange(iwr_18,4,5) &
inrange(yrschool,1,6)]
        ] & inrange(age_18,60,69)
replace iwr_z_18= 0.0       if [[inrange(iwr_18,4,5) & yrschool==0] | [inrange(iwr_18,5,6) &
inrange(yrschool,1,6)] | [inrange(iwr_18,5,7) & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace iwr_z_18= 1.0       if [[inrange(iwr_18,5,6) & yrschool==0] | [inrange(iwr_18,6,7) &
inrange(yrschool,1,6)]                                                                    ]
& inrange(age_18,60,69)
replace iwr_z_18= 1.5       if [

[inrange(iwr_18,7,8) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace iwr_z_18= 2.0       if [[inrange(iwr_18,6,7) & yrschool==0] | [inrange(iwr_18,7,8) &
inrange(yrschool,1,6)] | [inlist(iwr_18,8)    & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace iwr_z_18= 3.0       if [[inrange(iwr_18,7,8) & yrschool==0] | [inlist(iwr_18,8)    &
inrange(yrschool,1,6)]
] & inrange(age_18,60,69)

replace iwr_z_18=-3.0       if [
[inrange(iwr_18,0,2) & inrange(yrschool,1,6)] | [inrange(iwr_18,0,3) & inrange(yrschool,7,25)]]
& inrange(age_18,70,79)
```

```
replace iwr_z_18=-2.0       if [[inrange(iwr_18,0,1) & yrschool==0] |
                                                    [inrange(iwr_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace iwr_z_18=-1.5       if [[inrange(iwr_18,1,2) & yrschool==0] | [inrange(iwr_18,2,3) &
inrange(yrschool,1,6)]
        ] & inrange(age_18,70,79)
replace iwr_z_18=-1.0       if [[inrange(iwr_18,2,3) & yrschool==0] | [inrange(iwr_18,3,4) &
inrange(yrschool,1,6)] | [inrange(iwr_18,4,5) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace iwr_z_18= 0.0       if [[inrange(iwr_18,3,5) & yrschool==0] | [inrange(iwr_18,4,5) &
inrange(yrschool,1,6)] | [inrange(iwr_18,5,6) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace iwr_z_18= 1.0       if [[inrange(iwr_18,5,6) & yrschool==0] | [inrange(iwr_18,5,6) &
inrange(yrschool,1,6)] | [inrange(iwr_18,6,7) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace iwr_z_18= 1.5       if [
[inrange(iwr_18,6,7) & inrange(yrschool,1,6)] | [inrange(iwr_18,7,8) & inrange(yrschool,7,25)]]
& inrange(age_18,70,79)
replace iwr_z_18= 2.0       if [[inrange(iwr_18,6,7) & yrschool==0] | [inrange(iwr_18,7,8) &
inrange(yrschool,1,6)] | [ inlist(iwr_18,8)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace iwr_z_18= 3.0       if [[inrange(iwr_18,7,8) & yrschool==0] | [ inlist(iwr_18,8)   &
inrange(yrschool,1,6)]
] & inrange(age_18,70,79)

replace iwr_z_18=-3.0       if [
[inrange(iwr_18,0,2) & inrange(yrschool,1,6)] | [inrange(iwr_18,0,2) & inrange(yrschool,7,25)]]
& inrange(age_18,80,120)
replace iwr_z_18=-2.0       if [[inrange(iwr_18,0,1) & yrschool==0] |
                                                    [inrange(iwr_18,2,3) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
```

```
replace iwr_z_18=-1.5       if [[inrange(iwr_18,1,2) & yrschool==0] | [inrange(iwr_18,2,3) &
inrange(yrschool,1,6)]
] & inrange(age_18,80,120)
replace iwr_z_18=-1.0       if [[inrange(iwr_18,2,3) & yrschool==0] | [inrange(iwr_18,3,4) &
inrange(yrschool,1,6)] | [inrange(iwr_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace iwr_z_18= 0.0       if [[inrange(iwr_18,3,5) & yrschool==0] | [inrange(iwr_18,4,5) &
inrange(yrschool,1,6)] | [inrange(iwr_18,4,6) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace iwr_z_18= 1.0       if [[inrange(iwr_18,5,6) & yrschool==0] | [inrange(iwr_18,5,6) &
inrange(yrschool,1,6)] | [inrange(iwr_18,6,7) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace iwr_z_18= 1.5       if [
[inrange(iwr_18,6,7) & inrange(yrschool,1,6)]
                            ] & inrange(age_18,80,120)
replace iwr_z_18= 2.0       if [[inrange(iwr_18,6,7) & yrschool==0] | [inrange(iwr_18,7,8) &
inrange(yrschool,1,6)] | [inrange(iwr_18,7,8) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace iwr_z_18= 3.0       if [[inrange(iwr_18,7,8) & yrschool==0] | [ inlist(iwr_18,8)   &
inrange(yrschool,1,6)] | [ inlist(iwr_18,8)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace iwr_z_18= iwr_18 if mi(iwr_18) & iwr_z_18==.
replace iwr_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & iwr_z_18==.
replace iwr_z_18= .p if inlist(tipent_18,3,4)
label variable iwr_z_18 "MHAS 2018 Immediate Memory Z-Scores"

** Impairment Status **
gen iwr_imp_18 = .
label variable iwr_imp_18 "MHAS 2018 Immediate Memory Impairment"
* age_18 <59 & Years of Education=0
replace iwr_imp_18=1 if inrange(age_18,17,59) & yrschool==0 & inrange(iwr_18,0,3)
```

```
replace iwr_imp_18=0 if inrange(age_18,17,59) & yrschool==0 & inrange(iwr_18,3,8)
* age_18 <59 & Years of Education 1-6
replace iwr_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) & inrange(iwr_18,0,3)
replace iwr_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) & inrange(iwr_18,3,8)
* age_18 <59 & Years of Education 7+
replace iwr_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(iwr_18,0,5)
replace iwr_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(iwr_18,5,8)

* age_18 60-69 & Years of Education=0
replace iwr_imp_18=1 if inrange(age_18,60,69) & yrschool==0 & inrange(iwr_18,0,3)
replace iwr_imp_18=0 if inrange(age_18,60,69) & yrschool==0 & inrange(iwr_18,3,8)
* age_18 60-69 & Years of Education 1-6
replace iwr_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) & inrange(iwr_18,0,4)
replace iwr_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) & inrange(iwr_18,4,8)
* age_18 60-69 & Years of Education 7+
replace iwr_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(iwr_18,0,5)
replace iwr_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(iwr_18,5,8)

* age_18 70-79 & Years of Education=0
replace iwr_imp_18=1 if inrange(age_18,70,79) & yrschool==0 & inrange(iwr_18,0,2)
replace iwr_imp_18=0 if inrange(age_18,70,79) & yrschool==0 & inrange(iwr_18,2,8)
* age_18 70-79 & Years of Education 1-6
replace iwr_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) & inrange(iwr_18,0,3)
replace iwr_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) & inrange(iwr_18,3,8)
* age_18 70-79 & Years of Education 7+
replace iwr_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) & inrange(iwr_18,0,4)
replace iwr_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) & inrange(iwr_18,4,8)
```

```
* age_18 80+ & Years of Education=0
replace iwr_imp_18=1 if inrange(age_18,80,120) & yrschool==0 & inrange(iwr_18,0,2)
replace iwr_imp_18=0 if inrange(age_18,80,120) & yrschool==0 & inrange(iwr_18,2,8)
* age_18 80+ & Years of Education 1-6
replace iwr_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inrange(iwr_18,0,3)
replace iwr_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inrange(iwr_18,3,8)
* age_18 80+ & Years of Education 7+
replace iwr_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(iwr_18,0,3)
replace iwr_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(iwr_18,3,8)

replace iwr_imp_18= iwr_18 if mi(iwr_18) & iwr_imp_18==.
replace iwr_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & iwr_imp_18==.
replace iwr_imp_18= .p if inlist(tipent_18,3,4)
label define cognitive_18 0 "0.Normal" 1 "1.Impaired", replace
label values iwr_imp_18 cognitive_18


*** DELAYED MEMORY Z-SCORES ***
*** Repeat 8 words ***
** Z-SCORES **
gen dwr_z_18=.

replace dwr_z_18= 2.0       if [[inlist(dwr_18,8)   & yrschool==0] | [inlist(dwr_18,8)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,8)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18= 1.5       if [[inlist(dwr_18,7)   & yrschool==0] | [inlist(dwr_18,7)   &
inrange(yrschool,1,6)]
] & inrange(age_18,17,59)
```

```
replace dwr_z_18= 1.0       if [[inlist(dwr_18,5,6) & yrschool==0] | [inlist(dwr_18,5,6) &
inrange(yrschool,1,6)] | [inlist(dwr_18,7)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18= 0.0       if [[inlist(dwr_18,3,4) & yrschool==0] | [inlist(dwr_18,4)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,5,6) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18=-1.0       if [[inlist(dwr_18,2) & yrschool==0]   | [inlist(dwr_18,3)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,4)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18=-1.5       if [[inlist(dwr_18,1) & yrschool==0]   | [inlist(dwr_18,2)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,3)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18=-2.0       if [[inlist(dwr_18,0) & yrschool==0]   | [inlist(dwr_18,1)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,2)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace dwr_z_18=-3.0       if [
[inlist(dwr_18,0)   & inrange(yrschool,1,6)] | [inlist(dwr_18,0,1) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)

replace dwr_z_18= 2.0       if [[inlist(dwr_18,8)   & yrschool==0] | [inlist(dwr_18,7,8) &
inrange(yrschool,1,6)] |      [inlist(dwr_18,8)       & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace dwr_z_18= 1.5       if [[inlist(dwr_18,7)   & yrschool==0]

                             ] & inrange(age_18,60,69)
replace dwr_z_18= 1.0       if [[inlist(dwr_18,5,6) & yrschool==0] | [inlist(dwr_18,6)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,7)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace dwr_z_18= 0.0       if [[inlist(dwr_18,3,4) & yrschool==0] | [inlist(dwr_18,4,5) &
inrange(yrschool,1,6)] | [inlist(dwr_18,5,6) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace dwr_z_18=-1.0       if [[inlist(dwr_18,2)   & yrschool==0] | [inlist(dwr_18,3)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,4)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace dwr_z_18=-1.5       if [[inlist(dwr_18,1)   & yrschool==0] | [inlist(dwr_18,2)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,3)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace dwr_z_18=-2.0       if [[inlist(dwr_18,0)   & yrschool==0] | [inlist(dwr_18,0,1) &
inrange(yrschool,1,6)] | [inlist(dwr_18,2)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
```

```
replace dwr_z_18=-3.0       if [
                                                                    [inlist(dwr_18,0,1) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)

replace dwr_z_18= 3.0       if [[inlist(dwr_18,8)   & yrschool==0]

                                        ] & inrange(age_18,70,79)
replace dwr_z_18= 2.0       if [[inlist(dwr_18,7)   & yrschool==0] | [inlist(dwr_18,8)   &
inrange(yrschool,1,6)] |      [inlist(dwr_18,8)       & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace dwr_z_18= 1.5       if [[inlist(dwr_18,6)   & yrschool==0] | [inlist(dwr_18,7)   &
inrange(yrschool,1,6)]
] & inrange(age_18,70,79)
replace dwr_z_18= 1.0       if [[inlist(dwr_18,5)   & yrschool==0] | [inlist(dwr_18,5,6) &
inrange(yrschool,1,6)] | [inlist(dwr_18,6,7) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace dwr_z_18= 0.0       if [[inlist(dwr_18,3,4) & yrschool==0] | [inlist(dwr_18,3,4) &
inrange(yrschool,1,6)] | [inlist(dwr_18,4,5) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace dwr_z_18=-1.0       if [[inlist(dwr_18,1,2) & yrschool==0] | [inlist(dwr_18,2)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,3)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace dwr_z_18=-1.5       if [[inlist(dwr_18,0)   & yrschool==0] | [inlist(dwr_18,1)   &
inrange(yrschool,1,6)] | [inlist(dwr_18,2)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace dwr_z_18=-2.0       if [
[inlist(dwr_18,0)   & inrange(yrschool,1,6)] | [inlist(dwr_18,0,1) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)

replace dwr_z_18= 3.0       if [[inlist(dwr_18,8)   & yrschool==0] | [inlist(dwr_18,8)     &
inrange(yrschool,1,6)]
] & inrange(age_18,80,120)
replace dwr_z_18= 2.0       if [[inlist(dwr_18,6,7) & yrschool==0] | [inlist(dwr_18,7)             &
inrange(yrschool,1,6)] | [inlist(dwr_18,7,8) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
```

```
replace dwr_z_18= 1.5       if [[inlist(dwr_18,5)   & yrschool==0] | [inlist(dwr_18,6)     &
inrange(yrschool,1,6)]
] & inrange(age_18,80,120)
replace dwr_z_18= 1.0       if [[inlist(dwr_18,4)   & yrschool==0] | [inlist(dwr_18,5)     &
inrange(yrschool,1,6)] | [inlist(dwr_18,5,6) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace dwr_z_18= 0.0       if [[inlist(dwr_18,3)   & yrschool==0] | [inlist(dwr_18,2,3,4) &
inrange(yrschool,1,6)] | [inlist(dwr_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace dwr_z_18=-1.0       if [[inlist(dwr_18,1,2) & yrschool==0] | [inlist(dwr_18,1)     &
inrange(yrschool,1,6)] | [inlist(dwr_18,1,2) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace dwr_z_18=-1.5       if [[inlist(dwr_18,0)   & yrschool==0] | [inlist(dwr_18,0)     &
inrange(yrschool,1,6)] | [inlist(dwr_18,0)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace dwr_z_18= dwr_18 if mi(dwr_18) & dwr_z_18==.
replace dwr_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & dwr_z_18==.
replace dwr_z_18= .p if inlist(tipent_18,3,4)
label variable dwr_z_18 "MHAS 2018 Delayed Memory - Repeat 8 words Z-Scores"

** Impairment Status **
gen dwr_imp_18 = .
label variable dwr_imp_18 "MHAS 2018 Delayed Memory - Repeat 8 words Impairment"
* age_18 <59 & Years of Education=0
replace dwr_imp_18=1 if inrange(age_18,17,59) & yrschool==0 & inrange(dwr_18,0,1)
replace dwr_imp_18=0 if inrange(age_18,17,59) & yrschool==0 & inrange(dwr_18,2,8)
* age_18 <59 & Years of Education 1-6
replace dwr_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(dwr_18,0,2)
replace dwr_imp_18=0  if inrange(age_18,17,659) & inrange(yrschool,1,6) &
inrange(dwr_18,3,8)
```

```
* age_18 <59 & Years of Education 7+
replace dwr_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(dwr_18,0,3)
replace dwr_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(dwr_18,4,8)

* age_18 60-69 & Years of Education=0
replace dwr_imp_18=1 if inrange(age_18,60,69) & yrschool==0 & inrange(dwr_18,0,1)
replace dwr_imp_18=0 if inrange(age_18,60,69) & yrschool==0 & inrange(dwr_18,2,8)
* age_18 60-69 & Years of Education 1-6
replace dwr_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(dwr_18,0,2)
replace dwr_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(dwr_18,3,8)
* age_18 60-69 & Years of Education 7+
replace dwr_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(dwr_18,0,3)
replace dwr_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(dwr_18,4,8)

* age_18 70-79 & Years of Education=0
replace dwr_imp_18=1 if inrange(age_18,70,79) & yrschool==0 & inlist(dwr_18,0)
replace dwr_imp_18=0 if inrange(age_18,70,79) & yrschool==0 & inrange(dwr_18,1,8)
* age_18 70-79 & Years of Education 1-6
replace dwr_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(dwr_18,0,1)
replace dwr_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(dwr_18,2,8)
* age_18 70-79 & Years of Education 7+
replace dwr_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(dwr_18,0,2)
```

```
replace dwr_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(dwr_18,3,8)

* age_18 80+ & Years of Education=0
replace dwr_imp_18=1 if inrange(age_18,80,120) & yrschool==0 & inlist(dwr_18,0)
replace dwr_imp_18=0 if inrange(age_18,80,120) & yrschool==0 & inrange(dwr_18,1,8)
* age_18 80+ & Years of Education 1-6
replace dwr_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) & inlist(dwr_18,0)
replace dwr_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(dwr_18,1,8)
* age_18 80+ & Years of Education 7+
replace dwr_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) & inlist(dwr_18,0)
replace dwr_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(dwr_18,1,8)

replace dwr_imp_18= dwr_18 if mi(dwr_18) & dwr_imp_18==.
replace dwr_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & dwr_imp_18==.
replace dwr_imp_18= .p if inlist(tipent_18,3,4)
label values dwr_imp_18 cognitive_18


*** ATTENTION Z-SCORES ***

*** Visual Scan ***
** Z-SCORES **
gen visual_scan_z_18=.
replace visual_scan_z_18= 3.0     if [[inrange(visual_scan_18,51,60) & yrschool==0]

                                                                                ] &
inrange(age_18,17,59)
```

```
replace visual_scan_z_18= 2.0     if [[inrange(visual_scan_18,42,50) & yrschool==0] |
[inrange(visual_scan_18,55,60) & inrange(yrschool,1,6)]
                                                                  ] & inrange(age_18,17,59)
replace visual_scan_z_18= 1.5     if [[inrange(visual_scan_18,38,41) & yrschool==0] |
[inrange(visual_scan_18,50,54) & inrange(yrschool,1,6)]
                                                                  ] & inrange(age_18,17,59)
replace visual_scan_z_18= 1.0     if [[inrange(visual_scan_18,28,37) & yrschool==0] |
[inrange(visual_scan_18,38,49) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,50,60) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace visual_scan_z_18= 0.0     if [[inrange(visual_scan_18,17,27) & yrschool==0] |
[inrange(visual_scan_18,26,37) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,38,49) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace visual_scan_z_18=-1.0     if [[inrange(visual_scan_18,8,16)  & yrschool==0] |
[inrange(visual_scan_18,14,25) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,27,37) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace visual_scan_z_18=-1.5     if [[inrange(visual_scan_18,3,7)   & yrschool==0] |
[inrange(visual_scan_18,10,13) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,23,26) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace visual_scan_z_18=-2.0     if [[inrange(visual_scan_18,0,2)   & yrschool==0] |
[inrange(visual_scan_18,0,9)   & inrange(yrschool,1,6)] | [inrange(visual_scan_18,13,22) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace visual_scan_z_18=-3.0     if [

                        [inrange(visual_scan_18,0,12)  & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)

replace visual_scan_z_18= 3.0     if [[inrange(visual_scan_18,47,60) & yrschool==0] |
[inlist(visual_scan_18,60)     & inrange(yrschool,1,6)]
                                                        ] & inrange(age_18,60,69)
```

```
replace visual_scan_z_18= 2.0     if [[inrange(visual_scan_18,38,46) & yrschool==0] |
[inrange(visual_scan_18,50,59) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,59,60) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18= 1.5     if [[inrange(visual_scan_18,34,37) & yrschool==0] |
[inrange(visual_scan_18,46,49) & inrange(yrschool,1,6)]
                                                        ] & inrange(age_18,60,69)
replace visual_scan_z_18= 1.0     if [[inrange(visual_scan_18,23,33) & yrschool==0] |
[inrange(visual_scan_18,34,45) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,47,58) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18= 0.0     if [[inrange(visual_scan_18,13,22) & yrschool==0] |
[inrange(visual_scan_18,23,33) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,35,46) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18=-1.0     if [[inrange(visual_scan_18,2,12)  & yrschool==0] |
[inrange(visual_scan_18,11,22) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,23,34) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18=-1.5     if [[inrange(visual_scan_18,0,1)   & yrschool==0] |
[inrange(visual_scan_18,8,10)  & inrange(yrschool,1,6)] | [inrange(visual_scan_18,19,22) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18=-2.0     if [
                        [inrange(visual_scan_18,0,7)   & inrange(yrschool,1,6)] |
[inrange(visual_scan_18,8,18)  & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace visual_scan_z_18=-3.0     if [

                         [inrange(visual_scan_18,0,7)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)

replace visual_scan_z_18= 3.0     if [[inrange(visual_scan_18,38,60) & yrschool==0] |
        [inrange(visual_scan_18,53,60) & inrange(yrschool,1,6)]
                                                                ] & inrange(age_18,70,79)
```

```
replace visual_scan_z_18= 2.0     if [[inrange(visual_scan_18,30,38) & yrschool==0] |
[inrange(visual_scan_18,44,52) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,59,60) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18= 1.5     if [[inrange(visual_scan_18,27,31) & yrschool==0] |
[inrange(visual_scan_18,40,43) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,55,58) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18= 1.0     if [[inrange(visual_scan_18,19,27) & yrschool==0] |
[inrange(visual_scan_18,29,39) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,43,54) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18= 0.0     if [[inrange(visual_scan_18,10,21) & yrschool==0] |
[inrange(visual_scan_18,18,28) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,31,42) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18=-1.0     if [[inrange(visual_scan_18,2,9)   & yrschool==0] |
[inrange(visual_scan_18,7,17)  & inrange(yrschool,1,6)] | [inrange(visual_scan_18,19,30) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18=-1.5     if [[inrange(visual_scan_18,0,3)   & yrschool==0] |
[inrange(visual_scan_18,3,6)   & inrange(yrschool,1,6)] | [inrange(visual_scan_18,15,18) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18=-2.0     if [
                        [inrange(visual_scan_18,0,2)   & inrange(yrschool,1,6)] |
[inrange(visual_scan_18,5,14)  & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace visual_scan_z_18=-3.0     if [

                    [inrange(visual_scan_18,0,4)   & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)

replace visual_scan_z_18= 3.0     if [[inrange(visual_scan_18,28,60) & yrschool==0] |
[inrange(visual_scan_18,42,60) & inrange(yrschool,1,6)]
                                                        ] & inrange(age_18,80,120)
```

```
replace visual_scan_z_18= 2.0     if [[inrange(visual_scan_18,26,27) & yrschool==0] |
[inrange(visual_scan_18,34,41) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,49,60) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18= 1.5     if [[inrange(visual_scan_18,24,25) & yrschool==0] |
[inrange(visual_scan_18,31,33) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,44,48) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18= 1.0     if [[inrange(visual_scan_18,15,23) & yrschool==0] |
[inrange(visual_scan_18,21,30) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,31,43) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18= 0.0     if [[inrange(visual_scan_18,7,14)  & yrschool==0] |
[inrange(visual_scan_18,12,20) & inrange(yrschool,1,6)] | [inrange(visual_scan_18,19,30) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18=-1.0     if [[inrange(visual_scan_18,0,6)   & yrschool==0] |
[inrange(visual_scan_18,3,11)  & inrange(yrschool,1,6)] | [inrange(visual_scan_18,6,18)  &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18=-1.5     if [
                          [inrange(visual_scan_18,0,2)   & inrange(yrschool,1,6)] |
[inrange(visual_scan_18,2,5)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace visual_scan_z_18=-2.0     if [

                           [inrange(visual_scan_18,0,1)   & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)

replace visual_scan_z_18= visual_scan_18 if mi(visual_scan_18) & visual_scan_z_18==.
replace visual_scan_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
visual_scan_z_18==.
replace visual_scan_z_18= .p if inlist(tipent_18,3,4)
label variable visual_scan_z_18 "MHAS 2018 Attention - Visual Scan Z-Scores"

** Impairment Status **
gen visual_scan_imp_18 = .
```

```
label variable visual_scan_imp_18 "MHAS 2018 Attention - Visual Scan Impairment"

* age_18 <59 & Years of Education=0
replace visual_scan_imp_18=1 if inrange(age_18,17,69) & yrschool==0 &
inrange(visual_scan_18,0,7)
replace visual_scan_imp_18=0 if inrange(age_18,17,69) & yrschool==0 &
inrange(visual_scan_18,8,60)
* age_18 <59 & Years of Education 1-6
replace visual_scan_imp_18=1  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inrange(visual_scan_18,0,13)
replace visual_scan_imp_18=0  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inrange(visual_scan_18,14,60)
* age_18 <59 & Years of Education 7+
replace visual_scan_imp_18=1  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inrange(visual_scan_18,0,26)
replace visual_scan_imp_18=0  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inrange(visual_scan_18,27,60)

* age_18 60-69 & Years of Education=0
replace visual_scan_imp_18=1 if inrange(age_18,17,69) & yrschool==0 &
inrange(visual_scan_18,0,1)
replace visual_scan_imp_18=0 if inrange(age_18,17,69) & yrschool==0 &
inrange(visual_scan_18,2,60)
* age_18 60-69 & Years of Education 1-6
replace visual_scan_imp_18=1  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inrange(visual_scan_18,0,10)
replace visual_scan_imp_18=0  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inrange(visual_scan_18,11,60)
* age_18 60-69 & Years of Education 7+
replace visual_scan_imp_18=1  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inrange(visual_scan_18,0,22)
```

```
replace visual_scan_imp_18=0  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inrange(visual_scan_18,23,60)

* age_18 70-79 & Years of Education=0
replace visual_scan_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(visual_scan_18,0,1)
replace visual_scan_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(visual_scan_18,2,60)
* age_18 70-79 & Years of Education 1-6
replace visual_scan_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(visual_scan_18,0,6)
replace visual_scan_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(visual_scan_18,7,60)
* age_18 70-79 & Years of Education 7+
replace visual_scan_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(visual_scan_18,0,18)
replace visual_scan_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(visual_scan_18,19,60)

* age_18 80+ & Years of Education=0
//replace visual_scan_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inrange(visual_scan_18,)
replace visual_scan_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(visual_scan_18,0,60)
* age_18 80+ & Years of Education 1-6
replace visual_scan_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(visual_scan_18,0,2)
replace visual_scan_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(visual_scan_18,3,60)
* age_18 80+ & Years of Education 7+
```

```
replace visual_scan_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(visual_scan_18,0,5)
replace visual_scan_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(visual_scan_18,6,60)

replace visual_scan_imp_18= visual_scan_18 if mi(visual_scan_18) & visual_scan_imp_18==.
replace visual_scan_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
visual_scan_imp_18==.
replace visual_scan_imp_18= .p if inlist(tipent_18,3,4)
label values visual_scan_imp_18 cognitive_18


*** ORIENTATION Z-SCORES ***
gen orientation_z_18=.
replace         orientation_z_18= 3.0        if [
                  [inlist(orientation_18,2,3) & inrange(yrschool,1,6)] | [inlist(orientation_18,1,2,3)
& inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace         orientation_z_18= 2.0        if [[inlist(orientation_18,3) & yrschool==0]

                                                                                ] &
inrange(age_18,17,59)
replace         orientation_z_18= 1.5        if [
                  [inlist(orientation_18,1)   & inrange(yrschool,1,6)]
                                                                  ] & inrange(age_18,17,59)
replace         orientation_z_18= 1.0        if [[inlist(orientation_18,2) & yrschool==0]

                                                                                    ] &
inrange(age_18,17,59)
replace         orientation_z_18= 0.0        if [[inlist(orientation_18,1) & yrschool==0] |
[inlist(orientation_18,0)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,17,59)
```

```
replace         orientation_z_18=-1.0       if [[inlist(orientation_18,0) & yrschool==0]

                                                                                ] &
inrange(age_18,17,59)
replace         orientation_z_18=-2.0       if [

                  [inlist(orientation_18,0)     & inrange(yrschool,7,25)]] & inrange(age_18,17,59)

replace         orientation_z_18= 3.0       if [[inlist(orientation_18,3) & yrschool==0] |
[inlist(orientation_18,2,3) & inrange(yrschool,1,6)] | [inlist(orientation_18,1,2,3) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace         orientation_z_18= 1.5       if [[inlist(orientation_18,2) & yrschool==0]

                                                                                ] &
inrange(age_18,60,69)
replace         orientation_z_18= 1.0       if [
                    [inlist(orientation_18,1)   & inrange(yrschool,1,6)]
                                                                                ] & inrange(age_18,60,69)
replace         orientation_z_18= 0.0       if [[inlist(orientation_18,1) & yrschool==0] |
[inlist(orientation_18,0)   & inrange(yrschool,1,6)] | [inlist(orientation_18,0)     &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace         orientation_z_18=-1.0       if [[inlist(orientation_18,0) & yrschool==0]

                                                                                ] &
inrange(age_18,60,69)

replace         orientation_z_18= 3.0       if [
                    [inlist(orientation_18,2,3) & inrange(yrschool,1,6)] | [inlist(orientation_18,2,3)   &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
```

```
replace         orientation_z_18= 2.0        if [[inlist(orientation_18,3) & yrschool==0] |

[inlist(orientation_18,1)     & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace         orientation_z_18= 1.0        if [[inlist(orientation_18,2) & yrschool==0] |
[inlist(orientation_18,1)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,70,79)
replace         orientation_z_18= 0.0        if [[inlist(orientation_18,1) & yrschool==0] |
[inlist(orientation_18,0)   & inrange(yrschool,1,6)] | [inlist(orientation_18,0)     &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace         orientation_z_18=-1.0        if [[inlist(orientation_18,0) & yrschool==0]

                                                                                    ] &
inrange(age_18,70,79)

replace         orientation_z_18= 3.0        if [
                  [inlist(orientation_18,3)   & inrange(yrschool,1,6)] | [inlist(orientation_18,3)
        & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         orientation_z_18= 2.0        if [

            [inlist(orientation_18,2)          & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         orientation_z_18= 1.5        if [[inlist(orientation_18,3) & yrschool==0] |
[inlist(orientation_18,2)   & inrange(yrschool,1,6)]
                                                  ] & inrange(age_18,80,120)
replace         orientation_z_18= 1.0        if [[inlist(orientation_18,2) & yrschool==0] |

[inlist(orientation_18,1)       & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace         orientation_z_18= 0.0        if [[inlist(orientation_18,1) & yrschool==0] |

[inlist(orientation_18,0)       & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
```

```
replace orientation_z_18=-1.0 if [[inlist(orientation_18,0) & yrschool==0] |
[inlist(orientation_18,0,1) & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,80,120)

replace orientation_z_18= orientation_18 if mi(orientation_18) & orientation_z_18==.
replace orientation_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
orientation_z_18==.
replace orientation_z_18= .p if inlist(tipent_18,3,4)
label variable orientation_z_18 "MHAS 2018 Attention - Orientation Z-Scores"

** Impairment Status **
gen orientation_imp_18 = .
label variable orientation_imp_18 "MHAS 2018 Attention - Orientation Impairment"

* age_18 <59 & Years of Education=0
//replace orientation_imp_18=1 if inrange(age_18,17,59) & yrschool==0 &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,17,59) & yrschool==0 &
inrange(orientation_18,0,3)
* age_18 <59 & Years of Education 1-6
//replace orientation_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(orientation_18,)
replace orientation_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(orientation_18,0,3)
* age_18 <59 & Years of Education 7+
replace orientation_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inlist(orientation_18,0)
replace orientation_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(orientation_18,1,3)

* age_18 60-69 & Years of Education=0
```

```
//replace orientation_imp_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(orientation_18,0,3)
* age_18 60-79 & Years of Education 1-6
//replace orientation_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(orientation_18,0,3)
* age_18 60-69 & Years of Education 7+
//replace orientation_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(orientation_18,0,3)

* age_18 70-79 & Years of Education=0
//replace orientation_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(orientation_18,0,3)
* age_18 70-79 & Years of Education 1-6
//replace orientation_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(orientation_18,0,3)
* age_18 70-79 & Years of Education 7+
//replace orientation_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(orientation_18,0,3)
```

```
* age_18 80+ & Years of Education=0
//replace orientation_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inlist(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(orientation_18,0,3)
* age_18 80+ & Years of Education 1-6
//replace orientation_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(orientation_18,0,3)
* age_18 80+ & Years of Education 7+
//replace orientation_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(orientation_18,)
replace orientation_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(orientation_18,0,3)

replace orientation_imp_18= orientation_18 if mi(orientation_18) & orientation_imp_18==.
replace orientation_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
orientation_imp_18==.
replace orientation_imp_18= .p if inlist(tipent_18,3,4)
label values orientation_imp_18 cognitive_18


*** EXECUTIVE FUNCTION Z-SCORES ***
** Verbal fluency z-scores **
gen verbal_fluency_z_18=.
replace verbal_fluency_z_18= 3.0 if [[inrange(verbal_fluency_num_18,24,40) & yrschool==0] |
[inrange(verbal_fluency_num_18,27,40) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,27,40) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
```

```
replace verbal_fluency_z_18= 2.0 if [[inrange(verbal_fluency_num_18,20,23) & yrschool==0] |
[inrange(verbal_fluency_num_18,23,26) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,27,40) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18= 1.5 if [[ inlist(verbal_fluency_num_18,19)    & yrschool==0] | [
inlist(verbal_fluency_num_18,22)   & inrange(yrschool,1,6)] | [
inlist(verbal_fluency_num_18,26)    & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18= 1.0 if [[inrange(verbal_fluency_num_18,15,18) & yrschool==0] |
[inrange(verbal_fluency_num_18,18,21) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,22,25) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18= 0.0 if [[inrange(verbal_fluency_num_18,12,14) & yrschool==0] |
[inrange(verbal_fluency_num_18,14,17) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,17,21) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18=-1.0 if [[inrange(verbal_fluency_num_18,8,11)  & yrschool==0] |
[inrange(verbal_fluency_num_18,10,13) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,13,16) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18=-1.5 if [[ inlist(verbal_fluency_num_18,7)          & yrschool==0] | [
inlist(verbal_fluency_num_18,9)     & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,11,12) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18=-2.0 if [[inrange(verbal_fluency_num_18,4,6)   & yrschool==0] |
[inrange(verbal_fluency_num_18,5,8)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,7,10)  & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace verbal_fluency_z_18=-3.0 if [[inrange(verbal_fluency_num_18,0,3)   & yrschool==0] |
[inrange(verbal_fluency_num_18,0,4)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,0,6)   & inrange(yrschool,7,25)]] & inrange(age_18,17,59)

replace verbal_fluency_z_18= 3.0 if [[inrange(verbal_fluency_num_18,24,40) & yrschool==0] |
[inrange(verbal_fluency_num_18,27,40) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,31,40) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18= 2.0 if [[inrange(verbal_fluency_num_18,20,23) & yrschool==0] |
[inrange(verbal_fluency_num_18,24,26) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,27,30) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
```

```
replace verbal_fluency_z_18= 1.5 if [[ inlist(verbal_fluency_num_18,19)    & yrschool==0] |
[inrange(verbal_fluency_num_18,22,23) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,25,26) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18= 1.0 if [[inrange(verbal_fluency_num_18,15,18) & yrschool==0] |
[inrange(verbal_fluency_num_18,18,21) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,21,24) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18= 0.0 if [[inrange(verbal_fluency_num_18,11,14) & yrschool==0] |
[inrange(verbal_fluency_num_18,14,17) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,16,20) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18=-1.0 if [[inrange(verbal_fluency_num_18,8,10)  & yrschool==0] |
[inrange(verbal_fluency_num_18,10,13) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,12,15) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18=-1.5 if [[ inlist(verbal_fluency_num_18,7)     & yrschool==0] |
[inrange(verbal_fluency_num_18,8,9)   & inrange(yrschool,1,6)] | [
inlist(verbal_fluency_num_18,11)    & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18=-2.0 if [[inrange(verbal_fluency_num_18,4,6)   & yrschool==0] |
[inrange(verbal_fluency_num_18,5,7)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,7,10)  & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace verbal_fluency_z_18=-3.0 if [[inrange(verbal_fluency_num_18,0,3)   & yrschool==0] |
[inrange(verbal_fluency_num_18,0,4)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,0,6)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

replace verbal_fluency_z_18= 3.0 if [[inrange(verbal_fluency_num_18,23,50) & yrschool==0] |
[inrange(verbal_fluency_num_18,25,40) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,30,40) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18= 2.0 if [[inrange(verbal_fluency_num_18,20,22) & yrschool==0] |
[inrange(verbal_fluency_num_18,22,24) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,26,29) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18= 1.5 if [[ inlist(verbal_fluency_num_18,19)    & yrschool==0] | [
inlist(verbal_fluency_num_18,21)   & inrange(yrschool,1,6)] | [
inlist(verbal_fluency_num_18,25)    & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
```

```
replace verbal_fluency_z_18= 1.0 if [[inrange(verbal_fluency_num_18,15,18) & yrschool==0] |
[inrange(verbal_fluency_num_18,16,20) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,20,24) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18= 0.0 if [[inrange(verbal_fluency_num_18,11,14) & yrschool==0] |
[inrange(verbal_fluency_num_18,12,15) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,16,19) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18=-1.0 if [[inrange(verbal_fluency_num_18,7,10)  & yrschool==0] |
[inrange(verbal_fluency_num_18,8,11)  & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,11,15) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18=-1.5 if [[ inlist(verbal_fluency_num_18,6)          & yrschool==0] | [
inlist(verbal_fluency_num_18,7)     & inrange(yrschool,1,6)] | [
inlist(verbal_fluency_num_18,10)    & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18=-2.0 if [[inrange(verbal_fluency_num_18,3,5)   & yrschool==0] |
[inrange(verbal_fluency_num_18,4,6)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,6,9)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace verbal_fluency_z_18=-3.0 if [[inrange(verbal_fluency_num_18,0,2)   & yrschool==0] |
[inrange(verbal_fluency_num_18,0,3)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,0,5)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)

replace verbal_fluency_z_18= 3.0 if [[inrange(verbal_fluency_num_18,21,40) & yrschool==0] |
[inrange(verbal_fluency_num_18,21,40) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,27,40) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18= 2.0 if [[inrange(verbal_fluency_num_18,18,20) & yrschool==0] |
[inrange(verbal_fluency_num_18,18,20) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,23,26) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18= 1.5 if [[ inlist(verbal_fluency_num_18,17)    & yrschool==0] | [
inlist(verbal_fluency_num_18,17)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,21,22) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18= 1.0 if [[inrange(verbal_fluency_num_18,13,16) & yrschool==0] |
[inrange(verbal_fluency_num_18,13,16) & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,18,20) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
```

```
replace verbal_fluency_z_18= 0.0 if [[inrange(verbal_fluency_num_18,9,12)  & yrschool==0] |
[inrange(verbal_fluency_num_18,9,12)  & inrange(yrschool,1,6)] | [
inlist(verbal_fluency_num_18,17)    & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18=-1.0 if [[inrange(verbal_fluency_num_18,5,8)   & yrschool==0] |
[inrange(verbal_fluency_num_18,5,8)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,8,16)  & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18=-1.5 if [[ inlist(verbal_fluency_num_18,4)          & yrschool==0] | [
inlist(verbal_fluency_num_18,4)     & inrange(yrschool,1,6)] | [ inlist(verbal_fluency_num_18,7)
& inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18=-2.0 if [[inrange(verbal_fluency_num_18,1,3)   & yrschool==0] |
[inrange(verbal_fluency_num_18,1,3)   & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,3,6)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace verbal_fluency_z_18=-3.0 if [[ inlist(verbal_fluency_num_18,0)          & yrschool==0] | [
inlist(verbal_fluency_num_18,0)     & inrange(yrschool,1,6)] |
[inrange(verbal_fluency_num_18,0,2)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace verbal_fluency_z_18= verbal_fluency_num_18 if mi(verbal_fluency_num_18) &
verbal_fluency_z_18==.
replace verbal_fluency_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
verbal_fluency_z_18==.
replace verbal_fluency_z_18= .p if inlist(tipent_18,3,4)
label variable verbal_fluency_z_18 "MHAS 2018 Executive function - Verbal Fluency Z-
Scores"

** Impairment Status **
gen verbal_fluency_imp_18 = .
label variable verbal_fluency_imp_18 "MHAS 2018 Executive function - Verbal Fluency
Impairment"

* age_18 <59 & Years of Education=0
```

```
replace verbal_fluency_imp_18=1 if inrange(age_18,17,59) & yrschool==0 &
inrange(verbal_fluency_num_18,0,7)
replace verbal_fluency_imp_18=0 if inrange(age_18,17,59) & yrschool==0 &
inrange(verbal_fluency_num_18,8,40)
* age_18 <59 & Years of Education 1-6
replace verbal_fluency_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,0,9)
replace verbal_fluency_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,10,40)
* age_18 <59 & Years of Education 7+
replace verbal_fluency_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,0,12)
replace verbal_fluency_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,13,50)

* age_18 60-69 & Years of Education=0
replace verbal_fluency_imp_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(verbal_fluency_num_18,0,7)
replace verbal_fluency_imp_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(verbal_fluency_num_18,8,40)
* age_18 60-69 & Years of Education 1-6
replace verbal_fluency_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,0,9)
replace verbal_fluency_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,10,40)
* age_18 60-69 & Years of Education 7+
replace verbal_fluency_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,0,11)
replace verbal_fluency_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,12,50)
```

```
* age_18 70-79 & Years of Education=0
replace verbal_fluency_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(verbal_fluency_num_18,0,6)
replace verbal_fluency_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(verbal_fluency_num_18,7,50)
* age_18 70-79 & Years of Education 1-6
replace verbal_fluency_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,0,7)
replace verbal_fluency_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,8,50)
* age_18 70-79 & Years of Education 7+
replace verbal_fluency_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,0,10)
replace verbal_fluency_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,11,50)

* age_18 80+ & Years of Education=0
replace verbal_fluency_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inrange(verbal_fluency_num_18,0,4)
replace verbal_fluency_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(verbal_fluency_num_18,5,50)
* age_18 80+ & Years of Education 1-6
replace verbal_fluency_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,0,4)
replace verbal_fluency_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(verbal_fluency_num_18,5,50)
* age_18 80+ & Years of Education 7+
replace verbal_fluency_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,0,7)
replace verbal_fluency_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(verbal_fluency_num_18,8,50)
```

```
replace verbal_fluency_imp_18= verbal_fluency_num_18 if mi(verbal_fluency_num_18) &
verbal_fluency_imp_18==.
replace verbal_fluency_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
verbal_fluency_imp_18==.
replace verbal_fluency_imp_18= .p if inlist(tipent_18,3,4)
label values verbal_fluency_imp_18 cognitive_18


** Serial 7 z-scores **
gen serial7_z_18=.
//replace serial7_z_18= 3.0          if [

                                                                    ] &
inrange(age_18,17,59)
//replace serial7_z_18= 2.0          if [

                                                                        ] &
inrange(age_18,17,59)
replace serial7_z_18= 1.5  if [[inlist(serial7_18,5)   & yrschool==0]

                                                ] & inrange(age_18,17,59)
replace serial7_z_18= 1.0  if [[inlist(serial7_18,4)   & yrschool==0] | [inlist(serial7_18,5)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,5) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace serial7_z_18= 0.0  if [[inlist(serial7_18,2,3) & yrschool==0] | [inlist(serial7_18,3,4) &
inrange(yrschool,1,6)] | [inlist(serial7_18,4) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace serial7_z_18=-1.0  if [[inlist(serial7_18,1)   & yrschool==0] | [inlist(serial7_18,2)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,3) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace serial7_z_18=-1.5  if [
[inlist(serial7_18,1)   & inrange(yrschool,1,6)] | [inlist(serial7_18,2) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)
```

```
replace serial7_z_18=-2.0  if [[inlist(serial7_18,0)   & yrschool==0] | [inlist(serial7_18,0)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,1) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace serial7_z_18=-3.0  if [

        [inlist(serial7_18,0) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)

//replace serial7_z_18= 3.0          if [

                                                                                ] &
inrange(age_18,60,69)
replace serial7_z_18= 2.0  if [[inlist(serial7_18,5)   & yrschool==0]

                                                    ] & inrange(age_18,60,69)
replace serial7_z_18= 1.5  if [
[inlist(serial7_18,5) & inrange(yrschool,1,6)]
                                    ] & inrange(age_18,60,69)
replace serial7_z_18= 1.0  if [[inlist(serial7_18,3,4) & yrschool==0] | [inlist(serial7_18,4) &
inrange(yrschool,1,6)] | [inlist(serial7_18,5)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace serial7_z_18= 0.0  if [[inlist(serial7_18,2)   & yrschool==0] | [inlist(serial7_18,3) &
inrange(yrschool,1,6)] | [inlist(serial7_18,4)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace serial7_z_18=-1.0  if [[inlist(serial7_18,1)   & yrschool==0] | [inlist(serial7_18,2) &
inrange(yrschool,1,6)] | [inlist(serial7_18,3)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace serial7_z_18=-1.5  if [[inlist(serial7_18,0)   & yrschool==0] | [inlist(serial7_18,1) &
inrange(yrschool,1,6)] | [inlist(serial7_18,2)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
replace serial7_z_18=-2.0  if [
[inlist(serial7_18,0) & inrange(yrschool,1,6)] | [inlist(serial7_18,1)   & inrange(yrschool,7,25)]] &
inrange(age_18,60,69)
```

```
replace serial7_z_18=-3.0  if [

[inlist(serial7_18,0)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

//replace serial7_z_18= 3.0          if [

                                                                                ] &
inrange(age_18,70,79)
replace serial7_z_18= 2.0  if [[inlist(serial7_18,5) & yrschool==0]

                                                ] & inrange(age_18,70,79)
replace serial7_z_18= 1.5  if [[inlist(serial7_18,4) & yrschool==0] | [inlist(serial7_18,5)   &
inrange(yrschool,1,6)]
          ] & inrange(age_18,70,79)
replace serial7_z_18= 1.0  if [[inlist(serial7_18,3) & yrschool==0] | [inlist(serial7_18,4)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,5)   & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace serial7_z_18= 0.0  if [[inlist(serial7_18,2) & yrschool==0] | [inlist(serial7_18,2,3) &
inrange(yrschool,1,6)] | [inlist(serial7_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace serial7_z_18=-1.0  if [[inlist(serial7_18,1) & yrschool==0] | [inlist(serial7_18,1)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,2)   & inrange(yrschool,7,25)]] &
inrange(age_18,70,79)
replace serial7_z_18=-1.5  if [[inlist(serial7_18,0) & yrschool==0]

                                                ] & inrange(age_18,70,79)
replace serial7_z_18=-2.0  if [
[inlist(serial7_18,0)   & inrange(yrschool,1,6)] | [inlist(serial7_18,1)   & inrange(yrschool,7,25)]]
& inrange(age_18,70,79)
```

```
replace serial7_z_18=-3.0  if [

[inlist(serial7_18,0)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)

//replace serial7_z_18= 3.0          if [

                                                                                ] &
inrange(age_18,80,120)
replace serial7_z_18= 2.0  if [[inlist(serial7_18,5)   & yrschool==0]

                                                ] & inrange(age_18,80,120)
replace serial7_z_18= 1.5  if [[inlist(serial7_18,4)   & yrschool==0] | [inlist(serial7_18,5)   &
inrange(yrschool,1,6)]
                    ] & inrange(age_18,80,120)
replace serial7_z_18= 1.0  if [[inlist(serial7_18,3)   & yrschool==0] | [inlist(serial7_18,4)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,5)   & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace serial7_z_18= 0.0  if [[inlist(serial7_18,2)   & yrschool==0] | [inlist(serial7_18,2,3) &
inrange(yrschool,1,6)] | [inlist(serial7_18,3,4) & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace serial7_z_18=-1.0  if [[inlist(serial7_18,0,1) & yrschool==0] | [inlist(serial7_18,1)   &
inrange(yrschool,1,6)] | [inlist(serial7_18,2)   & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace serial7_z_18=-1.5  if [
[inlist(serial7_18,0)   & inrange(yrschool,1,6)] | [inlist(serial7_18,1)   & inrange(yrschool,7,25)]]
& inrange(age_18,80,120)
replace serial7_z_18=-2.0  if [

        [inlist(serial7_18,0)   & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
//replace serial7_z_18=-3.0          if [
```

```
                                                                    ] &
inrange(age_18,80,120)

replace serial7_z_18= serial7_18 if mi(serial7_18) & serial7_z_18==.
replace serial7_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] & serial7_z_18==.
replace serial7_z_18= .p if inlist(tipent_18,3,4)
label variable serial7_z_18 "MHAS Executive Function - Series of subtractions Z-Scores"

** Impairment Status **
gen serial7_imp_18 = .
label variable serial7_imp_18 "MHAS 2018 Executive function - Verbal Fluency Impairment"

* age_18 <59 & Years of Education=0
replace serial7_imp_18=1  if inrange(age_18,17,59) & yrschool==0 & inlist(serial7_18,0)
replace serial7_imp_18=0  if inrange(age_18,17,59) & yrschool==0 & inrange(serial7_18,1,5)
* age_18 <59 & Years of Education 1-6
replace serial7_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(serial7_18,0,1)
replace serial7_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(serial7_18,2,5)
* age_18 <59 & Years of Education 7+
replace serial7_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(serial7_18,0,2)
replace serial7_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(serial7_18,3,5)

* age_18 60-69 & Years of Education=0
replace serial7_imp_18=1  if inrange(age_18,60,69) & yrschool==0 & inlist(serial7_18,0)
replace serial7_imp_18=0  if inrange(age_18,60,69) & yrschool==0 & inrange(serial7_18,1,5)
* age_18 60-69 Years of Education 1-6
```

```
replace serial7_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(serial7_18,0,1)
replace serial7_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(serial7_18,2,5)
* age_18 60-69 & Years of Education 7+
replace serial7_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(serial7_18,0,2)
replace serial7_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(serial7_18,3,5)

* age_18 70-79 & Years of Education=0
replace serial7_imp_18=1  if inrange(age_18,70,79) & yrschool==0 & inlist(serial7_18,0)
replace serial7_imp_18=0  if inrange(age_18,70,79) & yrschool==0 & inrange(serial7_18,1,5)
* age_18 70-79 & Years of Education 1-6
replace serial7_imp_18=1  if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inlist(serial7_18,0)
replace serial7_imp_18=0  if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(serial7_18,1,5)
* age_18 70-79 & Years of Education 7+
replace serial7_imp_18=1  if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(serial7_18,0,1)
replace serial7_imp_18=0  if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(serial7_18,2,5)

* age_18 80+ & Years of Education=0
//replace serial7_imp_18=1  if inrange(age_18,80,120) & yrschool==0 & inlist(serial7_18,)
replace serial7_imp_18=0  if inrange(age_18,80,120) & yrschool==0 & inrange(serial7_18,0,5)
* age_18 80+ & Years of Education 1-6
replace serial7_imp_18=1  if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inlist(serial7_18,0)
```

```
replace serial7_imp_18=0  if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(serial7_18,1,5)
* age_18 80+ & Years of Education 7+
replace serial7_imp_18=1  if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(serial7_18,0,1)
replace serial7_imp_18=0  if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(serial7_18,2,5)

replace serial7_imp_18= serial7_18 if mi(serial7_18) & serial7_imp_18==.
replace serial7_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
serial7_imp_18==.
replace serial7_imp_18= .p if inlist(tipent_18,3,4)
label values serial7_imp_18 cognitive_18


***CONSTRUCTIONAL PRAXIS Z-SCORES ***
** Copy one figure z-scores **
gen construction_z_18=.
//replace construction_z_18= 3.0  if [


        ] & inrange(age_18,17,59)
//replace construction_z_18= 2.0  if [


        ] & inrange(age_18,17,59)
//replace construction_z_18= 1.5  if [


        ] & inrange(age_18,17,59)
```

```
replace construction_z_18= 1.0    if [[inlist(construction_18,6)   & yrschool==0]

                                                                                ] &
inrange(age_18,17,59)
replace construction_z_18= 0.0    if [[inlist(construction_18,5)   & yrschool==0] | [
inlist(construction_18,6)   & inrange(yrschool,1,6)] | [inlist(construction_18,6)    &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_z_18=-1.0    if [[inlist(construction_18,4)   & yrschool==0] | [
inlist(construction_18,5)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,17,59)
replace construction_z_18=-1.5    if [[inlist(construction_18,3)   & yrschool==0]

                                                                                ] &
inrange(age_18,17,59)
replace construction_z_18=-2.0    if [[inlist(construction_18,2)   & yrschool==0] | [
inlist(construction_18,4)   & inrange(yrschool,1,6)] | [inlist(construction_18,5)    &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_z_18=-3.0    if [[inlist(construction_18,0,1) & yrschool==0] |
[inrange(construction_18,0,3) & inrange(yrschool,1,6)] | [inrange(construction_18,0,4) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)

//replace construction_z_18= 3.0  if [



        ] & inrange(age_18,60,69)
//replace construction_z_18= 2.0  if [



        ] & inrange(age_18,60,69)
//replace construction_z_18= 1.5  if [
```

```
        ] & inrange(age_18,60,69)
replace construction_z_18= 1.0    if [[inlist(construction_18,6)   & yrschool==0]

                                                                                ] &
inrange(age_18,60,69)
replace construction_z_18= 0.0    if [[inlist(construction_18,4,5) & yrschool==0] | [
inlist(construction_18,6)   & inrange(yrschool,1,6)] | [ inlist(construction_18,6)   &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_z_18=-1.0    if [[inlist(construction_18,3)   & yrschool==0] | [
inlist(construction_18,5)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,60,69)
replace construction_z_18=-1.5    if [

                [ inlist(construction_18,5)   & inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_z_18=-2.0    if [[inlist(construction_18,2)   & yrschool==0] | [
inlist(construction_18,4)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,60,69)
replace construction_z_18=-3.0    if [[inlist(construction_18,0,1) & yrschool==0] |
[inrange(construction_18,0,3) & inrange(yrschool,1,6)] | [inrange(construction_18,0,4) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)

//replace construction_z_18= 3.0  if [


        ] & inrange(age_18,70,79)
//replace construction_z_18= 2.0  if [


        ] & inrange(age_18,70,79)
```

```
//replace construction_z_18= 1.5  if [


          ] & inrange(age_18,70,79)
replace construction_z_18= 1.0    if [[inlist(construction_18,6)   & yrschool==0]

                                                                                ] &
inrange(age_18,70,79)
replace construction_z_18= 0.0    if [[inlist(construction_18,4,5) & yrschool==0] | [
inlist(construction_18,5,6) & inrange(yrschool,1,6)] | [ inlist(construction_18,6)   &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_z_18=-1.0    if [[inlist(construction_18,3)   & yrschool==0] |
                                                                                [
inlist(construction_18,5)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_z_18=-1.5    if [
                  [ inlist(construction_18,4)   & inrange(yrschool,1,6)]
                                                                    ] & inrange(age_18,70,79)
replace construction_z_18=-2.0    if [[inlist(construction_18,2)   & yrschool==0] | [
inlist(construction_18,3)   & inrange(yrschool,1,6)] | [ inlist(construction_18,4)   &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_z_18=-3.0    if [[inlist(construction_18,0,1) & yrschool==0] |
[inrange(construction_18,0,2) & inrange(yrschool,1,6)] | [inrange(construction_18,0,3) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)

//replace construction_z_18= 3.0  if [


         ] & inrange(age_18,80,120)
//replace construction_z_18= 2.0  if [
```

```
        ] & inrange(age_18,80,120)
//replace construction_z_18= 1.5  if [



        ] & inrange(age_18,80,120)
replace construction_z_18= 1.0    if [[inlist(construction_18,6)   & yrschool==0]

                                                                                ] &
inrange(age_18,80,120)
replace construction_z_18= 0.0    if [[inlist(construction_18,4,5) & yrschool==0] |
[inlist(construction_18,5,6) & inrange(yrschool,1,6)] | [ inlist(construction_18,5,6) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_z_18=-1.0    if [[inlist(construction_18,2,3) & yrschool==0] |
[inlist(construction_18,4)   & inrange(yrschool,1,6)]
                                                ] & inrange(age_18,80,120)
replace construction_z_18=-1.5    if [
                [inlist(construction_18,3)   & inrange(yrschool,1,6)] | [ inlist(construction_18,4)
& inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_z_18=-2.0    if [
                [inlist(construction_18,2)   & inrange(yrschool,1,6)]
                                                                ] & inrange(age_18,80,120)
replace construction_z_18=-3.0    if [[inlist(construction_18,0,1) & yrschool==0] |
[inlist(construction_18,0,1) & inrange(yrschool,1,6)] | [inrange(construction_18,0,3) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace construction_z_18= construction_18 if mi(construction_18) & construction_z_18==.
replace construction_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_z_18==.
replace construction_z_18= .p if inlist(tipent_18,3,4)
label variable construction_z_18 "MHAS 2018 Constructional Praxis Z-Scores"
```

```
** Impairment Status (using 0-6 score) **
gen construction_imp_18 = .
label variable construction_imp_18 "MHAS 2018 Constructional Praxis Impairment"

* age_18 <59 & Years of Education=0
replace construction_imp_18=1 if inrange(age_18,17,59) & yrschool==0 &
inrange(construction_18,0,3)
replace construction_imp_18=0 if inrange(age_18,17,59) & yrschool==0 &
inrange(construction_18,4,6)
* age_18 <59 & Years of Education 1-6
replace construction_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(construction_18,0,4)
replace construction_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(construction_18,5,6)
* age_18 <59 & Years of Education 7+
replace construction_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(construction_18,0,5)
replace construction_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inlist(construction_18,6)

* age_18 60-69 & Years of Education=0
replace construction_imp_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(construction_18,0,2)
replace construction_imp_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(construction_18,3,6)
* age_18 60-69 & Years of Education 1-6
replace construction_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(construction_18,0,4)
replace construction_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(construction_18,5,6)
```

```
* age_18 60-69 & Years of Education 7+
replace construction_imp_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(construction_18,0,5)
replace construction_imp_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inlist(construction_18,6)

* age_18 70-79 & Years of Education=0
replace construction_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(construction_18,0,2)
replace construction_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(construction_18,3,6)
* age_18 70-79 & Years of Education 1-6
replace construction_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(construction_18,0,4)
replace construction_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(construction_18,5,6)
* age_18 70-79 & Years of Education 7+
replace construction_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(construction_18,0,4)
replace construction_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(construction_18,5,6)

* age_18 80+ & Years of Education=0
replace construction_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inrange(construction_18,0,1)
replace construction_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(construction_18,2,6)
* age_18 80+ & Years of Education 1-6
replace construction_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(construction_18,0,3)
```

```
replace construction_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(construction_18,4,6)
* age_18 80+ & Years of Education 7+
replace construction_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(construction_18,0,4)
replace construction_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(construction_18,5,6)

replace construction_imp_18= construction_18 if mi(construction_18) &
construction_imp_18==.
replace construction_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_imp_18==.
replace construction_imp_18= .p if inlist(tipent_18,3,4)
label values construction_imp_18 cognitive_18


** Copy one figure z-scores comparable with 2001 & 2003 **
gen construction_z_v01_18=.
replace construction_z_v01_18= 0.0       if [[inlist(construction_v01_18,2) & yrschool==0] |
[inlist(construction_v01_18,2)   & inrange(yrschool,1,6)] | [inlist(construction_v01_18,2)   &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_z_v01_18=-1.0       if [[inlist(construction_v01_18,1) & yrschool==0]


          ] & inrange(age_18,17,59)
//replace construction_z_v01_18=-1.5     if [


                                  ] & inrange(age_18,17,59)
//replace construction_z_v01_18=-2.0     if [
```

] & inrange(age_18,17,59)

replace construction_z_v01_18=-3.0 if [[inlist(construction_v01_18,0) & yrschool==0] | [inlist(construction_v01_18,0,1) & inrange(yrschool,1,6)] | [inlist(construction_v01_18,0,1) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)

replace construction_z_v01_18= 0.0 if [[inlist(construction_v01_18,2) & yrschool==0] | [inlist(construction_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_v01_18,2) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

replace construction_z_v01_18=-1.0 if [[inlist(construction_v01_18,1) & yrschool==0] | [inlist(construction_v01_18,1) & inrange(yrschool,1,6)]

] & inrange(age_18,60,69)

//replace construction_z_v01_18=-1.5 if [

] & inrange(age_18,60,69)

//replace construction_z_v01_18=-2.0 if [

] & inrange(age_18,60,69)

replace construction_z_v01_18=-3.0 if [[inlist(construction_v01_18,0) & yrschool==0] | [inlist(construction_v01_18,0) & inrange(yrschool,1,6)] | [inlist(construction_v01_18,0,1) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

replace construction_z_v01_18= 1.0 if [[inlist(construction_v01_18,2) & yrschool==0]

] & inrange(age_18,70,79)

replace construction_z_v01_18= 0.0 if [

[inlist(construction_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_v01_18,2) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)

```
replace construction_z_v01_18=-1.0      if [[inlist(construction_v01_18,1) & yrschool==0]

        ] & inrange(age_18,70,79)
//replace construction_z_v01_18=-1.5    if [

                                ] & inrange(age_18,70,79)
replace construction_z_v01_18=-2.0      if [[inlist(construction_v01_18,0) & yrschool==0] |
[inlist(construction_v01_18,1) & inrange(yrschool,1,6)]
                                                        ] & inrange(age_18,70,79)
replace construction_z_v01_18=-3.0      if [
                        [inlist(construction_v01_18,0) & inrange(yrschool,1,6)] |
[inlist(construction_v01_18,0,1) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)

replace construction_z_v01_18= 1.0      if [[inlist(construction_v01_18,2) & yrschool==0]


] & inrange(age_18,80,120)
replace construction_z_v01_18= 0.0      if [[inlist(construction_v01_18,1) & yrschool==0] |
[inlist(construction_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_v01_18,2) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_z_v01_18=-1.0      if [
                                [inlist(construction_v01_18,1) & inrange(yrschool,1,6)]
                                                                        ] &
inrange(age_18,80,120)
//replace construction_z_v01_18=-1.5    if [

                ] & inrange(age_18,80,120)
```

```
replace construction_z_v01_18=-2.0       if [[inlist(construction_v01_18,0) & yrschool==0] |

[inlist(construction_v01_18,1) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_z_v01_18=-3.0       if [
                                    [inlist(construction_v01_18,0) & inrange(yrschool,1,6)] |
[inlist(construction_v01_18,0) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace construction_z_v01_18= construction_v01_18 if mi(construction_v01_18) &
construction_z_v01_18==.
replace construction_z_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_z_v01_18==.
replace construction_z_v01_18= .p if inlist(tipent_18,3,4)
label variable construction_z_v01_18 "MHAS 2018 Constructional Praxis Z-Scores v2001-
2003"

** Impairment Status (using 0-2 score comparable to 2001 and 2003) **
gen construction_imp_v01_18 = .
label variable construction_imp_v01_18 "MHAS 2018 Constructional Praxis Impairment v2001-
2003"

* age_18 69< & Years of Education=0
replace construction_imp_v01_18=1 if inrange(age_18,17,69) & yrschool==0 &
inlist(construction_v01_18,0)
replace construction_imp_v01_18=0 if inrange(age_18,17,69) & yrschool==0 &
inlist(construction_v01_18,1,2)
* age_18 69< & Years of Education 1-6
replace construction_imp_v01_18=1  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inlist(construction_v01_18,0,1)
replace construction_imp_v01_18=0  if inrange(age_18,17,69) & inrange(yrschool,1,6) &
inlist(construction_v01_18,2)
* age_18 69< & Years of Education 7+
```

```
replace construction_imp_v01_18=1  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inlist(construction_v01_18,0,1)
replace construction_imp_v01_18=0  if inrange(age_18,17,69) & inrange(yrschool,7,25) &
inlist(construction_v01_18,2)

* age_18 70-79 & Years of Education=0
replace construction_imp_v01_18=1 if inrange(age_18,70,79) & yrschool==0 &
inlist(construction_v01_18,0)
replace construction_imp_v01_18=0 if inrange(age_18,70,79) & yrschool==0 &
inlist(construction_v01_18,1,2)
* age_18 70-79 & Years of Education 1-6
replace construction_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inlist(construction_v01_18,0,1)
replace construction_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inlist(construction_v01_18,2)
* age_18 70-79 & Years of Education 7+
replace construction_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inlist(construction_v01_18,0,1)
replace construction_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inlist(construction_v01_18,2)

* age_18 80+ & Years of Education=0
replace construction_imp_v01_18=1 if inrange(age_18,80,120) & yrschool==0 &
inlist(construction_v01_18,0)
replace construction_imp_v01_18=0 if inrange(age_18,80,120) & yrschool==0 &
inlist(construction_v01_18,1,2)
* age_18 80+ & Years of Education 1-6
replace construction_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inlist(construction_v01_18,0)
replace construction_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inlist(construction_v01_18,1,2)
```

```
* age_18 80+ & Years of Education 7+
replace construction_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inlist(construction_v01_18,0,1)
replace construction_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inlist(construction_v01_18,2)

replace construction_imp_v01_18= construction_v01_18 if mi(construction_v01_18) &
construction_imp_v01_18==.
replace construction_imp_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_imp_v01_18==.
replace construction_imp_v01_18= .p if inlist(tipent_18,3,4)
label values construction_imp_v01_18 cognitive_18


*** Delayed recall/copy of figure ***
** Z-SCORES **
gen construction_m_z_18=.
//replace construction_m_z_18= 3.0        if [


                                    ] & inrange(age_18,17,59)
//replace construction_m_z_18= 2.0        if [


                                ] & inrange(age_18,17,59)
//replace construction_m_z_18= 1.5        if [


                                    ] & inrange(age_18,17,59)
replace construction_m_z_18= 1.0          if [[inlist(construction_m_18,6)   & yrschool==0]
```

```
] & inrange(age_18,17,59)
replace construction_m_z_18= 0.0        if [[inlist(construction_m_18,4,5) & yrschool==0] |
[inlist(construction_m_18,5,6) & inrange(yrschool,1,6)] | [inlist(construction_m_18,5,6)  &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_m_z_18=-1.0        if [[inlist(construction_m_18,3)   & yrschool==0] |
[inlist(construction_m_18,4)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,4)    &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_m_z_18=-1.5        if [[inlist(construction_m_18,2)   & yrschool==0] |
[inlist(construction_m_18,3)   & inrange(yrschool,1,6)]
                                                    ] & inrange(age_18,17,59)
replace construction_m_z_18=-2.0        if [[inlist(construction_m_18,1)   & yrschool==0] |
[inlist(construction_m_18,2)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,3)    &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_m_z_18=-3.0        if [[inlist(construction_m_18,0)   & yrschool==0] |
[inlist(construction_m_18,0,1) & inrange(yrschool,1,6)] | [inrange(construction_m_18,0,2) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)

//replace construction_m_z_18= 3.0       if [



                                    ] & inrange(age_18,60,69)
//replace construction_m_z_18= 2.0       if [



                                    ] & inrange(age_18,60,69)
//replace construction_m_z_18= 1.5       if [



                                    ] & inrange(age_18,60,69)
```

```
//replace construction_m_z_18= 1.0       if [


                                    ] & inrange(age_18,60,69)
replace construction_m_z_18= 0.0        if [[inlist(construction_m_18,4,5,6) & yrschool==0] |
[inlist(construction_m_18,5,6) & inrange(yrschool,1,6)] | [inlist(construction_m_18,5,6)  &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_m_z_18=-1.0        if [[inlist(construction_m_18,2,3)   & yrschool==0] |
[inlist(construction_m_18,4)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,4)    &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_m_z_18=-1.5        if [
                          [inlist(construction_m_18,3)   & inrange(yrschool,1,6)]
                                                                                ] &
inrange(age_18,60,69)
replace construction_m_z_18=-2.0        if [[inlist(construction_m_18,0,1)   & yrschool==0] |
[inlist(construction_m_18,2)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,3)    &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_m_z_18=-3.0        if [
                                  [inlist(construction_m_18,0,1) & inrange(yrschool,1,6)] |
[inrange(construction_m_18,0,2) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

//replace construction_m_z_18= 3.0       if [


                          ] & inrange(age_18,70,79)
//replace construction_m_z_18= 2.0       if [


                          ] & inrange(age_18,70,79)
//replace construction_m_z_18= 1.5       if [
```

```
                                ] & inrange(age_18,70,79)
replace construction_m_z_18= 1.0        if [[inlist(construction_m_18,5,6) & yrschool==0]


] & inrange(age_18,70,79)
replace construction_m_z_18= 0.0        if [[inlist(construction_m_18,3,4) & yrschool==0] |
[inlist(construction_m_18,4,5,6) & inrange(yrschool,1,6)] | [inlist(construction_m_18,4,5,6) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_m_z_18=-1.0        if [[inlist(construction_m_18,2)   & yrschool==0] |
[inlist(construction_m_18,3)     & inrange(yrschool,1,6)]
                                                                ] & inrange(age_18,70,79)
replace construction_m_z_18=-1.5        if [[inlist(construction_m_18,1)   & yrschool==0]
                                                                                        |
[inlist(construction_m_18,3)     & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_m_z_18=-2.0        if [[inlist(construction_m_18,0)   & yrschool==0] |
[inlist(construction_m_18,1,2)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,2)     &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_m_z_18=-3.0        if [
                            [inlist(construction_m_18,0)    & inrange(yrschool,1,6)]  |
[inlist(construction_m_18,0,1)   & inrange(yrschool,7,25)]] & inrange(age_18,70,79)

//replace construction_m_z_18= 3.0       if [


                                ] & inrange(age_18,80,120)
//replace construction_m_z_18= 2.0       if [


                                ] & inrange(age_18,80,120)
```

```
replace construction_m_z_18= 1.5         if [[inlist(construction_m_18,6)     & yrschool==0]

                                                                                                ] &
inrange(age_18,80,120)
replace construction_m_z_18= 1.0         if [[inlist(construction_m_18,5)     & yrschool==0] |
[inlist(construction_m_18,6)     & inrange(yrschool,1,6)] | [inlist(construction_m_18,6)   &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_18= 0.0         if [[inlist(construction_m_18,2,3,4) & yrschool==0] |
[inlist(construction_m_18,3,4,5) & inrange(yrschool,1,6)] | [inlist(construction_m_18,4,5) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_18=-1.0         if [[inlist(construction_m_18,1)     & yrschool==0] |
[inlist(construction_m_18,2)     & inrange(yrschool,1,6)] | [inlist(construction_m_18,3)   &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_18=-1.5         if [

                                [inlist(construction_m_18,2)   & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)
replace construction_m_z_18=-2.0         if [[inlist(construction_m_18,0)     & yrschool==0] |
[inlist(construction_m_18,0,1)   & inrange(yrschool,1,6)] | [inlist(construction_m_18,1)   &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_18=-3.0         if [

                                [inlist(construction_m_18,0)   & inrange(yrschool,7,25)]] &
inrange(age_18,80,120)

replace construction_m_z_18= construction_m_18 if mi(construction_m_18) &
construction_m_z_18==.
replace construction_m_z_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_m_z_18==.
replace construction_m_z_18= .p if inlist(tipent_18,3,4)
```

```
label variable construction_m_z_18 "MHAS 2018 Delayed Memory - Delayed recall of figure Z-Scores"

** Impairment Status **
gen construction_m_imp_18 = .
label variable construction_m_imp_18 "MHAS 2018 Delayed Memory - Delayed recall of figure Impairment"

* age_18 <59 & Years of Education=0
replace construction_m_imp_18=1 if inrange(age_18,17,59) & yrschool==0 &
inrange(construction_m_18,0,2)
replace construction_m_imp_18=0 if inrange(age_18,17,69) & yrschool==0 &
inrange(construction_m_18,3,6)
* age_18 <59 & Years of Education 1-6
replace construction_m_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(construction_m_18,0,3)
replace construction_m_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inrange(construction_m_18,4,6)
* age_18 <59 & Years of Education 7+
replace construction_m_imp_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(construction_m_18,0,3)
replace construction_m_imp_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inrange(construction_m_18,4,6)

* age_18 60-69 & Years of Education=0
replace construction_m_imp_18=1 if inrange(age_18,60,69) & yrschool==0 &
inrange(construction_m_18,0,1)
replace construction_m_imp_18=0 if inrange(age_18,60,69) & yrschool==0 &
inrange(construction_m_18,2,6)
* age_18 60-69 & Years of Education 1-6
```

```
replace construction_m_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(construction_m_18,0,3)
replace construction_m_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inrange(construction_m_18,4,6)
* age_18 60-69 & Years of Education 7+
replace construction_m_imp_18=1 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(construction_m_18,0,3)
replace construction_m_imp_18=0 if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inrange(construction_m_18,4,6)

* age_18 70-79 & Years of Education=0
replace construction_m_imp_18=1 if inrange(age_18,70,79) & yrschool==0 &
inrange(construction_m_18,0,1)
replace construction_m_imp_18=0 if inrange(age_18,70,79) & yrschool==0 &
inrange(construction_m_18,2,6)
* age_18 70-79 & Years of Education 1-6
replace construction_m_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(construction_m_18,0,2)
replace construction_m_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inrange(construction_m_18,3,6)
* age_18 70-79 & Years of Education 7+
replace construction_m_imp_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(construction_m_18,0,3)
replace construction_m_imp_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inrange(construction_m_18,4,6)

* age_18 80+ & Years of Education=0
replace construction_m_imp_18=1 if inrange(age_18,80,120) & yrschool==0 &
inlist(construction_m_18,0)
replace construction_m_imp_18=0 if inrange(age_18,80,120) & yrschool==0 &
inrange(construction_m_18,1,6)
```

```
* age_18 80+ & Years of Education 1-6
replace construction_m_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(construction_m_18,0,1)
replace construction_m_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inrange(construction_m_18,2,6)
* age_18 80+ & Years of Education 7+
replace construction_m_imp_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(construction_m_18,0,2)
replace construction_m_imp_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inrange(construction_m_18,3,6)

replace construction_m_imp_18= construction_m_18 if mi(construction_m_18) &
construction_m_imp_18==.
replace construction_m_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_m_imp_18==.
replace construction_m_imp_18= .p if inlist(tipent_18,3,4)
label values construction_m_imp_18 cognitive_18


** Delayed recall/copy of figure z-scores comparable with 2001 & 2003 **
gen construction_m_z_v01_18=.
replace construction_m_z_v01_18= 1.0  if [[inlist(construction_m_v01_18,2) & yrschool==0]


        ] & inrange(age_18,17,59)
replace construction_m_z_v01_18= 0.0  if [[inlist(construction_m_v01_18,1) & yrschool==0] |
[inlist(construction_m_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_m_v01_18,2) &
inrange(yrschool,7,25)]] & inrange(age_18,17,59)
replace construction_m_z_v01_18=-1.0  if [
                                [inlist(construction_m_v01_18,1) & inrange(yrschool,1,6)]
```

```
        ] & inrange(age_18,17,59)
replace construction_m_z_v01_18=-2.0  if [[inlist(construction_m_v01_18,0) & yrschool==0]
        |
                [inlist(construction_m_v01_18,1) & inrange(yrschool,7,25)]] &
inrange(age_18,17,59)
replace construction_m_z_v01_18=-3.0  if [
                        [inlist(construction_m_v01_18,0) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_18,0) & inrange(yrschool,7,25)]] & inrange(age_18,17,59)

replace construction_m_z_v01_18= 1.0  if [[inlist(construction_m_v01_18,2) & yrschool==0]



                ] & inrange(age_18,60,69)
replace construction_m_z_v01_18= 0.0  if [[inlist(construction_m_v01_18,1) & yrschool==0] |
[inlist(construction_m_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_m_v01_18,2) &
inrange(yrschool,7,25)]] & inrange(age_18,60,69)
replace construction_m_z_v01_18=-1.0  if [
                            [inlist(construction_m_v01_18,1) & inrange(yrschool,1,6)]
                                                                                ] &
inrange(age_18,60,69)
replace construction_m_z_v01_18=-1.5  if [

                                    [inlist(construction_m_v01_18,1) & inrange(yrschool,7,25)]]
& inrange(age_18,60,69)
replace construction_m_z_v01_18=-2.0  if [[inlist(construction_m_v01_18,0) & yrschool==0]


        ] & inrange(age_18,60,69)
```

```
replace construction_m_z_v01_18=-3.0  if [
                                [inlist(construction_m_v01_18,0) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_18,0) & inrange(yrschool,7,25)]] & inrange(age_18,60,69)

replace construction_m_z_v01_18= 1.0  if [[inlist(construction_m_v01_18,2) & yrschool==0]



          ] & inrange(age_18,70,79)
replace construction_m_z_v01_18= 0.0  if [[inlist(construction_m_v01_18,1) & yrschool==0] |
[inlist(construction_m_v01_18,2) & inrange(yrschool,1,6)] | [inlist(construction_m_v01_18,2) &
inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_m_z_v01_18=-1.0  if [
                                [inlist(construction_m_v01_18,1) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_18,1) & inrange(yrschool,7,25)]] & inrange(age_18,70,79)
replace construction_m_z_v01_18=-1.5  if [[inlist(construction_m_v01_18,0) & yrschool==0]


] & inrange(age_18,70,79)
replace construction_m_z_v01_18=-2.0  if [
                                [inlist(construction_m_v01_18,0) & inrange(yrschool,1,6)]
                                                                                ] &
inrange(age_18,70,79)
replace construction_m_z_v01_18=-3.0  if [

                                [inlist(construction_m_v01_18,0) & inrange(yrschool,7,25)]]
& inrange(age_18,70,79)

replace construction_m_z_v01_18= 1.5  if [[inlist(construction_m_v01_18,2) & yrschool==0]


    ] & inrange(age_18,80,120)
```

```
replace construction_m_z_v01_18= 1.0  if [
                                    [inlist(construction_m_v01_18,2) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_18,2) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_v01_18= 0.0  if [[inlist(construction_m_v01_18,1) & yrschool==0] |
[inlist(construction_m_v01_18,1) & inrange(yrschool,1,6)] | [inlist(construction_m_v01_18,1) &
inrange(yrschool,7,25)]] & inrange(age_18,80,120)
replace construction_m_z_v01_18=-1.0  if [[inlist(construction_m_v01_18,0) & yrschool==0]



                ] & inrange(age_18,80,120)
replace construction_m_z_v01_18=-2.0  if [
                                    [inlist(construction_m_v01_18,0) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_18,0) & inrange(yrschool,7,25)]] & inrange(age_18,80,120)

replace construction_m_z_v01_18= construction_m_v01_18 if mi(construction_m_v01_18) &
construction_m_z_v01_18==.
replace construction_m_z_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_m_z_v01_18==.
replace construction_m_z_v01_18= .p if inlist(tipent_18,3,4)
label variable construction_m_z_v01_18 "MHAS 2018 Delayed Memory - Delayed recall of
figure Z-Scores v2001-2003"

** Impairment Status **
gen construction_m_imp_v01_18 = .
label variable construction_m_imp_v01_18 "MHAS 2018 Delayed Memory - Delayed recall of
figure Impairment v2001-2003"

* age_18 <59 & Years of Education=0
replace construction_m_imp_v01_18=1 if inrange(age_18,17,59) & yrschool==0 &
inlist(construction_m_v01_18,0)
```

```
replace construction_m_imp_v01_18=0 if inrange(age_18,17,59) & yrschool==0 &
inlist(construction_m_v01_18,1,2)
* age_18 <59 & Years of Education 1-6
replace construction_m_imp_v01_18=1  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0  if inrange(age_18,17,59) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,1,2)
* age_18 <59 & Years of Education 7+
replace construction_m_imp_v01_18=1  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,0,1)
replace construction_m_imp_v01_18=0  if inrange(age_18,17,59) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,2)

* age_18 60-69 & Years of Education=0
replace construction_m_imp_v01_18=1 if inrange(age_18,60,69) & yrschool==0 &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,60,69) & yrschool==0 &
inlist(construction_m_v01_18,1,2)
* age_18 60-69 & Years of Education 1-6
replace construction_m_imp_v01_18=1  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0  if inrange(age_18,60,69) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,1,2)
* age_18 60-69 & Years of Education 7+
replace construction_m_imp_v01_18=1  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,0,1)
replace construction_m_imp_v01_18=0  if inrange(age_18,60,69) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,2)

* age_18 70-79 & Years of Education=0
```

```
replace construction_m_imp_v01_18=1 if inrange(age_18,70,79) & yrschool==0 &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,70,79) & yrschool==0 &
inlist(construction_m_v01_18,1,2)
* age_18 70-79 & Years of Education 1-6
replace construction_m_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,1,2)
* age_18 70-79 & Years of Education 7+
replace construction_m_imp_v01_18=1 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,70,79) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,1,2)

* age_18 80+ & Years of Education=0
//replace construction_m_imp_v01_18=1 if inrange(age_18,80,120) & yrschool==0 &
inlist(construction_m_v01_18,)
replace construction_m_imp_v01_18=0 if inrange(age_18,80,120) & yrschool==0 &
inlist(construction_m_v01_18,0,1,2)
* age_18 80+ & Years of Education 1-6
replace construction_m_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,1,6) &
inlist(construction_m_v01_18,1,2)
* age_18 80+ & Years of Education 7+
replace construction_m_imp_v01_18=1 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,0)
replace construction_m_imp_v01_18=0 if inrange(age_18,80,120) & inrange(yrschool,7,25) &
inlist(construction_m_v01_18,1,2)
```

```
replace construction_m_imp_v01_18= construction_m_v01_18 if mi(construction_m_v01_18)
& construction_m_imp_v01_18==.
replace construction_m_imp_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
construction_m_imp_v01_18==.
replace construction_m_imp_v01_18= .p if inlist(tipent_18,3,4)
label values construction_m_imp_v01_18 cognitive_18


***  TOTAL NUMBER OF TASKS COMPLETED (max. 5) ***
egen numb_tasks_miss_v01_18= rowmiss(iwr_18 dwr_18 construction_m_18 visual_scan_18
construction_18) if inlist(tipent_18,1,2)

gen numb_tasks_comp_v01_18= 5-numb_tasks_miss_v01_18 if inlist(tipent_18,1,2)
replace numb_tasks_comp_v01_18= .p if inlist(tipent_18,3,4)
label variable numb_tasks_comp_v01_18 "MHAS 2018 Number of Tasks Completed (0-5)"

drop numb_tasks_miss_v01_18

***  TOTAL NUMBER OF TASKS WITH IMPAIRMENT IF COMPLETED 2 OR MORE
TASKS(max. 5) ***
egen numb_tasks_imp_v01_18= rowtotal(iwr_imp_18 dwr_imp_18
construction_m_imp_v01_18 visual_scan_imp_18 construction_imp_v01_18) if
inrange(numb_tasks_comp_v01_18,2,5)
replace numb_tasks_imp_v01_18= .i if inrange(numb_tasks_comp_v01_18,0,1)
replace numb_tasks_imp_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
mi(numb_tasks_imp_v01_18)
replace numb_tasks_imp_v01_18= .s if [iwr_imp_18==.s ///
                                                    & dwr_imp_18==.s ///
                                                    &
construction_m_imp_v01_18==.s ///
                                                    & visual_scan_imp_18==.s ///
```

```
                                                    & construction_imp_v01_18==.s]
///
                                                    & mi(numb_tasks_imp_v01_18)
replace numb_tasks_imp_v01_18= .p if inlist(tipent_18,3,4)
label variable numb_tasks_imp_v01_18 "MHAS 2018 Number of Tasks with Impairment if
completed 2+ (0-5)"

gen two_more_tasks_imp_v01_18=.
replace two_more_tasks_imp_v01_18=0 if inrange(numb_tasks_imp_v01_18,0,1)
replace two_more_tasks_imp_v01_18=1 if inrange(numb_tasks_imp_v01_18,2,5)
replace two_more_tasks_imp_v01_18= .i if inrange(numb_tasks_comp_v01_18,0,1)
replace two_more_tasks_imp_v01_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
mi(two_more_tasks_imp_v01_18)
replace two_more_tasks_imp_v01_18= .s if [iwr_imp_18==.s ///
                                                    & dwr_imp_18==.s ///
                                                    &
construction_m_imp_v01_18==.s ///
                                                    & visual_scan_imp_18==.s
///
                                                    &
construction_imp_v01_18==.s] ///
                                                    &
mi(two_more_tasks_imp_v01_18)
replace two_more_tasks_imp_v01_18= .p if inlist(tipent_18,3,4)
label variable two_more_tasks_imp_v01_18 "MHAS 2018 Dummy of 2+/5 Tasks with
Impairment if completed 2+ (0-1)"


***  TOTAL NUMER OF TASKS COMPLETED (max. 8) ***
egen numb_tasks_miss_18= rowmiss(iwr_18 dwr_18 construction_m_18 visual_scan_18
construction_18 orientation_18 verbal_fluency_18 serial7_18) if inlist(tipent_18,1,2)
```

```
gen numb_tasks_comp_18= 8-numb_tasks_miss_18 if inlist(tipent_18,1,2)
replace numb_tasks_comp_18= .p if inlist(tipent_18,3,4)
label variable numb_tasks_comp_18 "MHAS 2018 Number of Tasks Completed (0-7)"

drop numb_tasks_miss_18

***  TOTAL NUMBER OF TASKS WITH IMPAIRMENT IF COMPLETED 2 OR MORE TASKS(max. 8) ***
egen numb_tasks_imp_18= rowtotal(iwr_imp_18 dwr_imp_18 construction_m_imp_18 visual_scan_imp_18 construction_imp_18 orientation_imp_18 verbal_fluency_imp_18 serial7_imp_18) ///
                                                    if inrange(numb_tasks_comp_18,2,8)
replace numb_tasks_imp_18= .i if inrange(numb_tasks_comp_18,0,1)
replace numb_tasks_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)]
replace numb_tasks_imp_18= .s if [iwr_imp_18==.s ///
                                                    & dwr_imp_18==.s ///
                                                    & construction_m_imp_18==.s ///
                                                    & visual_scan_imp_18==.s ///
                                                    & construction_imp_18==.s ///
                                                    & orientation_imp_18==.s ///
                                                    & verbal_fluency_imp_18==.s] ///
                                                    & mi(numb_tasks_imp_18)
replace numb_tasks_imp_18= .p if inlist(tipent_18,3,4)
label variable numb_tasks_imp_18 "MHAS 2012 Number of Tasks with Impairment if completed 2+ (0-8)"

gen two_more_tasks_imp_18=.
replace two_more_tasks_imp_18=0 if inrange(numb_tasks_imp_18,0,1)
replace two_more_tasks_imp_18=1 if inrange(numb_tasks_imp_18,2,8)
replace two_more_tasks_imp_18= .i if inrange(numb_tasks_comp_18,0,1)
```

```
replace two_more_tasks_imp_18= .m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m)] &
mi(two_more_tasks_imp_18)
replace two_more_tasks_imp_18= .s if [iwr_imp_18==.s ///
                                      & dwr_imp_18==.s ///
                                      & construction_m_imp_18==.s ///
                                      & visual_scan_imp_18==.s ///
                                      & construction_imp_18==.s ///
                                      & orientation_imp_18==.s ///
                                      & verbal_fluency_imp_18==.s] ///
                                      & mi(two_more_tasks_imp_18)
replace two_more_tasks_imp_18= .p if inlist(tipent_18,3,4)
label variable two_more_tasks_imp_18 "MHAS 2018 Dummy of 2+/8 Tasks with Impairment if
completed 2+ (0-1)"


**********************************************************************
********* MHAS 2018 INSTRUMENTAL ACTIVITIES OF DAILY LIVING *********
**********************************************************************

*** IADL ***
** Managing money **
gen iadl_money_18 =.
replace iadl_money_18 = .m if inlist(tipent_18,1,2,3,4) & h29a_18 == .i
replace iadl_money_18 = .d if h29a_18 == 9
replace iadl_money_18 = .r if h29a_18 == 8
replace iadl_money_18 = .p if inlist(tipent_18,3,4)
replace iadl_money_18 = .x if h29a_18 == 7 & inlist(h29b_18,8,9)
replace iadl_money_18 = 0 if h29a_18 == 2 | (h29a_18 == 7 & h29b_18 == 2)
replace iadl_money_18 = 1 if inlist(h29a_18,1,6) | (h29a_18 == 7 & h29b_18 == 1)
label variable iadl_money_18 "IADLs: Difficulty Managing Money 2018"
```

```
** Taking medicines **
gen iadl_medicines_18 =.
replace iadl_medicines_18 = .m if inlist(tipent_18,1,2,3,4) & h28a_18 == .i
replace iadl_medicines_18 = .d if h28a_18 == 9
replace iadl_medicines_18 = .r if h28a_18 == 8
replace iadl_medicines_18 = .p if inlist(tipent_18,3,4)
replace iadl_medicines_18 = .x if h28a_18 == 7 & inlist(h28b_18,8,9)
replace iadl_medicines_18 = 0 if h28a_18 == 2 | (h28a_18 == 7 & h28b_18 == 2)
replace iadl_medicines_18 = 1 if inlist(h28a_18,1,6) | (h28a_18 == 7 & h28b_18 == 1)
label variable iadl_medicines_18 "IADLs: Difficulty Taking Medicines 2018"

** Shopping for groceries **
gen iadl_shopping_18 =.
replace iadl_shopping_18 = .m if inlist(tipent_18,1,2,3,4) & h27a_18 == .i
replace iadl_shopping_18 = .d if h27a_18 == 9
replace iadl_shopping_18 = .r if h27a_18 == 8
replace iadl_shopping_18 = .p if inlist(tipent_18,3,4)
replace iadl_shopping_18 = .x if h27a_18 == 7 & inlist(h27b_18,8,9)
replace iadl_shopping_18 = 0 if h27a_18 == 2 | (h27a_18 == 7 & h27b_18 == 2)
replace iadl_shopping_18 = 1 if inlist(h27a_18,1,6) | (h27a_18 == 7 & h27b_18 == 1)
label variable iadl_shopping_18 "IADLs: Difficulty Shopping for Groceries 2018"

** Cooking a hot meal **
gen iadl_cooking_18 =.
replace iadl_cooking_18 = .m if inlist(tipent_18,1,2,3,4) & h26a_18 == .i
replace iadl_cooking_18 = .d if h26a_18 == 9
replace iadl_cooking_18 = .r if h26a_18 == 8
replace iadl_cooking_18 = .p if inlist(tipent_18,3,4)
replace iadl_cooking_18 = .x if h26a_18 == 7 & inlist(h26b_18,8,9)
replace iadl_cooking_18 = 0 if h26a_18 == 2 | (h26a_18 == 7 & h26b_18 == 2)
replace iadl_cooking_18 = 1 if inlist(h26a_18,1,6) | (h26a_18 == 7 & h26b_18 == 1)
```

```
label variable iadl_cooking_18 "IADLs: Difficulty Cooking a Meal 2018"
label values iadl_cooking_18 adls

** Number of Limitations with IADLs (0-4) **
egen iadls_18 = rowtotal(iadl_money_18 iadl_medicines_18 iadl_shopping_18
iadl_cooking_18) if inlist(tipent_18,1,2)
replace iadls_18 = .m if inlist(tipent_18,1,2,3,4)  & h26a_18 == .i & h27a_18 == .i & h28a_18
== .i & h29a_18 == .i & iadls_18==.
replace iadls_18 = .d if iadl_money_18 == .d & iadl_medicines_18 == .d & iadl_shopping_18
== .d & iadl_cooking_18 == .d
replace iadls_18 = .r if iadl_money_18 == .r & iadl_medicines_18 == .r & iadl_shopping_18 ==
.r & iadl_cooking_18 == .r
replace iadls_18 = .x if iadl_money_18 == .x & iadl_medicines_18 == .x & iadl_shopping_18
== .x & iadl_cooking_18 == .x
replace iadls_18 = .x if iadl_money_18 == .x & iadl_medicines_18 == .x & iadl_shopping_18
== .x & iadl_cooking_18 == .x
replace iadls_18 = .p if inlist(tipent_18,3,4)
label variable iadls_18 "Number of Limitations with IADLs 2018 (0–4)"



*********************************************************
********* MHAS 2018 COGNITIVE STATUS ASSESSMENT  *********
*********************************************************

* COGNITIVE STATUS CLASIFICATION: 1) CIND: impairment in 2+ cognitive tasks and NO
IADLs difficulties
                    /// 2) Dementia: 2a) Proxy subjects who had a score equal or above 3.4 in
the IQCODE
                                        ///          2b) Direct interview subjects impaired in 2+
cognitive tasks and
```

```
                                        ///             had difficulty in one or more IADLs
                                    /// 3) Normal with IADLS:
                                    ///         3a) Direct interview subjects with
no impairment
                                    ///             or with impairment in only one
cognitive task and who had difficulty in one or more IADLs
                                    /// 4) Normal NO IADLS:
                                    ///         4a) Proxy respondents with a score
below 3.4 in the IQCODE
                                    ///         4b) Direct interview subjects with
no impairment
                                    ///             or with impairment in only one
cognitive task and who did not have difficulty with IADLs

** Cognitive Status Classification using 8 tasks (if 2 or more completed) **
gen cognitive_status_18=.
replace cognitive_status_18=3 if cog_imp_iqcode_18==1 | [inrange(numb_tasks_imp_18,2,8)
& inrange(iadls_18,1,4)]
replace cognitive_status_18=2 if inrange(numb_tasks_imp_18,2,8) & iadls_18==0
replace cognitive_status_18=1 if inrange(numb_tasks_imp_18,0,1) & inrange(iadls_18,1,4)
replace cognitive_status_18=0 if cog_imp_iqcode_18==0 | [inrange(numb_tasks_imp_18,0,1)
& iadls_18==0]
replace cognitive_status_18=.i if [inrange(numb_tasks_comp_18,0,1) |
cog_imp_iqcode_18==.i] & cognitive_status_18==.
replace cognitive_status_18=.d if [iadls_18==.d | cog_imp_iqcode_18==.d] &
cognitive_status_18==.
replace cognitive_status_18=.r if [iadls_18==.r | cog_imp_iqcode_18==.r] &
cognitive_status_18==.
replace cognitive_status_18=.m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m) |
inlist(iadls_18,.m,.)] & cognitive_status_18==.
label variable cognitive_status_18 "MHAS 2018 Cognitive Status (using 8 tasks)"
```

```
label define cognitive_status_18 0 "0.Normal" 1 "1.Normal with Instrumental Impairment" 2
"2.CIND" 3 "3.Dementia", replace
label values cognitive_status_18 cognitive_status_18

** Cognitive Status Classification using 5 tasks (if 2 or more completed) **
gen cognitive_status_v01_18=.
replace cognitive_status_v01_18=3 if cog_imp_iqcode_18==1 |
[inrange(numb_tasks_imp_v01_18,2,5) & inrange(iadls_18,1,4)]
replace cognitive_status_v01_18=2 if inrange(numb_tasks_imp_v01_18,2,5) & iadls_18==0
replace cognitive_status_v01_18=1 if inrange(numb_tasks_imp_v01_18,0,1) &
inrange(iadls_18,1,4)
replace cognitive_status_v01_18=0 if cog_imp_iqcode_18==0 |
[inrange(numb_tasks_imp_v01_18,0,1) & iadls_18==0]
replace cognitive_status_v01_18=.i if [inrange(numb_tasks_comp_v01_18,0,1) |
cog_imp_iqcode_18==.i] & cognitive_status_v01_18==.
replace cognitive_status_v01_18=.d if [iadls_18==.d | cog_imp_iqcode_18==.d] &
cognitive_status_v01_18==.
replace cognitive_status_v01_18=.r if [iadls_18==.r | cog_imp_iqcode_18==.r] &
cognitive_status_v01_18==.
replace cognitive_status_v01_18=.m if [inlist(age_18,888,999) | inlist(yrschool,.d,.r,.m) |
inlist(iadls_18,.m,.)] & cognitive_status_v01_18==.
label variable cognitive_status_v01_18 "MHAS 2018 Cognitive Status v2001-2003"
label values cognitive_status_v01_18 cognitive_status_18
```

Appendix III. STATA Codes for the Creation of the 2021 Cognitive Function Variables (Direct and Proxy Interviews)

The following STATA codes can be used to construct the score variable for each task included in the 2021 wave.

```
************************************************************
**** MHAS 2021 COGNITIVE EXERCISES - CCCE (SECTION E) ****
************************************************************

label data "Version 1. April 2025"

**** CREATING VARIABLES BY TASK AND DOMAIN ****

* NOTE: Special missing values were assigned as follows:
* .p indicates proxy interview, .d don't know, .s skip, .r refusal, and .l physical limitation or visual limitation,


************************************************************
**** MHAS 2021 COGNITIVE EXERCISES - CCCE (SECTION E) ****
************************************************************


**** CREATING VARIABLES BY TASK AND DOMAIN ****

* NOTE: Special missing values were assigned as follows:
* .p indicates proxy interview, .d don't know, .skip, .r refusal and .v indicates visual limination, and .l physical limitation


*** INMEDIATE MEMORY ***
*** Repeat 8 words (score 0-8) ***
recode e7a1_tot_21 e7a2_tot_21 e7a3_tot_21 e7b1_tot_21 e7b2_tot_21 e7b3_tot_21 (9=0)
```

```
egen iwr_21=rowmean (e7a1_tot_21 e7a2_tot_21 e7a3_tot_21 e7b1_tot_21 e7b2_tot_21 e7b3_tot_21)
if [inrange(e7a1_tot_21,0,8) | inrange(e7a2_tot_21,0,8) | inrange(e7a3_tot_21,0,8) |
inrange(e7b1_tot_21,0,8) | inrange(e7b2_tot_21,0,8) | inrange(e7b3_tot_21,0,8)] & inlist(tipent_21,1,2)
replace iwr_21 = .r if [e7_1_21 == 9 | e7_2_21 == 9 | e7_3_21 == 9] & mi(iwr_21)
replace iwr_21 = .p if inlist(tipent_21,3,4)
label variable iwr_21 "MHAS 2021 Immediate Memory - Repeat 8 words (0-8)"

*** ATTENTION ***
*** Visual Scan (score 0-60) ***
gen visual_scan_21=.
replace visual_scan_21 = .s if e10a_21 == .s
replace visual_scan_21 = .r if e10a_21 == 8
replace visual_scan_21 = .p if inlist(tipent_21,3,4)
replace visual_scan_21 = 0 if e10a_21 == 2
replace visual_scan_21 = e10_c_21 if inrange(e10_c_21, 0,60)
label variable visual_scan_21 "MHAS 2021 Attention - Visual Scan (0-60)"

*** ORIENTATION ***
*** Day of the month (score 0-1) ***
gen name_day_21=.
replace name_day_21 = .d if e11a_21 == 9
replace name_day_21 = .r if e11a_21 == 8
replace name_day_21 = .p if inlist(tipent_21,3,4)
replace name_day_21 = 1 if e11a_21 == 1
replace name_day_21 = 0 if e11a_21 == 2

*** Month (score 0-1) ***
gen name_month_21=.
replace name_month_21 = .d if e11b_21 == 9
replace name_month_21 = .r if e11b_21 == 8
replace name_month_21 = .p if inlist(tipent_21,3,4)
replace name_month_21 = 1 if e11b_21 == 1
replace name_month_21 = 0 if e11b_21 == 2
```

```
*** Year (score 0-1) ***
gen name_year_21=.
replace name_year_21 = .d if e11c_21 == 9
replace name_year_21 = .r if e11c_21 == 8
replace name_year_21 = .p if inlist(tipent_21,3,4)
replace name_year_21 = 1 if e11c_21 == 1
replace name_year_21 = 0 if e11c_21 == 2

*** Orientation (score 0-3)
egen orientation_21= rowtotal(name_day_21 name_month_21 name_year_21) if inlist(tipent_21,1,2)
replace orientation_21 = .d if e11a_21 == 9 & e11b_21 == 9 & e11c_21 == 9
replace orientation_21 = .r if e11a_21 == 8 | e11b_21 == 8 | e11c_21 == 8
replace orientation_21 = .p if inlist(tipent_21,3,4)
label variable orientation_21 "MHAS 2021 Orientation (0-3)"

*** EXECUTIVE FUNCTION ***
*** Verbal fluency (score 0-4) ***
** Number of repeated animals
gen verbal_fluency_rep_21= e9b_c_21 if inrange(e9b_c_21,0,60)
replace verbal_fluency_rep_21 = 0 if e9a_21==2
replace verbal_fluency_rep_21 = .r if e9a_21==8
replace verbal_fluency_rep_21 = .p if inlist(tipent_21,3,4)
label variable verbal_fluency_rep_21 "MHAS 2021 Executive function - Verbal Fluency: number of repeted animals (0-50)"

** Number of different animals (88 "No response" recorded as a missing)
gen verbal_fluency_num_21=e9a_c_21
replace verbal_fluency_num_21 = 0 if e9a_21==2
replace verbal_fluency_num_21 = .r if e9a_21==8
replace verbal_fluency_num_21 = .p if inlist(tipent_21,3,4)
label variable verbal_fluency_num_21 "MHAS 2021 Executive function - Verbal Fluency: number of animals (0-50)"

** Categorical variable (using S. Mejia 2014 SPM)
```

```
gen verbal_fluency_21=.
replace verbal_fluency_21=0 if e9a_21 == 2
replace verbal_fluency_21=1 if inrange(verbal_fluency_num_21,0,8)
replace verbal_fluency_21=2 if inrange(verbal_fluency_num_21,9,18)
replace verbal_fluency_21=3 if inrange(verbal_fluency_num_21,19,24)
replace verbal_fluency_21=4 if inrange(verbal_fluency_num_21,25,50)
replace verbal_fluency_21=.r if e9a_21==8
replace verbal_fluency_21=.p if inlist(tipent_21,3,4)
label variable verbal_fluency_21 "MHAS 2021 Executive function - Verbal Fluency (0-4)"



*** Serial 3 (score 0-5) ***
** Correct/Incorrect
gen serial3_1=.
replace serial3_1=.p if inlist(tipent_21,3,4)
replace serial3_1=.r if e15_21==88 | e15a_21==88
replace serial3_1=1 if inlist(e15a_21,93,100)
replace serial3_1=0 if !inlist(e15a_21,93,100,.) | e15_21==99 | e15a_21==99

gen serial3_2=.
replace serial3_2=.p if inlist(tipent_21,3,4)
replace serial3_2=.r if e15_21==88 | e15b_21==88
replace serial3_2=1 if e15b_21==e15a_21-7 & e15b_21!=.
replace serial3_2=0 if !inlist(e15b_21,e15a_21-7,.) | e15_21==99 | e15b_21==99

gen serial3_3=.
replace serial3_3=.p if inlist(tipent_21,3,4)
replace serial3_3=.r if e15_21==88 | e15c_21==88
replace serial3_3=1 if e15c_21==e15b_21-7 & e15c_21!=.
replace serial3_3=0 if !inlist(e15c_21,e15b_21-7,.) | e15_21==99 | e15c_21==99

gen serial3_4=.
replace serial3_4=.p if inlist(tipent_21,3,4)
replace serial3_4=.r if e15_21==88 | e15d_21==88
```

```
replace serial3_4=1 if e15d_21==e15c_21-7 & e15d_21!=.
replace serial3_4=0 if !inlist(e15d_21,e15c_21-7,.) | e15_21==99 | e15d_21==99

gen serial3_5=.
replace serial3_5=.p if inlist(tipent_21,3,4)
replace serial3_5=.r if e15_21==88 | e15e_21==88
replace serial3_5=1 if e15e_21==e15d_21-7 & e15e_21!=.
replace serial3_5=0 if !inlist(e15e_21,e15d_21-7,.) | e15_21==99 | e15e_21==00


** Serial 3 score variable
egen serial3_21 = rowtotal(serial3_1 serial3_2 serial3_3 serial3_4 serial3_5) if inlist(serial3_1,0,1) |
inlist(serial3_2,0,1) | inlist(serial3_3,0,1) | inlist(serial3_4,0,1) | inlist(serial3_5,0,1)
replace serial3_21=.r if e15_21==88
replace serial3_21=.p if inlist(tipent_21,3,4)
label variable serial3_21 "MHAS Executive Function - Series of subtractions 3 from 20 (0-5)"


*** CONSTRUCTIONAL PRAXIS ***
*** Copy one figure (score 0-6) ***
gen construction_21= .
replace construction_21 = .s if e8a_21==.s
replace construction_21 = .r if e8a_21==8
replace construction_21 = .p if inlist(tipent_21,3,4)
replace construction_21 = e8a_c_21 if inrange(e8a_c_21,0,6)
replace construction_21 = 0 if e8a_21==2
label variable construction_21 "MHAS 2021 Constructional Praxis (0-6)"

*** Copy one figure comparable to 2021 & 2003 (score 0-2) ***
gen construction_v01_21= .
replace construction_v01_21 = .s if e8a_21==.s
replace construction_v01_21 = .r if e8a_21==8
replace construction_v01_21 = .p if inlist(tipent_21,3,4)
replace construction_v01_21 = 0 if inrange(e8a_c_21,0,2)
```

```
replace construction_v01_21 = 0 if e8a_21==2
replace construction_v01_21 = 1 if inrange(e8a_c_21,3,4)
replace construction_v01_21 = 2 if inrange(e8a_c_21,5,6)
label variable construction_v01_21 "MHAS 2021 Constructional Praxis v2001-2003 (0-2)"


*** DELAYED MEMORY ***
*** Repeat 8 words (score 0-8) ***
egen dwr_21=rowtotal (e14a_tot_21 e14b_tot_21) if inlist(tipent_21,1,2)
replace dwr_21 = 0 if e14a_tot_21==9 | e14b_tot_21==9
replace dwr_21 = .r if e14_21==8
replace dwr_21 = .p if inlist(tipent_21,3,4)
label variable dwr_21 "MHAS 2021 Delayed Memory - Repeat 8 words (0-8)"

*** Delayed recall/copy of figure (score 0-6) ***
gen construction_m_21= .
replace construction_m_21 = .s if e13_21==.s
replace construction_m_21 = .r if e13_21==8
replace construction_m_21 = .p if inlist(tipent_21,3,4)
replace construction_m_21 = e13_c_21 if inrange(e13_c_21,0,6)
replace construction_m_21 = 0 if e13_21==2
label variable construction_m_21 "MHAS 2021 Delayed Memory - Delayed recall of figure (0-6)"

*** Delayed recall/copy of figure comparable to 2001 & 2003 (score 0-2) ***
gen construction_m_v01_21= .
replace construction_m_v01_21 = .s if e13_21==.s
replace construction_m_v01_21 = .r if e13_21 == 8
replace construction_m_v01_21 = .p if inlist(tipent_21,3,4)
replace construction_m_v01_21 = 0 if inrange(e13_c_21,0,2)
replace construction_m_v01_21 = 0 if e13_21==2
replace construction_m_v01_21 = 1 if inrange(e13_c_21,3,4)
replace construction_m_v01_21 = 2 if inrange(e13_c_21,5,6)
label variable construction_m_v01_21 "MHAS 2021 Delayed Memory - Delayed recall of figure v2001-2003 (0-2)"
```

```
*** TOTAL CCCE SCORE ***
*** NUMBER of Tasks missing (out of 8 tasks) ***
egen ccce_miss_21=rowmiss(iwr_21 dwr_21 visual_scan_21 orientation_21 verbal_fluency_21 construction_21 construction_m_21 serial3_21) ///
                                        if inlist(tipent_21,1,2)
replace ccce_miss_21=.p if inlist(tipent_21,3,4)
label variable ccce_miss_21 "MHAS 2021 Number of CCCE Tasks missing (0-8)"


*** TOTAL MHAS 2021 CCCE SCORE (score 0-99) - Including all 2021 Items ***
egen ccce_21 = rowtotal(iwr_21 dwr_21 visual_scan_21 orientation_21 verbal_fluency_21 construction_21 construction_m_21 serial3_21) if inlist(tipent_21,1,2)
replace ccce_21 = .d  if [iwr_21==.d ///
                                        | dwr_21==.d ///
                                        | visual_scan_21==.d ///
                                        | orientation_21==.d ///
                                        | verbal_fluency_21==.d ///
                                        | construction_21==.d ///
                                        | construction_m_21==.d ///
                                        | serial3_21==.d] & ccce_21==.
replace ccce_21 = .r   if [iwr_21==.r ///
                                        | dwr_21==.r ///
                                        | visual_scan_21==.r ///
                                        | orientation_21==.r ///
                                        | verbal_fluency_21==.r ///
                                        | construction_21==.r ///
                                        | construction_m_21==.r ///
                                        | serial3_21==.r] & ccce_21==.
replace ccce_21 = .s if [e3a_21 == 2 | inlist(e5_21,1,2,3) | inlist(e8a_21,2)] & ccce_21==.
replace ccce_21 = .r if resul_ec_21==4 & ccce_21==.
replace ccce_21 = .i if resul_ec_21==3 & ccce_21==.
replace ccce_21 = .p if inlist(tipent_21,3,4)
```

```
replace ccce_21 = .m if ccce_miss_21 == 8

label variable ccce_21 "MHAS 2021 CCCE Score (0-99)"



*** NUMBER of Tasks missing (out of 5 tasks) ***
egen ccce_miss_v01_21=rowmiss(iwr_21 dwr_21 visual_scan_21 construction_v01_21 construction_m_v01_21) if inlist(tipent_21,1,2)
replace ccce_miss_v01_21=.p if inlist(tipent_21,3,4)
label variable ccce_miss_v01_21 "MHAS 2021 Number of CCCE Tasks missing - Comparable to 2001 (0-5)"

*** TOTAL MHAS 2021 CCCE SCORE (score 0-80) - With only the 5 items included since 2001***
egen ccce_v01_21 = rowtotal(iwr_21 dwr_21 visual_scan_21 construction_v01_21 construction_m_v01_21) ///
                                        if inlist(tipent_21,1,2)
replace ccce_v01_21 = .d  if [iwr_21==.d ///
                                                | dwr_21==.d ///
                                                | visual_scan_21==.d ///
                                                | construction_v01_21==.d ///
                                                | construction_m_v01_21==.d] & ccce_v01_21==.
replace ccce_v01_21 = .r   if [iwr_21==.r ///
                                                | dwr_21==.r ///
                                                | visual_scan_21==.r ///
                                                | construction_v01_21==.r ///
                                                | construction_m_v01_21==.r] & ccce_v01_21==.

replace ccce_v01_21 = .s if [e3a_21 == 2 | inlist(e5_21,1,2,3) | inlist(e8a_21,2)] & ccce_v01_21==.
replace ccce_v01_21 = .r if resul_ec_21==4 & ccce_v01_21==.
replace ccce_v01_21 = .i if resul_ec_21==3 & ccce_v01_21==.
replace ccce_v01_21 = .p if inlist(tipent_21,3,4)
replace ccce_v01_21 = .m if ccce_miss_v01_21 == 5

label variable ccce_v01_21 "MHAS 2021 CCCE Score - Comparable to 2001 (0-80)"
```

```
************************************************************
******* MHAS 2021 IQCODE SHORT -Jorm (SECTION PC) ********
************************************************************

* Remembering things about family & friends
gen iqcode1_21=.
replace iqcode1_21=.r if pc5_21==8 | pc6_21==8 | pc7_21==8
replace iqcode1_21=.d if pc5_21==9 | pc6_21==9 | pc7_21==9
replace iqcode1_21=3 if pc5_21==2
replace iqcode1_21=pc6_21 if inlist(pc6_21,1,2)
replace iqcode1_21=4 if pc7_21==2
replace iqcode1_21=5 if pc7_21==1

* Remembering things that happened recently
gen iqcode2_21=.
replace iqcode2_21=.r if pc8_21==8 | pc9_21==8 | pc10_21==8
replace iqcode2_21=.d if pc8_21==9 | pc9_21==9 | pc10_21==9
replace iqcode2_21=3 if pc8_21==2
replace iqcode2_21=pc9_21 if inlist(pc9_21,1,2)
replace iqcode2_21=4 if pc10_21==2
replace iqcode2_21=5 if pc10_21==1

* Recalling conversations a few name_day_21s later
gen iqcode3_21=.
replace iqcode3_21=.r if pc11_21==8 | pc12_21==8 | pc13_21==8
replace iqcode3_21=.d if pc11_21==9 | pc12_21==9 | pc13_21==9
replace iqcode3_21=3 if pc11_21==2
replace iqcode3_21=pc12_21 if inlist(pc12_21,1,2)
replace iqcode3_21=4 if pc13_21==2
replace iqcode3_21=5 if pc13_21==1

* Remembering his/her address & phone number
```

```
gen iqcode4_21=.
replace iqcode4_21=.r if pc14_21==8 | pc15_21==8 | pc16_21==8
replace iqcode4_21=.d if pc14_21==9 | pc15_21==9 | pc16_21==9
replace iqcode4_21=3 if pc14_21==2
replace iqcode4_21=pc15_21 if inlist(pc15_21,1,2)
replace iqcode4_21=4 if pc16_21==2
replace iqcode4_21=5 if pc16_21==1

* Remembering what name_day_21 and name_month_21 it is
gen iqcode5_21=.
replace iqcode5_21=.r if pc17_21==8 | pc18_21==8 | pc19_21==8
replace iqcode5_21=.d if pc17_21==9 | pc18_21==9 | pc19_21==9
replace iqcode5_21=3 if pc17_21==2
replace iqcode5_21=pc18_21 if inlist(pc18_21,1,2)
replace iqcode5_21=4 if pc19_21==2
replace iqcode5_21=5 if pc19_21==1

* Remembering where things are usually kept
gen iqcode6_21=.
replace iqcode6_21=.r if pc20_21==8 | pc21_21==8 | pc22_21==8
replace iqcode6_21=.d if pc20_21==9 | pc21_21==9 | pc22_21==9
replace iqcode6_21=3 if pc20_21==2
replace iqcode6_21=pc21_21 if inlist(pc21_21,1,2)
replace iqcode6_21=4 if pc22_21==2
replace iqcode6_21=5 if pc22_21==1

* Remembering where to find things which have been put in a different place from usual
gen iqcode7_21=.
replace iqcode7_21=.r if pc23_21==8 | pc24_21==8 | pc25_21==8
replace iqcode7_21=.d if pc23_21==9 | pc24_21==9 | pc25_21==9
replace iqcode7_21=3 if pc23_21==2
replace iqcode7_21=pc24_21 if inlist(pc24_21,1,2)
replace iqcode7_21=4 if pc25_21==2
replace iqcode7_21=5 if pc25_21==1
```

```
* Knowing how to work familiar machines around the house
gen iqcode8a_c_21=.
replace iqcode8a_c_21=.r if pc26_21==8 | pc27_21==8 | pc28_21==8
replace iqcode8a_c_21=.d if pc26_21==9 | pc27_21==9 | pc28_21==9
replace iqcode8a_c_21=3 if pc26_21==2
replace iqcode8a_c_21=pc27_21 if inlist(pc27_21,1,2)
replace iqcode8a_c_21=4 if pc28_21==2
replace iqcode8a_c_21=5 if pc28_21==1

* Learning to use a new gadget or machine around the house
gen iqcode9_21=.
replace iqcode9_21=.r if pc29_21==8 | pc30_21==8 | pc31_21==8
replace iqcode9_21=.d if pc29_21==9 | pc30_21==9 | pc31_21==9
replace iqcode9_21=3 if pc29_21==2
replace iqcode9_21=pc30_21 if inlist(pc30_21,1,2)
replace iqcode9_21=4 if pc31_21==2
replace iqcode9_21=5 if pc31_21==1

* Learning new things in general
gen iqcode10_c_21=.
replace iqcode10_c_21=.r if pc32_21==8 | pc33_21==8 | pc34_21==8
replace iqcode10_c_21=.d if pc32_21==9 | pc33_21==9 | pc34_21==9
replace iqcode10_c_21=3 if pc32_21==2
replace iqcode10_c_21=pc33_21 if inlist(pc33_21,1,2)
replace iqcode10_c_21=4 if pc34_21==2
replace iqcode10_c_21=5 if pc34_21==1

* Following a story in a book or TV
gen iqcode11_21=.
replace iqcode11_21=.r if pc35_21==8 | pc36_21==8 | pc37_21==8
replace iqcode11_21=.d if pc35_21==9 | pc36_21==9 | pc37_21==9
replace iqcode11_21=3 if pc35_21==2
replace iqcode11_21=pc36_21 if inlist(pc36_21,1,2)
```

```
replace iqcode11_21=4 if pc37_21==2
replace iqcode11_21=5 if pc37_21==1

* Making decision on everyname_day_21 matters
gen iqcode12_21=.
replace iqcode12_21=.r if pc38_21==8 | pc39_21==8 | pc40_21==8
replace iqcode12_21=.d if pc38_21==9 | pc39_21==9 | pc40_21==9
replace iqcode12_21=3 if pc38_21==2
replace iqcode12_21=pc39_21 if inlist(pc39_21,1,2)
replace iqcode12_21=4 if pc40_21==2
replace iqcode12_21=5 if pc40_21==1

* Handling money for shopping
gen iqcode13_21=.
replace iqcode13_21=.r if pc41_21==8 | pc42_21==8 | pc43_21==8
replace iqcode13_21=.d if pc41_21==9 | pc42_21==9 | pc43_21==9
replace iqcode13_21=3 if pc41_21==2
replace iqcode13_21=pc42_21 if inlist(pc42_21,1,2)
replace iqcode13_21=4 if pc43_21==2
replace iqcode13_21=5 if pc43_21==1

* Handling financial matters
gen iqcode14_21=.
replace iqcode14_21=.r if pc44_21==8 | pc45_21==8 | pc46_21==8
replace iqcode14_21=.d if pc44_21==9 | pc45_21==9 | pc46_21==9
replace iqcode14_21=3 if pc44_21==2
replace iqcode14_21=pc45_21 if inlist(pc45_21,1,2)
replace iqcode14_21=4 if pc46_21==2
replace iqcode14_21=5 if pc46_21==1

* Handling other everyname_day_21 arithmetic problems
gen iqcode15_21=.
replace iqcode15_21=.r if pc47_21==8 | pc48_21==8 | pc49_21==8
replace iqcode15_21=.d if pc47_21==9 | pc48_21==9 | pc49_21==9
```

```
replace iqcode15_21=3 if pc47_21==2
replace iqcode15_21=pc48_21 if inlist(pc48_21,1,2)
replace iqcode15_21=4 if pc49_21==2
replace iqcode15_21=5 if pc49_21==1

* Using his/her intelligence to understand what's going on and to reason things through
gen iqcode16_21=.
replace iqcode16_21=.r if pc50_21==8 | pc51_21==8 | pc52_21==8
replace iqcode16_21=.d if pc50_21==9 | pc51_21==9 | pc52_21==9
replace iqcode16_21=3 if pc50_21==2
replace iqcode16_21=pc51_21 if inlist(pc51_21,1,2)
replace iqcode16_21=4 if pc52_21==2
replace iqcode16_21=5 if pc52_21==1


***  TOTAL IQCODE SCORE (score 1-5) ***
egen iqcode= rowmean(iqcode1_21 iqcode2_21 iqcode3_21 iqcode4_21 iqcode5_21 iqcode6_21 iqcode7_21 iqcode8a_c_21 iqcode9_21 iqcode10_c_21 iqcode11_21 iqcode12_21 iqcode13_21 iqcode14_21 iqcode15_21 iqcode16_21) if inlist(tipent_21,3,4)

gen iqcode_21= round(iqcode,0.1)
replace iqcode_21 = .d if  [iqcode1_21==.d | ///
                                                iqcode2_21==.d | ///
                                                iqcode3_21==.d | ///
                                                iqcode4_21==.d | ///
                                                iqcode5_21==.d | ///
                                                iqcode6_21==.d | ///
                                                iqcode7_21==.d | ///
                                                iqcode8a_c_21==.d | ///
                                                iqcode9_21==.d | ///
                                                iqcode10_c_21==.d | ///
                                                iqcode11_21==.d | ///
                                                iqcode12_21==.d | ///
                                                iqcode13_21==.d | ///
```

```
                                                    iqcode14_21==.d | ///
                                                    iqcode15_21==.d | ///
                                                    iqcode16_21==.d]

replace iqcode_21 = .r if  [iqcode1_21==.r | ///
                                                    iqcode2_21==.r | ///
                                                    iqcode3_21==.r | ///
                                                    iqcode4_21==.r | ///
                                                    iqcode5_21==.r | ///
                                                    iqcode6_21==.r | ///
                                                    iqcode7_21==.r | ///
                                                    iqcode8a_c_21==.r | ///
                                                    iqcode9_21==.r | ///
                                                    iqcode10_c_21==.r | ///
                                                    iqcode11_21==.r | ///
                                                    iqcode12_21==.r | ///
                                                    iqcode13_21==.r | ///
                                                    iqcode14_21==.r | ///
                                                    iqcode15_21==.r | ///
                                                    iqcode16_21==.r]
replace iqcode_21 = .p if inlist(tipent_21,1,2)
label variable iqcode_21 "MHAS 2021 IQCODE Score (1-5)"


*** COGNITIVE IMPAIRMENT - IQCODE (only proxy interviews) ***
gen cog_imp_iqcode_21 = .
replace cog_imp_iqcode_21=0 if inrange(iqcode_21,1,3.4)
replace cog_imp_iqcode_21=1 if inrange(iqcode_21,3.4,5)
replace cog_imp_iqcode_21 = .d if iqcode_21==.d
replace cog_imp_iqcode_21 = .r if iqcode_21==.r
replace cog_imp_iqcode_21 = .n if inlist(tipent_21,1,2)
replace cog_imp_iqcode_21 = .i if inlist(tipent_21,3,4) & pc1_21==.i
label variable cog_imp_iqcode_21 "MHAS 2021 IQ CODE Cognitive Impairment - Normal, Impaired"
label define imp_iqcode 0 "Normal" 1 "Impaired", replace
```

```
label values cog_imp_iqcode_21 imp_iqcode
```

Appendix IV. STATA Codes for the Creation of the 2021 Z-Scores and the Classification of Subjects

The following STATA codes can be used to construct the z-scores for each task and total cognition score included in the 2021 wave. In addition, we provide the code for the cognitive function classification (categories Normal, CIND, Dementia).

```
*********************************************************
**** MHAS 2021 COGNITIVE EXERCISES - CCCE (SECTION E) ****
*********************************************************

label data "Version 1. April 2025"

*** CREATING Z-SCORES & IMPAIREMENT STATUS VARIABLES ***
***          BY TOTAL SCORE, TASK & DOMAIN           ***

*** CCCE Total Score (0-80) - Including only the items in 2001 *** ***
** Z-SCORES **
gen ccce_z_score_v01_21=.
replace        ccce_z_score_v01_21=-3.0     if [
                           [inrange(ccce_v01_21,0,10)                    &
inrange(yrschool,1,6)]       |       [inrange(ccce_v01_21,0,25)             &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace        ccce_z_score_v01_21=-2.0     if [[inrange(ccce_v01_21,0,12)  &
yrschool==0]   |   [inrange(ccce_v01_21,10,22)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,25,36)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace        ccce_z_score_v01_21=-1.5     if [[inrange(ccce_v01_21,12,16) &
yrschool==0]   |   [inrange(ccce_v01_21,22,26)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,36,40)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=-1.0    if    [[inrange(ccce_v01_21,16,28)    &
yrschool==0]   |   [inrange(ccce_v01_21,26,39)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,40,53)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=0.0          if [[inrange(ccce_v01_21,28,40) &
yrschool==0]   |   [inrange(ccce_v01_21,39,52)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,53,65)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=1.0          if [[inrange(ccce_v01_21,40,51) &
yrschool==0]   |   [inrange(ccce_v01_21,52,65)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,65,77)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=1.5          if [[inrange(ccce_v01_21,51,55) &
yrschool==0]   |   [inrange(ccce_v01_21,65,69)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,77,80)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=2.0          if [[inrange(ccce_v01_21,55,66) &
yrschool==0] | [inrange(ccce_v01_21,69,80) & inrange(yrschool,1,6)]
```

```
                                                                    ]        &
inrange(age_21,17,59)
replace    ccce_z_score_v01_21=3.0          if [[inrange(ccce_v01_21,66,80) &
yrschool==0]

            ] & inrange(age_21,17,59)

replace        ccce_z_score_v01_21=-3.0     if [
                           [inrange(ccce_v01_21,0,7)                        &
inrange(yrschool,1,6)]       |       [inrange(ccce_v01_21,0,21)              &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace        ccce_z_score_v01_21=-2.0     if [[inrange(ccce_v01_21,0,7)   &
yrschool==0]   |  [inrange(ccce_v01_21,7,18)     &  inrange(yrschool,1,6)]  |
[inrange(ccce_v01_21,21,32)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace        ccce_z_score_v01_21=-1.5     if [[inrange(ccce_v01_21,7,11)  &
yrschool==0]   |  [inrange(ccce_v01_21,18,23)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,32,36)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace    ccce_z_score_v01_21=-1.0    if    [[inrange(ccce_v01_21,11,23)    &
yrschool==0]   |  [inrange(ccce_v01_21,23,35)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,36,49)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace    ccce_z_score_v01_21=0.0          if [[inrange(ccce_v01_21,23,35) &
yrschool==0]   |  [inrange(ccce_v01_21,35,48)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,49,62)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace    ccce_z_score_v01_21=1.0          if [[inrange(ccce_v01_21,35,47) &
yrschool==0]   |  [inrange(ccce_v01_21,48,60)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,62,75)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace    ccce_z_score_v01_21=1.5          if [[inrange(ccce_v01_21,47,51) &
yrschool==0]   |  [inrange(ccce_v01_21,60,64)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,75,80)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,60,69)
replace    ccce_z_score_v01_21=2.0          if [[inrange(ccce_v01_21,51,63) &
yrschool==0]   |  [inrange(ccce_v01_21,64,75)   &  inrange(yrschool,1,6)]   |
[inlist(ccce_v01_21,80)    & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_v01_21=3.0          if [[inrange(ccce_v01_21,63,80) &
yrschool==0] | [inrange(ccce_v01_21,75,80) & inrange(yrschool,1,6)]
                                                                    ]        &
inrange(age_21,60,69)

//replace  ccce_z_score_v01_21=-3.0   if [


            ] & inrange(age_21,70,79)
replace        ccce_z_score_v01_21=-2.0     if [[inrange(ccce_v01_21,0,5)   &
yrschool==0]  |  [inrange(ccce_v01_21,0,7)      &  inrange(yrschool,1,6)]  |
[inrange(ccce_v01_21,0,10)            &      inrange(yrschool,7,25)]]      &
inrange(age_21,70,79)
```

```
replace        ccce_z_score_v01_21=-1.5     if [[inrange(ccce_v01_21,5,9)   &
yrschool==0]   |  [inrange(ccce_v01_21,7,10)     &  inrange(yrschool,1,6)]  |
[inrange(ccce_v01_21,10,15)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=-1.0    if   [[inrange(ccce_v01_21,9,19)      &
yrschool==0]   |  [inrange(ccce_v01_21,10,26)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,15,36)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=0.0          if [[inrange(ccce_v01_21,19,29) &
yrschool==0]   |  [inrange(ccce_v01_21,26,32)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,36,44)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=1.0          if [[inrange(ccce_v01_21,29,39) &
yrschool==0]   |  [inrange(ccce_v01_21,32,43)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,44,58)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=1.5          if [[inrange(ccce_v01_21,39,42) &
yrschool==0]   |  [inrange(ccce_v01_21,43,46)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,58,61)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=2.0          if [[inrange(ccce_v01_21,42,51) &
yrschool==0]   |  [inrange(ccce_v01_21,46,56)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,61,80)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,70,79)
replace    ccce_z_score_v01_21=3.0          if [[inrange(ccce_v01_21,51,80) &
yrschool==0] | [inrange(ccce_v01_21,56,80) & inrange(yrschool,1,6)]
                                                                  ]        &
inrange(age_21,70,79)

//replace  ccce_z_score_v01_21=-3.0   if [


                ] & inrange(age_21,80,120)
replace        ccce_z_score_v01_21=-2.0     if [
                           [inrange(ccce_v01_21,0,7)                        &
inrange(yrschool,1,6)]       |       [inrange(ccce_v01_21,0,10)             &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace        ccce_z_score_v01_21=-1.5     if [[inrange(ccce_v01_21,0,5)   &
yrschool==0]   |  [inrange(ccce_v01_21,7,10)     &  inrange(yrschool,1,6)]  |
[inrange(ccce_v01_21,10,15)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=-1.0    if   [[inrange(ccce_v01_21,5,14)      &
yrschool==0]   |  [inrange(ccce_v01_21,10,26)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,15,36)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=0.0          if [[inrange(ccce_v01_21,14,24) &
yrschool==0]   |  [inrange(ccce_v01_21,26,32)   &  inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,36,44)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=1.0          if [[inrange(ccce_v01_21,24,33) &
yrschool==0]   |  [inrange(ccce_v01_21,32,43)   &  inrange(yrschool,1,6)]   |
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[inrange(ccce_v01_21,44,58)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=1.5          if [[inrange(ccce_v01_21,33,37) &
yrschool==0]   |   [inrange(ccce_v01_21,43,46)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,58,62)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=2.0          if [[inrange(ccce_v01_21,37,45) &
yrschool==0]   |   [inrange(ccce_v01_21,46,56)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,62,75)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace    ccce_z_score_v01_21=3.0          if [[inrange(ccce_v01_21,45,80) &
yrschool==0]   |   [inrange(ccce_v01_21,56,80)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_v01_21,75,80)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)

replace    ccce_z_score_v01_21=    ccce_v01_21    if    mi(ccce_v01_21)    &
ccce_z_score_v01_21==.
replace    ccce_z_score_v01_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & ccce_z_score_v01_21==.
replace    ccce_z_score_v01_21= .p if inlist(tipent_21,3,4)
label variable ccce_z_score_v01_21 "MHAS 2021 CCCE Z-Scores - Comparable to
2001"

** Impairment Status  **
gen cognitive_imp_v01_21 = .
label variable cognitive_imp_v01_21 "MHAS 2021 Cognitive Impairment - Normal,
MCI, Impaired Comparable to 2001"
** Age <59 & Years of Education=0
replace  cognitive_imp_v01_21=2  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_v01_21,0,12)
replace  cognitive_imp_v01_21=1  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_v01_21,12,16)
replace  cognitive_imp_v01_21=0  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_v01_21,16,80)

** Age <59 & Years of Education 1-6
replace    cognitive_imp_v01_21=2          if    inrange(age_21,17,59)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,0,22)
replace    cognitive_imp_v01_21=1          if    inrange(age_21,17,59)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,22,26)
replace    cognitive_imp_v01_21=0          if    inrange(age_21,17,59)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,26,80)

** Age <59 & Years of Education 7+
replace    cognitive_imp_v01_21=2          if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,0,36)
replace    cognitive_imp_v01_21=1          if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,36,40)
replace    cognitive_imp_v01_21=0          if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,40,80)
```

```
** Age 60-69 & Years of Education=0
replace  cognitive_imp_v01_21=2  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_v01_21,0,7)
replace  cognitive_imp_v01_21=1  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_v01_21,7,11)
replace  cognitive_imp_v01_21=0  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_v01_21,11,80)

** Age 60-69 & Years of Education 1-6
replace    cognitive_imp_v01_21=2         if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,0,18)
replace    cognitive_imp_v01_21=1         if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,18,23)
replace    cognitive_imp_v01_21=0         if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inrange(ccce_v01_21,23,80)

** Age 60-69 & Years of Education 7+
replace    cognitive_imp_v01_21=2         if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,0,32)
replace    cognitive_imp_v01_21=1         if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,32,36)
replace    cognitive_imp_v01_21=0         if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inrange(ccce_v01_21,36,80)


** Age 70-79 & Years of Education=0
replace  cognitive_imp_v01_21=2  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_v01_21,0,5)
replace  cognitive_imp_v01_21=1  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_v01_21,5,9)
replace  cognitive_imp_v01_21=0  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_v01_21,9,80)

** Age 70-79 & Years of Education 1-6
replace     cognitive_imp_v01_21=2     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inrange(ccce_v01_21,0,13)
replace     cognitive_imp_v01_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inrange(ccce_v01_21,13,17)
replace     cognitive_imp_v01_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inrange(ccce_v01_21,17,80)

** Age 70-79 & Years of Education 7+
replace     cognitive_imp_v01_21=2     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inrange(ccce_v01_21,0,27)
replace     cognitive_imp_v01_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inrange(ccce_v01_21,27,31)
replace     cognitive_imp_v01_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inrange(ccce_v01_21,31,80)


** Age 80+ & Years of Education=0
```

```
//replace cognitive_imp_v01_21=2 if inrange(age_21,80,120) & yrschool==0 &
inrange(ccce_v01_21,,)
replace cognitive_imp_v01_21=1 if inrange(age_21,80,120) & yrschool==0 &
inrange(ccce_v01_21,0,5)
replace cognitive_imp_v01_21=0 if inrange(age_21,80,120) & yrschool==0 &
inrange(ccce_v01_21,5,80)

** Age 80+ & Years of Education 1-6
replace cognitive_imp_v01_21=2 if inrange(age_21,80,120) &
inrange(yrschool,1,6) & inrange(ccce_v01_21,0,7)
replace cognitive_imp_v01_21=1 if inrange(age_21,80,120) &
inrange(yrschool,1,6) & inrange(ccce_v01_21,7,10)
replace cognitive_imp_v01_21=0 if inrange(age_21,80,120) &
inrange(yrschool,1,6) & inrange(ccce_v01_21,10,80)

** Age 80+ & Years of Education 7+
replace cognitive_imp_v01_21=2 if inrange(age_21,80,120) &
inrange(yrschool,7,25) & inrange(ccce_v01_21,0,10)
replace cognitive_imp_v01_21=1 if inrange(age_21,80,120) &
inrange(yrschool,7,25) & inrange(ccce_v01_21,10,15)
replace cognitive_imp_v01_21=0 if inrange(age_21,80,120) &
inrange(yrschool,7,25) & inrange(ccce_v01_21,15,80)

replace cognitive_imp_v01_21= ccce_v01_21 if mi(ccce_v01_21) &
cognitive_imp_v01_21==.
replace cognitive_imp_v01_21= .m if [inlist(age_21,888,999) |
inlist(yrschool,.d,.r,.m)] & cognitive_imp_v01_21==.
replace cognitive_imp_v01_21= .p if inlist(tipent_21,3,4)
label define cognitive_imp_v01_21 0 "0.Normal" 1 "1.MCI" 2 "2.Impaired",
replace
label values cognitive_imp_v01_21 cognitive_imp_v01_21


*** CCCE Total Score (0-100) ***
** Z-SCORES **
gen ccce_z_score_21=.
replace ccce_z_score_21=-3.0 if [[inrange(ccce_21,0,13) &
yrschool==0] | [inrange(ccce_21,0,22) & inrange(yrschool,1,6)] |
[inrange(ccce_21,0,36) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace ccce_z_score_21=-2.0 if [[inrange(ccce_21,13,23) &
yrschool==0] | [inrange(ccce_21,22,34) & inrange(yrschool,1,6)] |
[inrange(ccce_21,36,49) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace ccce_z_score_21=-1.5 if [[inrange(ccce_21,23,28) &
yrschool==0] | [inrange(ccce_21,34,40) & inrange(yrschool,1,6)] |
[inrange(ccce_21,49,54) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace ccce_z_score_21=-1.0 if [[inrange(ccce_21,28,37) & yrschool==0]
| [inrange(ccce_21,40,50) & inrange(yrschool,1,6)] | [inrange(ccce_21,54,64)
& inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace ccce_z_score_21=0.0 if [[inrange(ccce_21,37,46) &
yrschool==0] | [inrange(ccce_21,50,58) & inrange(yrschool,1,6)] |
[inrange(ccce_21,64,72) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
```

```
replace    ccce_z_score_21=1.0        if    [[inrange(ccce_21,46,61)        &
yrschool==0]   |   [inrange(ccce_21,58,75)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,72,89)  & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace    ccce_z_score_21=1.5        if    [[inrange(ccce_21,61,66)        &
yrschool==0]   |   [inrange(ccce_21,75,81)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,89,94)  & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace    ccce_z_score_21=2.0        if    [[inrange(ccce_21,66,79)        &
yrschool==0]   |   [inrange(ccce_21,81,93)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,94,100) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace    ccce_z_score_21=3.0        if     [[inrange(ccce_21,79,100)      &
yrschool==0] | [inrange(ccce_21,93,100) & inrange(yrschool,1,6)]
                                                        ]                   &
inrange(age_21,17,59)

replace        ccce_z_score_21=-3.0   if   [[inrange(ccce_21,0,8)           &
yrschool==0]   |   [inrange(ccce_21,0,18)        &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,0,31)   & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace        ccce_z_score_21=-2.0   if   [[inrange(ccce_21,8,18)          &
yrschool==0]   |   [inrange(ccce_21,18,31)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,31,44)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace        ccce_z_score_21=-1.5   if    [[inrange(ccce_21,18,23)        &
yrschool==0]   |   [inrange(ccce_21,31,36)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,44,51)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=-1.0  if [[inrange(ccce_21,23,33)   & yrschool==0]
| [inrange(ccce_21,36,46)  & inrange(yrschool,1,6)] | [inrange(ccce_21,51,61)
& inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=0.0        if    [[inrange(ccce_21,33,49)        &
yrschool==0]   |   [inrange(ccce_21,46,62)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,61,76)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=1.0        if    [[inrange(ccce_21,49,56)        &
yrschool==0]   |   [inrange(ccce_21,62,72)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,76,87)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=1.5        if    [[inrange(ccce_21,56,64)        &
yrschool==0]   |   [inrange(ccce_21,72,78)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,87,93)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=2.0        if    [[inrange(ccce_21,64,74)        &
yrschool==0]   |   [inrange(ccce_21,78,91)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,93,100) & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace    ccce_z_score_21=3.0        if     [[inrange(ccce_21,74,100)      &
yrschool==0] | [inrange(ccce_21,91,100) & inrange(yrschool,1,6)]
                                                        ]                   &
inrange(age_21,60,69)

replace        ccce_z_score_21=-3.0   if [
                   [inrange(ccce_21,0,10)        &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,0,21)   & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace        ccce_z_score_21=-2.0   if   [[inrange(ccce_21,0,13)          &
yrschool==0]   |   [inrange(ccce_21,10,23)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,21,35)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace        ccce_z_score_21=-1.5   if    [[inrange(ccce_21,13,18)        &
yrschool==0]   |   [inrange(ccce_21,23,28)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,35,41)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
```

```
replace    ccce_z_score_21=-1.0  if [[inrange(ccce_21,18,27)  & yrschool==0]
| [inrange(ccce_21,28,38)  & inrange(yrschool,1,6)] | [inrange(ccce_21,41,52)
& inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace    ccce_z_score_21=0.0        if    [[inrange(ccce_21,27,40)       &
yrschool==0]   |   [inrange(ccce_21,38,53)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,52,69)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace    ccce_z_score_21=1.0        if    [[inrange(ccce_21,40,49)       &
yrschool==0]   |   [inrange(ccce_21,53,62)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,69,78)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace    ccce_z_score_21=1.5        if    [[inrange(ccce_21,49,54)       &
yrschool==0]   |   [inrange(ccce_21,62,68)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,78,86)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace    ccce_z_score_21=2.0        if    [[inrange(ccce_21,54,67)       &
yrschool==0]   |   [inrange(ccce_21,68,81)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,86,93)  & inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace    ccce_z_score_21=3.0        if     [[inrange(ccce_21,67,100)     &
yrschool==0]   |   [inrange(ccce_21,81,100)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,93,100) & inrange(yrschool,7,25)]] & inrange(age_21,70,79)

replace         ccce_z_score_21=-3.0  if [
                  [inrange(ccce_21,0,5)         &  inrange(yrschool,1,6)]   |
[inrange(ccce_21,0,13)   & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace        ccce_z_score_21=-2.0   if   [[inrange(ccce_21,0,7)          &
yrschool==0]   |   [inrange(ccce_21,5,13)        &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,13,24)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace        ccce_z_score_21=-1.5   if   [[inrange(ccce_21,7,12)          &
yrschool==0]   |   [inrange(ccce_21,13,18)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,24,29)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=-1.0  if [[inrange(ccce_21,12,21)  & yrschool==0]
| [inrange(ccce_21,18,28)  & inrange(yrschool,1,6)] | [inrange(ccce_21,29,40)
& inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=0.0        if    [[inrange(ccce_21,21,28)       &
yrschool==0]   |   [inrange(ccce_21,28,43)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,40,57)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=1.0        if    [[inrange(ccce_21,28,44)       &
yrschool==0]   |   [inrange(ccce_21,43,52)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,57,68)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=1.5        if    [[inrange(ccce_21,44,50)       &
yrschool==0]   |   [inrange(ccce_21,52,58)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,68,75)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=2.0        if    [[inrange(ccce_21,50,61)       &
yrschool==0]   |   [inrange(ccce_21,58,70)      &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,75,84)  & inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace    ccce_z_score_21=3.0        if     [[inrange(ccce_21,61,100)     &
yrschool==0]   |   [inrange(ccce_21,70,100)   &   inrange(yrschool,1,6)]   |
[inrange(ccce_21,84,100) & inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace    ccce_z_score_21= ccce_21 if mi(ccce_21) & ccce_z_score_21==.
replace    ccce_z_score_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & ccce_z_score_21==.
replace    ccce_z_score_21= .p if inlist(tipent_21,3,4)
label variable ccce_z_score_21 "MHAS 2021 CCCE Z-Scores"
```

```
** Impairment Status  **
gen cognitive_imp_21 = .
label variable cognitive_imp_21 "MHAS 2021 Cognitive Impairment - Normal, MCI,
Impaired"
** Age <59 & Years of Education=0
replace  cognitive_imp_21=2  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_21,0,23)
replace  cognitive_imp_21=1  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_21,23,27)
replace  cognitive_imp_21=0  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(ccce_21,27,100)

** Age <59 & Years of Education 1-6
replace cognitive_imp_21=2  if inrange(age_21,17,59) & inrange(yrschool,1,6)
& inrange(ccce_21,0,35)
replace cognitive_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,1,6)
& inrange(ccce_21,35,41)
replace cognitive_imp_21=0  if inrange(age_21,17,59) & inrange(yrschool,1,6)
& inrange(ccce_21,41,100)

** Age <59 & Years of Education 7+
replace cognitive_imp_21=2  if inrange(age_21,17,59) & inrange(yrschool,7,25)
& inrange(ccce_21,0,50)
replace cognitive_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,7,25)
& inrange(ccce_21,50,54)
replace cognitive_imp_21=0  if inrange(age_21,17,59) & inrange(yrschool,7,25)
& inrange(ccce_21,54,100)


** Age 60-69 & Years of Education=0
replace  cognitive_imp_21=2  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_21,0,17)
replace  cognitive_imp_21=1  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_21,17,21)
replace  cognitive_imp_21=0  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(ccce_21,21,100)

** Age 60-69 & Years of Education 1-6
replace cognitive_imp_21=2  if inrange(age_21,60,69) & inrange(yrschool,1,6)
& inrange(ccce_21,0,30)
replace cognitive_imp_21=1  if inrange(age_21,60,69) & inrange(yrschool,1,6)
& inrange(ccce_21,30,35)
replace cognitive_imp_21=0  if inrange(age_21,60,69) & inrange(yrschool,1,6)
& inrange(ccce_21,35,100)

** Age 60-69 & Years of Education 7+
replace cognitive_imp_21=2  if inrange(age_21,60,69) & inrange(yrschool,7,25)
& inrange(ccce_21,0,39)
replace cognitive_imp_21=1  if inrange(age_21,60,69) & inrange(yrschool,7,25)
& inrange(ccce_21,39,44)
```

```
replace cognitive_imp_21=0  if inrange(age_21,60,69) & inrange(yrschool,7,25)
& inrange(ccce_21,44,100)


** Age 70-79 & Years of Education=0
replace  cognitive_imp_21=2  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_21,0,15)
replace  cognitive_imp_21=1  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_21,15,18)
replace  cognitive_imp_21=0  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(ccce_21,18,100)

** Age 70-79 & Years of Education 1-6
replace cognitive_imp_21=2 if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(ccce_21,0,23)
replace cognitive_imp_21=1 if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(ccce_21,23,28)
replace cognitive_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(ccce_21,28,100)

** Age 70-79 & Years of Education 7+
replace cognitive_imp_21=2 if inrange(age_21,70,79) & inrange(yrschool,7,25)
& inrange(ccce_21,0,39)
replace cognitive_imp_21=1 if inrange(age_21,70,79) & inrange(yrschool,7,25)
& inrange(ccce_21,39,44)
replace cognitive_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,7,25)
& inrange(ccce_21,44,100)


** Age 80+ & Years of Education=0
replace  cognitive_imp_21=2  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(ccce_21,0,10)
replace  cognitive_imp_21=1  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(ccce_21,10,13)
replace  cognitive_imp_21=0  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(ccce_21,13,100)

** Age 80+ & Years of Education 1-6
replace cognitive_imp_21=2 if inrange(age_21,80,120) & inrange(yrschool,1,6)
& inrange(ccce_21,0,16)
replace cognitive_imp_21=1 if inrange(age_21,80,120) & inrange(yrschool,1,6)
& inrange(ccce_21,16,20)
replace cognitive_imp_21=0 if inrange(age_21,80,120) & inrange(yrschool,1,6)
& inrange(ccce_21,20,100)

** Age 80+ & Years of Education 7+
replace cognitive_imp_21=2 if inrange(age_21,80,120) & inrange(yrschool,7,25)
& inrange(ccce_21,0,26)
replace cognitive_imp_21=1 if inrange(age_21,80,120) & inrange(yrschool,7,25)
& inrange(ccce_21,26,42)
replace cognitive_imp_21=0 if inrange(age_21,80,120) & inrange(yrschool,7,25)
& inrange(ccce_21,42,100)
```

```
replace cognitive_imp_21= ccce_21 if mi(ccce_21) & cognitive_imp_21==.
replace     cognitive_imp_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & cognitive_imp_21==.
replace cognitive_imp_21= .p if inlist(tipent_21,3,4)
label define cognitive_imp_21 0 "0.Normal" 1 "1.MCI" 2 "2.Impaired", replace
label values cognitive_imp_21 cognitive_imp_21


*** INMEDIATE MEMORY Z-SCORES ***
** Z-SCORES **
gen iwr_z_21=.
replace iwr_z_21=-3.0 if     [[inrange(iwr_21,0,2)     &     yrschool==0]     |
[inrange(iwr_21,0,2)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,0,3)   &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21=-2.0 if     [[inrange(iwr_21,2,3)     &     yrschool==0]     |
[inrange(iwr_21,2,3)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,3,4)   &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21=-1.5 if [

[inrange(iwr_21,4,5) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21=-1.0 if     [[inrange(iwr_21,3,4)     &     yrschool==0]     |
[inrange(iwr_21,3,5) & inrange(yrschool,1,6)]
                                ] & inrange(age_21,17,59)
replace iwr_z_21= 0.0 if     [[inrange(iwr_21,4,6)     &     yrschool==0]     |
[inrange(iwr_21,5,6)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,5,7)   &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21= 1.0 if     [[inrange(iwr_21,6,7)     &     yrschool==0]     |
[inrange(iwr_21,6,7) & inrange(yrschool,1,6)]
                                ] & inrange(age_21,17,59)
replace iwr_z_21= 1.5 if [

[inrange(iwr_21,7,8) & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21= 2.0 if     [[inrange(iwr_21,7,8)     &     yrschool==0]     |
[inrange(iwr_21,7,8)   &   inrange(yrschool,1,6)]   |   [   inlist(iwr_21,8)      &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace iwr_z_21= 3.0 if   [[   inlist(iwr_21,8)        &   yrschool==0]   |   [
inlist(iwr_21,8)   & inrange(yrschool,1,6)]
                                ] & inrange(age_21,17,59)

replace iwr_z_21=-3.0 if     [[inrange(iwr_21,0,2)     &     yrschool==0]     |
[inrange(iwr_21,0,3)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,0,3)   &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21=-2.0 if [
[inrange(iwr_21,3,4)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,3,4)   &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21=-1.5 if [[inrange(iwr_21,2,3) & yrschool==0] |
                                                  [inrange(iwr_21,4,5)     &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21=-1.0 if     [[inrange(iwr_21,3,4)     &     yrschool==0]     |
[inrange(iwr_21,4,5) & inrange(yrschool,1,6)]
                                ] & inrange(age_21,60,69)
```

```
replace iwr_z_21= 0.0 if     [[inrange(iwr_21,4,5)     &     yrschool==0]     |
[inrange(iwr_21,5,6)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,5,7)   &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21= 1.0 if     [[inrange(iwr_21,5,6)     &     yrschool==0]     |
[inrange(iwr_21,6,7) & inrange(yrschool,1,6)]
                            ] & inrange(age_21,60,69)
replace iwr_z_21= 1.5 if [

[inrange(iwr_21,7,8) & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21= 2.0 if     [[inrange(iwr_21,6,7)     &     yrschool==0]     |
[inrange(iwr_21,7,8)   &   inrange(yrschool,1,6)]   |   [inlist(iwr_21,8)        &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace iwr_z_21= 3.0 if     [[inrange(iwr_21,7,8)     &     yrschool==0]     |
[inlist(iwr_21,8)    & inrange(yrschool,1,6)]
                             ] & inrange(age_21,60,69)

replace iwr_z_21=-3.0 if [
[inrange(iwr_21,0,2)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,0,3)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21=-2.0 if [[inrange(iwr_21,0,1) & yrschool==0] |
                                                [inrange(iwr_21,3,4)      &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21=-1.5 if     [[inrange(iwr_21,1,2)     &     yrschool==0]     |
[inrange(iwr_21,2,3) & inrange(yrschool,1,6)]
                               ] & inrange(age_21,70,79)
replace iwr_z_21=-1.0 if     [[inrange(iwr_21,2,3)     &     yrschool==0]     |
[inrange(iwr_21,3,4)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,4,5)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21= 0.0 if     [[inrange(iwr_21,3,5)     &     yrschool==0]     |
[inrange(iwr_21,4,5)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,5,6)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21= 1.0 if     [[inrange(iwr_21,5,6)     &     yrschool==0]     |
[inrange(iwr_21,5,6)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,6,7)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21= 1.5 if [
[inrange(iwr_21,6,7)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,7,8)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21= 2.0 if     [[inrange(iwr_21,6,7)     &     yrschool==0]     |
[inrange(iwr_21,7,8)   &   inrange(yrschool,1,6)]   |   [   inlist(iwr_21,8)      &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace iwr_z_21= 3.0 if    [[inrange(iwr_21,7,8)    &    yrschool==0]    |    [
inlist(iwr_21,8)    & inrange(yrschool,1,6)]
                             ] & inrange(age_21,70,79)

replace iwr_z_21=-3.0 if [
[inrange(iwr_21,0,2)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,0,2)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace iwr_z_21=-2.0 if [[inrange(iwr_21,0,1) & yrschool==0] |
                                                  [inrange(iwr_21,2,3)     &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
```

```
replace iwr_z_21=-1.5 if     [[inrange(iwr_21,1,2)     &     yrschool==0]     |
[inrange(iwr_21,2,3) & inrange(yrschool,1,6)]
                                ] & inrange(age_21,80,120)
replace iwr_z_21=-1.0 if     [[inrange(iwr_21,2,3)     &     yrschool==0]     |
[inrange(iwr_21,3,4)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,3,4)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace iwr_z_21= 0.0 if     [[inrange(iwr_21,3,5)     &     yrschool==0]     |
[inrange(iwr_21,4,5)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,4,6)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace iwr_z_21= 1.0 if     [[inrange(iwr_21,5,6)     &     yrschool==0]     |
[inrange(iwr_21,5,6)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,6,7)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace iwr_z_21= 1.5 if [
[inrange(iwr_21,6,7) & inrange(yrschool,1,6)]
                                  ] & inrange(age_21,80,120)
replace iwr_z_21= 2.0 if     [[inrange(iwr_21,6,7)     &     yrschool==0]     |
[inrange(iwr_21,7,8)   &   inrange(yrschool,1,6)]   |   [inrange(iwr_21,7,8)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace iwr_z_21= 3.0 if    [[inrange(iwr_21,7,8)    &    yrschool==0]    |    [
inlist(iwr_21,8)      &   inrange(yrschool,1,6)]   |   [   inlist(iwr_21,8)      &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace iwr_z_21= iwr_21 if mi(iwr_21) & iwr_z_21==.
replace iwr_z_21= .m if [inlist(age_21,888,999) | inlist(yrschool,.d,.r,.m)]
& iwr_z_21==.
replace iwr_z_21= .p if inlist(tipent_21,3,4)
label variable iwr_z_21 "MHAS 2021 Immediate Memory Z-Scores"

** Impairment Status **
gen iwr_imp_21 = .
label variable iwr_imp_21 "MHAS 2021 Immediate Memory Impairment"
* age_21 <59 & Years of Education=0
replace    iwr_imp_21=1    if    inrange(age_21,17,59)    &    yrschool==0    &
inrange(iwr_21,0,3)
replace    iwr_imp_21=0    if    inrange(age_21,17,59)    &    yrschool==0    &
inrange(iwr_21,3,8)
* age_21 <59 & Years of Education 1-6
replace  iwr_imp_21=1    if  inrange(age_21,17,59)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,0,3)
replace  iwr_imp_21=0    if  inrange(age_21,17,59)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,3,8)
* age_21 <59 & Years of Education 7+
replace  iwr_imp_21=1   if  inrange(age_21,17,59)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,0,5)
replace  iwr_imp_21=0   if  inrange(age_21,17,59)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,5,8)

* age_21 60-69 & Years of Education=0
replace    iwr_imp_21=1    if    inrange(age_21,60,69)    &    yrschool==0    &
inrange(iwr_21,0,3)
replace    iwr_imp_21=0    if    inrange(age_21,60,69)    &    yrschool==0    &
inrange(iwr_21,3,8)
```

```
* age_21 60-69 & Years of Education 1-6
replace  iwr_imp_21=1   if  inrange(age_21,60,69)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,0,4)
replace  iwr_imp_21=0   if  inrange(age_21,60,69)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,4,8)
* age_21 60-69 & Years of Education 7+
replace  iwr_imp_21=1   if  inrange(age_21,60,69)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,0,5)
replace  iwr_imp_21=0   if  inrange(age_21,60,69)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,5,8)

* age_21 70-79 & Years of Education=0
replace   iwr_imp_21=1   if   inrange(age_21,70,79)   &   yrschool==0   &
inrange(iwr_21,0,2)
replace   iwr_imp_21=0   if   inrange(age_21,70,79)   &   yrschool==0   &
inrange(iwr_21,2,8)
* age_21 70-79 & Years of Education 1-6
replace  iwr_imp_21=1  if  inrange(age_21,70,79)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,0,3)
replace  iwr_imp_21=0  if  inrange(age_21,70,79)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,3,8)
* age_21 70-79 & Years of Education 7+
replace  iwr_imp_21=1  if  inrange(age_21,70,79)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,0,4)
replace  iwr_imp_21=0  if  inrange(age_21,70,79)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,4,8)

* age_21 80+ & Years of Education=0
replace   iwr_imp_21=1   if   inrange(age_21,80,120)   &   yrschool==0   &
inrange(iwr_21,0,2)
replace   iwr_imp_21=0   if   inrange(age_21,80,120)   &   yrschool==0   &
inrange(iwr_21,2,8)
* age_21 80+ & Years of Education 1-6
replace  iwr_imp_21=1  if  inrange(age_21,80,120)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,0,3)
replace  iwr_imp_21=0  if  inrange(age_21,80,120)  &  inrange(yrschool,1,6)  &
inrange(iwr_21,3,8)
* age_21 80+ & Years of Education 7+
replace  iwr_imp_21=1  if  inrange(age_21,80,120)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,0,3)
replace  iwr_imp_21=0  if  inrange(age_21,80,120)  &  inrange(yrschool,7,25)  &
inrange(iwr_21,3,8)

replace iwr_imp_21= iwr_21 if mi(iwr_21) & iwr_imp_21==.
replace      iwr_imp_21=      .m      if      [inlist(age_21,888,999)      |
inlist(yrschool,.d,.r,.m)] & iwr_imp_21==.
replace iwr_imp_21= .p if inlist(tipent_21,3,4)
label define cognitive_21 0 "0.Normal" 1 "1.Impaired", replace
label values iwr_imp_21 cognitive_21


*** DELAYED MEMORY Z-SCORES ***
```

```
*** Repeat 8 words ***
** Z-SCORES **
gen dwr_z_21=.

replace dwr_z_21= 2.0 if [[inlist(dwr_21,8)       &       yrschool==0]       |
[inlist(dwr_21,8)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,8)      &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21= 1.5 if [[inlist(dwr_21,7)       &       yrschool==0]       |
[inlist(dwr_21,7)   & inrange(yrschool,1,6)]
                              ] & inrange(age_21,17,59)
replace dwr_z_21= 1.0 if     [[inlist(dwr_21,5,6)     &     yrschool==0]     |
[inlist(dwr_21,5,6)  &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,7)       &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21= 0.0 if     [[inlist(dwr_21,3,4)     &     yrschool==0]     |
[inlist(dwr_21,4)       &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,5,6)  &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21=-1.0 if    [[inlist(dwr_21,2)    &    yrschool==0]           |
[inlist(dwr_21,3)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,4)      &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21=-1.5 if    [[inlist(dwr_21,1)    &    yrschool==0]           |
[inlist(dwr_21,2)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,3)      &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21=-2.0 if    [[inlist(dwr_21,0)    &    yrschool==0]           |
[inlist(dwr_21,1)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,2)      &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace dwr_z_21=-3.0 if [
[inlist(dwr_21,0)       &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,0,1)  &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)

replace dwr_z_21= 2.0 if    [[inlist(dwr_21,8)          &    yrschool==0]    |
[inlist(dwr_21,7,8) & inrange(yrschool,1,6)] |   [inlist(dwr_21,8)     &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace dwr_z_21= 1.5 if [[inlist(dwr_21,7)   & yrschool==0]

                              ] & inrange(age_21,60,69)
replace dwr_z_21= 1.0 if     [[inlist(dwr_21,5,6)     &     yrschool==0]     |
[inlist(dwr_21,6)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,7)      &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace dwr_z_21= 0.0 if     [[inlist(dwr_21,3,4)     &     yrschool==0]     |
[inlist(dwr_21,4,5)   &   inrange(yrschool,1,6)]   |   [inlist(dwr_21,5,6)   &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace dwr_z_21=-1.0 if    [[inlist(dwr_21,2)          &    yrschool==0]    |
[inlist(dwr_21,3)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,4)      &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace dwr_z_21=-1.5 if    [[inlist(dwr_21,1)          &    yrschool==0]    |
[inlist(dwr_21,2)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,3)      &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace dwr_z_21=-2.0 if    [[inlist(dwr_21,0)          &    yrschool==0]    |
[inlist(dwr_21,0,1)  &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,2)       &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
```

```
replace dwr_z_21=-3.0 if [

     [inlist(dwr_21,0,1) & inrange(yrschool,7,25)]] & inrange(age_21,60,69)

replace dwr_z_21= 3.0 if [[inlist(dwr_21,8)   & yrschool==0]

                                    ] & inrange(age_21,70,79)
replace dwr_z_21= 2.0 if   [[inlist(dwr_21,7)          &   yrschool==0]   |
[inlist(dwr_21,8)   & inrange(yrschool,1,6)] |   [inlist(dwr_21,8)     &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace dwr_z_21= 1.5 if   [[inlist(dwr_21,6)          &   yrschool==0]   |
[inlist(dwr_21,7)   & inrange(yrschool,1,6)]
                           ] & inrange(age_21,70,79)
replace dwr_z_21= 1.0 if   [[inlist(dwr_21,5)          &   yrschool==0]   |
[inlist(dwr_21,5,6)   &   inrange(yrschool,1,6)]   |   [inlist(dwr_21,6,7)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace dwr_z_21= 0.0 if     [[inlist(dwr_21,3,4)     &     yrschool==0]     |
[inlist(dwr_21,3,4)   &   inrange(yrschool,1,6)]   |   [inlist(dwr_21,4,5)   &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace dwr_z_21=-1.0 if     [[inlist(dwr_21,1,2)     &     yrschool==0]     |
[inlist(dwr_21,2)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,3)      &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace dwr_z_21=-1.5 if   [[inlist(dwr_21,0)          &   yrschool==0]   |
[inlist(dwr_21,1)      &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,2)      &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace dwr_z_21=-2.0 if [
[inlist(dwr_21,0)       &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,0,1)  &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)

replace dwr_z_21= 3.0 if   [[inlist(dwr_21,8)          &   yrschool==0]   |
[inlist(dwr_21,8)        & inrange(yrschool,1,6)]
                                  ] & inrange(age_21,80,120)
replace dwr_z_21= 2.0 if     [[inlist(dwr_21,6,7)     &     yrschool==0]     |
[inlist(dwr_21,7)        &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,7,8)  &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace dwr_z_21= 1.5 if   [[inlist(dwr_21,5)          &   yrschool==0]   |
[inlist(dwr_21,6)        & inrange(yrschool,1,6)]
                                    ] & inrange(age_21,80,120)
replace dwr_z_21= 1.0 if   [[inlist(dwr_21,4)          &   yrschool==0]   |
[inlist(dwr_21,5)        &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,5,6)  &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace dwr_z_21= 0.0 if   [[inlist(dwr_21,3)          &   yrschool==0]   |
[inlist(dwr_21,2,3,4)   &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,3,4)  &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace dwr_z_21=-1.0 if     [[inlist(dwr_21,1,2)     &     yrschool==0]     |
[inlist(dwr_21,1)        &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,1,2)  &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace dwr_z_21=-1.5 if   [[inlist(dwr_21,0)          &   yrschool==0]   |
[inlist(dwr_21,0)        &  inrange(yrschool,1,6)]  |  [inlist(dwr_21,0)     &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace dwr_z_21= dwr_21 if mi(dwr_21) & dwr_z_21==.
```

```
replace dwr_z_21= .m if [inlist(age_21,888,999) | inlist(yrschool,.d,.r,.m)]
& dwr_z_21==.
replace dwr_z_21= .p if inlist(tipent_21,3,4)
label variable dwr_z_21 "MHAS 2021 Delayed Memory - Repeat 8 words Z-Scores"

** Impairment Status **
gen dwr_imp_21 = .
label variable dwr_imp_21 "MHAS 2021 Delayed Memory - Repeat 8 words
Impairment"
* age_21 <59 & Years of Education=0
replace   dwr_imp_21=1   if   inrange(age_21,17,59)   &   yrschool==0   &
inrange(dwr_21,0,1)
replace   dwr_imp_21=0   if   inrange(age_21,17,59)   &   yrschool==0   &
inrange(dwr_21,2,8)
* age_21 <59 & Years of Education 1-6
replace dwr_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,1,6) &
inrange(dwr_21,0,2)
replace dwr_imp_21=0  if inrange(age_21,17,659) & inrange(yrschool,1,6) &
inrange(dwr_21,3,8)
* age_21 <59 & Years of Education 7+
replace dwr_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,7,25) &
inrange(dwr_21,0,3)
replace dwr_imp_21=0  if inrange(age_21,17,59) & inrange(yrschool,7,25) &
inrange(dwr_21,4,8)

* age_21 60-69 & Years of Education=0
replace   dwr_imp_21=1   if   inrange(age_21,60,69)   &   yrschool==0   &
inrange(dwr_21,0,1)
replace   dwr_imp_21=0   if   inrange(age_21,60,69)   &   yrschool==0   &
inrange(dwr_21,2,8)
* age_21 60-69 & Years of Education 1-6
replace dwr_imp_21=1 if inrange(age_21,60,69) & inrange(yrschool,1,6) &
inrange(dwr_21,0,2)
replace dwr_imp_21=0 if inrange(age_21,60,69) & inrange(yrschool,1,6) &
inrange(dwr_21,3,8)
* age_21 60-69 & Years of Education 7+
replace dwr_imp_21=1 if inrange(age_21,60,69) & inrange(yrschool,7,25) &
inrange(dwr_21,0,3)
replace dwr_imp_21=0 if inrange(age_21,60,69) & inrange(yrschool,7,25) &
inrange(dwr_21,4,8)

* age_21 70-79 & Years of Education=0
replace dwr_imp_21=1 if inrange(age_21,70,79) & yrschool==0 & inlist(dwr_21,0)
replace   dwr_imp_21=0   if   inrange(age_21,70,79)   &   yrschool==0   &
inrange(dwr_21,1,8)
* age_21 70-79 & Years of Education 1-6
replace dwr_imp_21=1 if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(dwr_21,0,1)
replace dwr_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(dwr_21,2,8)
* age_21 70-79 & Years of Education 7+
```

```
replace dwr_imp_21=1 if inrange(age_21,70,79) & inrange(yrschool,7,25) &
inrange(dwr_21,0,2)
replace dwr_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,7,25) &
inrange(dwr_21,3,8)

* age_21 80+ & Years of Education=0
replace dwr_imp_21=1 if inrange(age_21,80,120) & yrschool==0 &
inlist(dwr_21,0)
replace dwr_imp_21=0 if inrange(age_21,80,120) & yrschool==0 &
inrange(dwr_21,1,8)
* age_21 80+ & Years of Education 1-6
replace dwr_imp_21=1 if inrange(age_21,80,120) & inrange(yrschool,1,6) &
inlist(dwr_21,0)
replace dwr_imp_21=0 if inrange(age_21,80,120) & inrange(yrschool,1,6) &
inrange(dwr_21,1,8)
* age_21 80+ & Years of Education 7+
replace dwr_imp_21=1 if inrange(age_21,80,120) & inrange(yrschool,7,25) &
inlist(dwr_21,0)
replace dwr_imp_21=0 if inrange(age_21,80,120) & inrange(yrschool,7,25) &
inrange(dwr_21,1,8)

replace dwr_imp_21= dwr_21 if mi(dwr_21) & dwr_imp_21==.
replace dwr_imp_21= .m if [inlist(age_21,888,999) |
inlist(yrschool,.d,.r,.m)] & dwr_imp_21==.
replace dwr_imp_21= .p if inlist(tipent_21,3,4)
label values dwr_imp_21 cognitive_21


*** ATTENTION Z-SCORES ***

*** Visual Scan ***
** Z-SCORES **
gen visual_scan_z_21=.
replace visual_scan_z_21= 3.0 if [[inrange(visual_scan_21,51,60) &
yrschool==0]

                ] & inrange(age_21,17,59)
replace visual_scan_z_21= 2.0 if [[inrange(visual_scan_21,42,50) &
yrschool==0] | [inrange(visual_scan_21,55,60) & inrange(yrschool,1,6)]
                                                                ] &
inrange(age_21,17,59)
replace visual_scan_z_21= 1.5 if [[inrange(visual_scan_21,38,41) &
yrschool==0] | [inrange(visual_scan_21,50,54) & inrange(yrschool,1,6)]
                                                                ] &
inrange(age_21,17,59)
replace visual_scan_z_21= 1.0 if [[inrange(visual_scan_21,28,37) &
yrschool==0] | [inrange(visual_scan_21,38,49) & inrange(yrschool,1,6)] |
[inrange(visual_scan_21,50,60) & inrange(yrschool,7,25)]] &
inrange(age_21,17,59)
replace visual_scan_z_21= 0.0 if [[inrange(visual_scan_21,17,27) &
yrschool==0] | [inrange(visual_scan_21,26,37) & inrange(yrschool,1,6)] |
```

```
[inrange(visual_scan_21,38,49)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)
replace visual_scan_z_21=-1.0    if    [[inrange(visual_scan_21,8,16)       &
yrschool==0]  |  [inrange(visual_scan_21,14,25)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,27,37)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)
replace visual_scan_z_21=-1.5    if   [[inrange(visual_scan_21,3,7)        &
yrschool==0]  |  [inrange(visual_scan_21,10,13)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,23,26)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)
replace visual_scan_z_21=-2.0    if   [[inrange(visual_scan_21,0,2)        &
yrschool==0]  |  [inrange(visual_scan_21,0,9)     &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,13,22)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)
replace visual_scan_z_21=-3.0    if [

                        [inrange(visual_scan_21,0,12)                       &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)

replace visual_scan_z_21= 3.0    if     [[inrange(visual_scan_21,47,60)     &
yrschool==0] | [inlist(visual_scan_21,60)     & inrange(yrschool,1,6)]
                                                                          ]  &
inrange(age_21,60,69)
replace visual_scan_z_21= 2.0    if     [[inrange(visual_scan_21,38,46)     &
yrschool==0]  |  [inrange(visual_scan_21,50,59)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,59,60)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace visual_scan_z_21= 1.5    if     [[inrange(visual_scan_21,34,37)     &
yrschool==0] | [inrange(visual_scan_21,46,49) & inrange(yrschool,1,6)]
                                                                          ]  &
inrange(age_21,60,69)
replace visual_scan_z_21= 1.0    if     [[inrange(visual_scan_21,23,33)     &
yrschool==0]  |  [inrange(visual_scan_21,34,45)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,47,58)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace visual_scan_z_21= 0.0    if     [[inrange(visual_scan_21,13,22)     &
yrschool==0]  |  [inrange(visual_scan_21,23,33)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,35,46)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace visual_scan_z_21=-1.0    if    [[inrange(visual_scan_21,2,12)        &
yrschool==0]  |  [inrange(visual_scan_21,11,22)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,23,34)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace visual_scan_z_21=-1.5    if   [[inrange(visual_scan_21,0,1)        &
yrschool==0]  |  [inrange(visual_scan_21,8,10)    &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,19,22)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace visual_scan_z_21=-2.0    if [
                     [inrange(visual_scan_21,0,7)   & inrange(yrschool,1,6)]
|    [inrange(visual_scan_21,8,18)        &    inrange(yrschool,7,25)]]    &
inrange(age_21,60,69)
```

```
replace visual_scan_z_21=-3.0    if [

                        [inrange(visual_scan_21,0,7)                                    &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)

replace visual_scan_z_21= 3.0    if     [[inrange(visual_scan_21,38,60)     &
yrschool==0] |  [inrange(visual_scan_21,53,60) & inrange(yrschool,1,6)]

] & inrange(age_21,70,79)
replace visual_scan_z_21= 2.0    if     [[inrange(visual_scan_21,30,38)     &
yrschool==0]  |  [inrange(visual_scan_21,44,52)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,59,60)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21= 1.5    if     [[inrange(visual_scan_21,27,31)     &
yrschool==0]  |  [inrange(visual_scan_21,40,43)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,55,58)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21= 1.0    if     [[inrange(visual_scan_21,19,27)     &
yrschool==0]  |  [inrange(visual_scan_21,29,39)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,43,54)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21= 0.0    if     [[inrange(visual_scan_21,10,21)     &
yrschool==0]  |  [inrange(visual_scan_21,18,28)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,31,42)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21=-1.0    if   [[inrange(visual_scan_21,2,9)          &
yrschool==0]  |  [inrange(visual_scan_21,7,17)    &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,19,30)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21=-1.5    if   [[inrange(visual_scan_21,0,3)          &
yrschool==0]  |  [inrange(visual_scan_21,3,6)     &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,15,18)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)
replace visual_scan_z_21=-2.0    if [
                     [inrange(visual_scan_21,0,2)   & inrange(yrschool,1,6)]
|    [inrange(visual_scan_21,5,14)        &    inrange(yrschool,7,25)]]    &
inrange(age_21,70,79)
replace visual_scan_z_21=-3.0    if [

                       [inrange(visual_scan_21,0,4)                                 &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)

replace visual_scan_z_21= 3.0    if     [[inrange(visual_scan_21,28,60)     &
yrschool==0] | [inrange(visual_scan_21,42,60) & inrange(yrschool,1,6)]
                                                                              ] &
inrange(age_21,80,120)
replace visual_scan_z_21= 2.0    if     [[inrange(visual_scan_21,26,27)     &
yrschool==0]  |  [inrange(visual_scan_21,34,41)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,49,60)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace visual_scan_z_21= 1.5    if     [[inrange(visual_scan_21,24,25)     &
yrschool==0]  |  [inrange(visual_scan_21,31,33)  &  inrange(yrschool,1,6)]  |
```

```
[inrange(visual_scan_21,44,48)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace visual_scan_z_21= 1.0    if     [[inrange(visual_scan_21,15,23)     &
yrschool==0]  |  [inrange(visual_scan_21,21,30)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,31,43)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace visual_scan_z_21= 0.0    if    [[inrange(visual_scan_21,7,14)       &
yrschool==0]  |  [inrange(visual_scan_21,12,20)  &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,19,30)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace visual_scan_z_21=-1.0    if   [[inrange(visual_scan_21,0,6)          &
yrschool==0]  |  [inrange(visual_scan_21,3,11)    &  inrange(yrschool,1,6)]  |
[inrange(visual_scan_21,6,18)           &     inrange(yrschool,7,25)]]     &
inrange(age_21,80,120)
replace visual_scan_z_21=-1.5    if [
                     [inrange(visual_scan_21,0,2)   & inrange(yrschool,1,6)]
|    [inrange(visual_scan_21,2,5)           &    inrange(yrschool,7,25)]]    &
inrange(age_21,80,120)
replace visual_scan_z_21=-2.0    if [

                        [inrange(visual_scan_21,0,1)                         &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace   visual_scan_z_21=   visual_scan_21   if   mi(visual_scan_21)    &
visual_scan_z_21==.
replace     visual_scan_z_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & visual_scan_z_21==.
replace visual_scan_z_21= .p if inlist(tipent_21,3,4)
label variable visual_scan_z_21 "MHAS 2021 Attention - Visual Scan Z-Scores"

** Impairment Status **
gen visual_scan_imp_21 = .
label  variable  visual_scan_imp_21  "MHAS  2021  Attention  -  Visual  Scan
Impairment"

* age_21 <59 & Years of Education=0
replace  visual_scan_imp_21=1  if  inrange(age_21,17,69)  &  yrschool==0  &
inrange(visual_scan_21,0,7)
replace  visual_scan_imp_21=0  if  inrange(age_21,17,69)  &  yrschool==0  &
inrange(visual_scan_21,8,60)
* age_21 <59 & Years of Education 1-6
replace     visual_scan_imp_21=1          if     inrange(age_21,17,69)     &
inrange(yrschool,1,6) & inrange(visual_scan_21,0,13)
replace     visual_scan_imp_21=0          if     inrange(age_21,17,69)     &
inrange(yrschool,1,6) & inrange(visual_scan_21,14,60)
* age_21 <59 & Years of Education 7+
replace     visual_scan_imp_21=1          if     inrange(age_21,17,69)     &
inrange(yrschool,7,25) & inrange(visual_scan_21,0,26)
replace     visual_scan_imp_21=0          if     inrange(age_21,17,69)     &
inrange(yrschool,7,25) & inrange(visual_scan_21,27,60)

* age_21 60-69 & Years of Education=0
```

```
replace visual_scan_imp_21=1 if inrange(age_21,17,69) & yrschool==0 &
inrange(visual_scan_21,0,1)
replace visual_scan_imp_21=0 if inrange(age_21,17,69) & yrschool==0 &
inrange(visual_scan_21,2,60)
* age_21 60-69 & Years of Education 1-6
replace visual_scan_imp_21=1 if inrange(age_21,17,69) &
inrange(yrschool,1,6) & inrange(visual_scan_21,0,10)
replace visual_scan_imp_21=0 if inrange(age_21,17,69) &
inrange(yrschool,1,6) & inrange(visual_scan_21,11,60)
* age_21 60-69 & Years of Education 7+
replace visual_scan_imp_21=1 if inrange(age_21,17,69) &
inrange(yrschool,7,25) & inrange(visual_scan_21,0,22)
replace visual_scan_imp_21=0 if inrange(age_21,17,69) &
inrange(yrschool,7,25) & inrange(visual_scan_21,23,60)

* age_21 70-79 & Years of Education=0
replace visual_scan_imp_21=1 if inrange(age_21,70,79) & yrschool==0 &
inrange(visual_scan_21,0,1)
replace visual_scan_imp_21=0 if inrange(age_21,70,79) & yrschool==0 &
inrange(visual_scan_21,2,60)
* age_21 70-79 & Years of Education 1-6
replace visual_scan_imp_21=1 if inrange(age_21,70,79) & inrange(yrschool,1,6)
& inrange(visual_scan_21,0,6)
replace visual_scan_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,1,6)
& inrange(visual_scan_21,7,60)
* age_21 70-79 & Years of Education 7+
replace visual_scan_imp_21=1 if inrange(age_21,70,79) &
inrange(yrschool,7,25) & inrange(visual_scan_21,0,18)
replace visual_scan_imp_21=0 if inrange(age_21,70,79) &
inrange(yrschool,7,25) & inrange(visual_scan_21,19,60)

* age_21 80+ & Years of Education=0
//replace visual_scan_imp_21=1 if inrange(age_21,80,120) & yrschool==0 &
inrange(visual_scan_21,)
replace visual_scan_imp_21=0 if inrange(age_21,80,120) & yrschool==0 &
inrange(visual_scan_21,0,60)
* age_21 80+ & Years of Education 1-6
replace visual_scan_imp_21=1 if inrange(age_21,80,120) &
inrange(yrschool,1,6) & inrange(visual_scan_21,0,2)
replace visual_scan_imp_21=0 if inrange(age_21,80,120) &
inrange(yrschool,1,6) & inrange(visual_scan_21,3,60)
* age_21 80+ & Years of Education 7+
replace visual_scan_imp_21=1 if inrange(age_21,80,120) &
inrange(yrschool,7,25) & inrange(visual_scan_21,0,5)
replace visual_scan_imp_21=0 if inrange(age_21,80,120) &
inrange(yrschool,7,25) & inrange(visual_scan_21,6,60)

replace visual_scan_imp_21= visual_scan_21 if mi(visual_scan_21) &
visual_scan_imp_21==.
replace visual_scan_imp_21= .m if [inlist(age_21,888,999) |
inlist(yrschool,.d,.r,.m)] & visual_scan_imp_21==.
replace visual_scan_imp_21= .p if inlist(tipent_21,3,4)
```

```
label values visual_scan_imp_21 cognitive_21


*** ORIENTATION Z-SCORES ***
gen orientation_z_21=.
replace    orientation_z_21= 3.0 if [
             [inlist(orientation_21,2,3)    &    inrange(yrschool,1,6)]    |
[inlist(orientation_21,1,2,3)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)
replace    orientation_z_21= 2.0 if [[inlist(orientation_21,3) & yrschool==0]

                                                                          ]
& inrange(age_21,17,59)
replace    orientation_z_21= 1.5 if [
             [inlist(orientation_21,1)   & inrange(yrschool,1,6)]
                                                                  ]         &
inrange(age_21,17,59)
replace    orientation_z_21= 1.0 if [[inlist(orientation_21,2) & yrschool==0]


] & inrange(age_21,17,59)
replace    orientation_z_21= 0.0 if [[inlist(orientation_21,1) & yrschool==0]
| [inlist(orientation_21,0)   & inrange(yrschool,1,6)]
                                                        ]                   &
inrange(age_21,17,59)
replace    orientation_z_21=-1.0 if [[inlist(orientation_21,0) & yrschool==0]


] & inrange(age_21,17,59)
replace    orientation_z_21=-2.0 if [

              [inlist(orientation_21,0)         &  inrange(yrschool,7,25)]]  &
inrange(age_21,17,59)

replace    orientation_z_21= 3.0 if [[inlist(orientation_21,3) & yrschool==0]
|      [inlist(orientation_21,2,3)      &      inrange(yrschool,1,6)]      |
[inlist(orientation_21,1,2,3)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)
replace    orientation_z_21= 1.5 if [[inlist(orientation_21,2) & yrschool==0]

                                                                        ]   &
inrange(age_21,60,69)
replace    orientation_z_21= 1.0 if [
             [inlist(orientation_21,1)   & inrange(yrschool,1,6)]
                                                                    ]   &
inrange(age_21,60,69)
replace    orientation_z_21= 0.0 if [[inlist(orientation_21,1) & yrschool==0]
|     [inlist(orientation_21,0)             &     inrange(yrschool,1,6)]     |
[inlist(orientation_21,0)                  &    inrange(yrschool,7,25)]]    &
inrange(age_21,60,69)
replace    orientation_z_21=-1.0 if [[inlist(orientation_21,0) & yrschool==0]
```

```
] & inrange(age_21,60,69)

replace    orientation_z_21= 3.0 if [
              [inlist(orientation_21,2,3)    &    inrange(yrschool,1,6)]    |
[inlist(orientation_21,2,3)              &     inrange(yrschool,7,25)]]    &
inrange(age_21,70,79)
replace    orientation_z_21= 2.0 if [[inlist(orientation_21,3) & yrschool==0]
|
[inlist(orientation_21,1)                   &    inrange(yrschool,7,25)]]    &
inrange(age_21,70,79)
replace    orientation_z_21= 1.0 if [[inlist(orientation_21,2) & yrschool==0]
| [inlist(orientation_21,1)   & inrange(yrschool,1,6)]
                                                          ]                   &
inrange(age_21,70,79)
replace    orientation_z_21= 0.0 if [[inlist(orientation_21,1) & yrschool==0]
|     [inlist(orientation_21,0)              &     inrange(yrschool,1,6)]     |
[inlist(orientation_21,0)                   &    inrange(yrschool,7,25)]]    &
inrange(age_21,70,79)
replace    orientation_z_21=-1.0 if [[inlist(orientation_21,0) & yrschool==0]


] & inrange(age_21,70,79)

replace    orientation_z_21= 3.0 if [
              [inlist(orientation_21,3)        &   inrange(yrschool,1,6)]   |
[inlist(orientation_21,3)        &         inrange(yrschool,7,25)]]         &
inrange(age_21,80,120)
replace    orientation_z_21= 2.0 if [

            [inlist(orientation_21,2)  &      inrange(yrschool,7,25)]]      &
inrange(age_21,80,120)
replace    orientation_z_21= 1.5 if [[inlist(orientation_21,3) & yrschool==0]
| [inlist(orientation_21,2)   & inrange(yrschool,1,6)]
                                                          ]                   &
inrange(age_21,80,120)
replace    orientation_z_21= 1.0 if [[inlist(orientation_21,2) & yrschool==0]
|
[inlist(orientation_21,1)   &           inrange(yrschool,7,25)]]           &
inrange(age_21,80,120)
replace    orientation_z_21= 0.0 if [[inlist(orientation_21,1) & yrschool==0]
|
[inlist(orientation_21,0)   &           inrange(yrschool,7,25)]]           &
inrange(age_21,80,120)
replace    orientation_z_21=-1.0 if [[inlist(orientation_21,0) & yrschool==0]
| [inlist(orientation_21,0,1) & inrange(yrschool,1,6)]
                                                        ]                     &
inrange(age_21,80,120)

replace   orientation_z_21=   orientation_21   if   mi(orientation_21)   &
orientation_z_21==.
```

```
replace     orientation_z_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & orientation_z_21==.
replace orientation_z_21= .p if inlist(tipent_21,3,4)
label variable orientation_z_21 "MHAS 2021 Attention - Orientation Z-Scores"

** Impairment Status **
gen orientation_imp_21 = .
label  variable  orientation_imp_21  "MHAS  2021  Attention  -  Orientation
Impairment"

* age_21 <59 & Years of Education=0
//replace  orientation_imp_21=1  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(orientation_21,)
replace  orientation_imp_21=0  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(orientation_21,0,3)
* age_21 <59 & Years of Education 1-6
//replace     orientation_imp_21=1          if     inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(orientation_21,)
replace     orientation_imp_21=0          if     inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(orientation_21,0,3)
* age_21 <59 & Years of Education 7+
replace     orientation_imp_21=1          if     inrange(age_21,17,59)     &
inrange(yrschool,7,25) & inlist(orientation_21,0)
replace     orientation_imp_21=0          if     inrange(age_21,17,59)     &
inrange(yrschool,7,25) & inrange(orientation_21,1,3)

* age_21 60-69 & Years of Education=0
//replace  orientation_imp_21=1  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(orientation_21,)
replace  orientation_imp_21=0  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(orientation_21,0,3)
* age_21 60-79 & Years of Education 1-6
//replace      orientation_imp_21=1      if      inrange(age_21,60,69)      &
inrange(yrschool,1,6) & inrange(orientation_21,)
replace orientation_imp_21=0 if inrange(age_21,60,69) & inrange(yrschool,1,6)
& inrange(orientation_21,0,3)
* age_21 60-69 & Years of Education 7+
//replace      orientation_imp_21=1      if      inrange(age_21,60,69)      &
inrange(yrschool,7,25) & inrange(orientation_21,)
replace      orientation_imp_21=0      if      inrange(age_21,60,69)      &
inrange(yrschool,7,25) & inrange(orientation_21,0,3)

* age_21 70-79 & Years of Education=0
//replace  orientation_imp_21=1  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(orientation_21,)
replace  orientation_imp_21=0  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(orientation_21,0,3)
* age_21 70-79 & Years of Education 1-6
//replace      orientation_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,1,6) & inrange(orientation_21,)
replace orientation_imp_21=0 if inrange(age_21,70,79) & inrange(yrschool,1,6)
& inrange(orientation_21,0,3)
```

```
* age_21 70-79 & Years of Education 7+
//replace      orientation_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(orientation_21,)
replace      orientation_imp_21=0      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(orientation_21,0,3)

* age_21 80+ & Years of Education=0
//replace  orientation_imp_21=1  if  inrange(age_21,80,120)  &  yrschool==0  &
inlist(orientation_21,)
replace  orientation_imp_21=0  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(orientation_21,0,3)
* age_21 80+ & Years of Education 1-6
//replace      orientation_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(orientation_21,)
replace      orientation_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(orientation_21,0,3)
* age_21 80+ & Years of Education 7+
//replace      orientation_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(orientation_21,)
replace      orientation_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(orientation_21,0,3)

replace   orientation_imp_21=   orientation_21   if   mi(orientation_21)   &
orientation_imp_21==.
replace    orientation_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & orientation_imp_21==.
replace orientation_imp_21= .p if inlist(tipent_21,3,4)
label values orientation_imp_21 cognitive_21


*** EXECUTIVE FUNCTION Z-SCORES ***
** Verbal fluency z-scores **
gen verbal_fluency_z_21=.
replace verbal_fluency_z_21= 3.0 if [[inrange(verbal_fluency_num_21,24,40) &
yrschool==0] | [inrange(verbal_fluency_num_21,27,40) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,27,40)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21= 2.0 if [[inrange(verbal_fluency_num_21,20,23) &
yrschool==0] | [inrange(verbal_fluency_num_21,23,26) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,27,40)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21= 1.5 if [[ inlist(verbal_fluency_num_21,19)    &
yrschool==0] | [ inlist(verbal_fluency_num_21,22)  &  inrange(yrschool,1,6)]
|  [  inlist(verbal_fluency_num_21,26)        &  inrange(yrschool,7,25)]]  &
inrange(age_21,17,59)
replace verbal_fluency_z_21= 1.0 if [[inrange(verbal_fluency_num_21,15,18) &
yrschool==0] | [inrange(verbal_fluency_num_21,18,21) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,22,25)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21= 0.0 if [[inrange(verbal_fluency_num_21,12,14) &
yrschool==0] | [inrange(verbal_fluency_num_21,14,17) & inrange(yrschool,1,6)]
```

```
|   [inrange(verbal_fluency_num_21,17,21)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21=-1.0 if [[inrange(verbal_fluency_num_21,8,11)   &
yrschool==0] | [inrange(verbal_fluency_num_21,10,13) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,13,16)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21=-1.5 if [[ inlist(verbal_fluency_num_21,7)
& yrschool==0] | [ inlist(verbal_fluency_num_21,9)       &
inrange(yrschool,1,6)]     |     [inrange(verbal_fluency_num_21,11,12)     &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace verbal_fluency_z_21=-2.0 if [[inrange(verbal_fluency_num_21,4,6)   &
yrschool==0] | [inrange(verbal_fluency_num_21,5,8)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,7,10)     &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)
replace verbal_fluency_z_21=-3.0 if [[inrange(verbal_fluency_num_21,0,3)   &
yrschool==0] | [inrange(verbal_fluency_num_21,0,4)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,0,6)       &   inrange(yrschool,7,25)]]   &
inrange(age_21,17,59)

replace verbal_fluency_z_21= 3.0 if [[inrange(verbal_fluency_num_21,24,40) &
yrschool==0] | [inrange(verbal_fluency_num_21,27,40) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,31,40)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21= 2.0 if [[inrange(verbal_fluency_num_21,20,23) &
yrschool==0] | [inrange(verbal_fluency_num_21,24,26) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,27,30)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21= 1.5 if [[ inlist(verbal_fluency_num_21,19)    &
yrschool==0] | [inrange(verbal_fluency_num_21,22,23) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,25,26)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21= 1.0 if [[inrange(verbal_fluency_num_21,15,18) &
yrschool==0] | [inrange(verbal_fluency_num_21,18,21) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,21,24)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21= 0.0 if [[inrange(verbal_fluency_num_21,11,14) &
yrschool==0] | [inrange(verbal_fluency_num_21,14,17) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,16,20)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21=-1.0 if [[inrange(verbal_fluency_num_21,8,10)   &
yrschool==0] | [inrange(verbal_fluency_num_21,10,13) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,12,15)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
replace verbal_fluency_z_21=-1.5 if [[ inlist(verbal_fluency_num_21,7)     &
yrschool==0] | [inrange(verbal_fluency_num_21,8,9)   & inrange(yrschool,1,6)]
|  [  inlist(verbal_fluency_num_21,11)        &  inrange(yrschool,7,25)]]  &
inrange(age_21,60,69)
replace verbal_fluency_z_21=-2.0 if [[inrange(verbal_fluency_num_21,4,6)   &
yrschool==0] | [inrange(verbal_fluency_num_21,5,7)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,7,10)     &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)
```

```
replace verbal_fluency_z_21=-3.0 if [[inrange(verbal_fluency_num_21,0,3)    &
yrschool==0] | [inrange(verbal_fluency_num_21,0,4)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,0,6)       &   inrange(yrschool,7,25)]]   &
inrange(age_21,60,69)

replace verbal_fluency_z_21= 3.0 if [[inrange(verbal_fluency_num_21,23,50) &
yrschool==0] | [inrange(verbal_fluency_num_21,25,40) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,30,40)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21= 2.0 if [[inrange(verbal_fluency_num_21,20,22) &
yrschool==0] | [inrange(verbal_fluency_num_21,22,24) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,26,29)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21= 1.5 if [[ inlist(verbal_fluency_num_21,19)    &
yrschool==0] | [ inlist(verbal_fluency_num_21,21)       &
inrange(yrschool,1,6)]   |   [   inlist(verbal_fluency_num_21,25)           &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace verbal_fluency_z_21= 1.0 if [[inrange(verbal_fluency_num_21,15,18) &
yrschool==0] | [inrange(verbal_fluency_num_21,16,20) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,20,24)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21= 0.0 if [[inrange(verbal_fluency_num_21,11,14) &
yrschool==0] | [inrange(verbal_fluency_num_21,12,15) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,16,19)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21=-1.0 if [[inrange(verbal_fluency_num_21,7,10)  &
yrschool==0] | [inrange(verbal_fluency_num_21,8,11)  & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,11,15)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21=-1.5 if [[ inlist(verbal_fluency_num_21,6)
& yrschool==0] | [ inlist(verbal_fluency_num_21,7)       &
inrange(yrschool,1,6)]   |   [   inlist(verbal_fluency_num_21,10)           &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace verbal_fluency_z_21=-2.0 if [[inrange(verbal_fluency_num_21,3,5)   &
yrschool==0] | [inrange(verbal_fluency_num_21,4,6)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,6,9)       &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace verbal_fluency_z_21=-3.0 if [[inrange(verbal_fluency_num_21,0,2)   &
yrschool==0] | [inrange(verbal_fluency_num_21,0,3)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,0,5)       &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)

replace verbal_fluency_z_21= 3.0 if [[inrange(verbal_fluency_num_21,21,40) &
yrschool==0] | [inrange(verbal_fluency_num_21,21,40) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,27,40)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,80,120)
replace verbal_fluency_z_21= 2.0 if [[inrange(verbal_fluency_num_21,18,20) &
yrschool==0] | [inrange(verbal_fluency_num_21,18,20) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,23,26)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,80,120)
replace verbal_fluency_z_21= 1.5 if [[ inlist(verbal_fluency_num_21,17)    &
yrschool==0] | [ inlist(verbal_fluency_num_21,17)       &
```

```
inrange(yrschool,1,6)]     |     [inrange(verbal_fluency_num_21,21,22)     &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace verbal_fluency_z_21= 1.0 if [[inrange(verbal_fluency_num_21,13,16) &
yrschool==0] | [inrange(verbal_fluency_num_21,13,16) & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,18,20)   &   inrange(yrschool,7,25)]]   &
inrange(age_21,80,120)
replace verbal_fluency_z_21= 0.0 if [[inrange(verbal_fluency_num_21,9,12)  &
yrschool==0] | [inrange(verbal_fluency_num_21,9,12)  & inrange(yrschool,1,6)]
|  [  inlist(verbal_fluency_num_21,17)        &  inrange(yrschool,7,25)]]  &
inrange(age_21,80,120)
replace verbal_fluency_z_21=-1.0 if [[inrange(verbal_fluency_num_21,5,8)   &
yrschool==0] | [inrange(verbal_fluency_num_21,5,8)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,8,16)     &   inrange(yrschool,7,25)]]   &
inrange(age_21,80,120)
replace verbal_fluency_z_21=-1.5 if [[ inlist(verbal_fluency_num_21,4)
& yrschool==0] | [ inlist(verbal_fluency_num_21,4)       &
inrange(yrschool,1,6)] | [ inlist(verbal_fluency_num_21,7)     &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace verbal_fluency_z_21=-2.0 if [[inrange(verbal_fluency_num_21,1,3)   &
yrschool==0] | [inrange(verbal_fluency_num_21,1,3)   & inrange(yrschool,1,6)]
|   [inrange(verbal_fluency_num_21,3,6)       &   inrange(yrschool,7,25)]]   &
inrange(age_21,80,120)
replace verbal_fluency_z_21=-3.0 if [[ inlist(verbal_fluency_num_21,0)
& yrschool==0] | [ inlist(verbal_fluency_num_21,0)       &
inrange(yrschool,1,6)]    |    [inrange(verbal_fluency_num_21,0,2)          &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace         verbal_fluency_z_21=         verbal_fluency_num_21         if
mi(verbal_fluency_num_21) & verbal_fluency_z_21==.
replace    verbal_fluency_z_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & verbal_fluency_z_21==.
replace verbal_fluency_z_21= .p if inlist(tipent_21,3,4)
label variable verbal_fluency_z_21 "MHAS 2021 Executive function - Verbal
Fluency Z-Scores"

** Impairment Status **
gen verbal_fluency_imp_21 = .
label variable verbal_fluency_imp_21 "MHAS 2021 Executive function - Verbal
Fluency Impairment"

* age_21 <59 & Years of Education=0
replace  verbal_fluency_imp_21=1  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(verbal_fluency_num_21,0,7)
replace  verbal_fluency_imp_21=0  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(verbal_fluency_num_21,8,40)
* age_21 <59 & Years of Education 1-6
replace    verbal_fluency_imp_21=1         if    inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,0,9)
replace    verbal_fluency_imp_21=0         if    inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,10,40)
* age_21 <59 & Years of Education 7+
```

```
replace    verbal_fluency_imp_21=1         if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,0,12)
replace    verbal_fluency_imp_21=0         if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,13,50)

* age_21 60-69 & Years of Education=0
replace  verbal_fluency_imp_21=1  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(verbal_fluency_num_21,0,7)
replace  verbal_fluency_imp_21=0  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(verbal_fluency_num_21,8,40)
* age_21 60-69 & Years of Education 1-6
replace    verbal_fluency_imp_21=1         if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,0,9)
replace    verbal_fluency_imp_21=0         if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,10,40)
* age_21 60-69 & Years of Education 7+
replace    verbal_fluency_imp_21=1         if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,0,11)
replace    verbal_fluency_imp_21=0         if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,12,50)

* age_21 70-79 & Years of Education=0
replace  verbal_fluency_imp_21=1  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(verbal_fluency_num_21,0,6)
replace  verbal_fluency_imp_21=0  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(verbal_fluency_num_21,7,50)
* age_21 70-79 & Years of Education 1-6
replace     verbal_fluency_imp_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,0,7)
replace     verbal_fluency_imp_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,8,50)
* age_21 70-79 & Years of Education 7+
replace     verbal_fluency_imp_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,0,10)
replace     verbal_fluency_imp_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,11,50)

* age_21 80+ & Years of Education=0
replace  verbal_fluency_imp_21=1  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(verbal_fluency_num_21,0,4)
replace  verbal_fluency_imp_21=0  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(verbal_fluency_num_21,5,50)
* age_21 80+ & Years of Education 1-6
replace     verbal_fluency_imp_21=1     if     inrange(age_21,80,120)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,0,4)
replace     verbal_fluency_imp_21=0     if     inrange(age_21,80,120)     &
inrange(yrschool,1,6) & inrange(verbal_fluency_num_21,5,50)
* age_21 80+ & Years of Education 7+
replace     verbal_fluency_imp_21=1     if     inrange(age_21,80,120)     &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,0,7)
replace     verbal_fluency_imp_21=0     if     inrange(age_21,80,120)     &
inrange(yrschool,7,25) & inrange(verbal_fluency_num_21,8,50)
```

```
replace        verbal_fluency_imp_21=        verbal_fluency_num_21        if
mi(verbal_fluency_num_21) & verbal_fluency_imp_21==.
replace    verbal_fluency_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & verbal_fluency_imp_21==.
replace verbal_fluency_imp_21= .p if inlist(tipent_21,3,4)
label values verbal_fluency_imp_21 cognitive_21


** Serial 3 z-scores **
gen serial3_z_21=.
//replace serial3_z_21= 3.0 if [

                                                                    ]             &
inrange(age_21,17,59)
//replace serial3_z_21= 2.0 if [

                                                                        ]         &
inrange(age_21,17,59)
//replace serial3_z_21= 1.5 if [

                                                                        ]         &
inrange(age_21,17,59)
replace serial3_z_21= 1.0  if  [[inlist(serial3_21,5)      &  yrschool==0]  |
[inlist(serial3_21,5)   & inrange(yrschool,1,6)] | [inlist(serial3_21,5)
& inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace serial3_z_21= 0.0  if  [[inlist(serial3_21,3,4)  &  yrschool==0]  |
[inlist(serial3_21,4)       & inrange(yrschool,1,6)] | [inlist(serial3_21,4)
       & inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace serial3_z_21=-1.0  if  [[inlist(serial3_21,2)      &  yrschool==0]  |
[inlist(serial3_21,3)   & inrange(yrschool,1,6)]
                                        ] & inrange(age_21,17,59)
//replace serial3_z_21=-1.5 if [

                                                                        ]         &
inrange(age_21,17,59)
replace serial3_z_21=-2.0  if  [[inlist(serial3_21,1)      &  yrschool==0]  |
[inlist(serial3_21,2)   & inrange(yrschool,1,6)] | [inlist(serial3_21,3)
& inrange(yrschool,7,25)]] & inrange(age_21,17,59)
replace serial3_z_21=-3.0  if  [[inlist(serial3_21,0)      &  yrschool==0]  |
[inlist(serial3_21,0,1)  & inrange(yrschool,1,6)] | [inrange(serial3_21,0,2)
& inrange(yrschool,7,25)]] & inrange(age_21,17,59)

//replace serial3_z_21= 3.0 if [

                                                                    ]             &
inrange(age_21,60,69)
//replace serial3_z_21= 2.0 if [

                                                                      ]           &
inrange(age_21,60,69)
```

```
//replace serial3_z_21= 1.5 if [

                                                                                ]      &
inrange(age_21,60,69)
replace serial3_z_21= 1.0  if [[inlist(serial3_21,5)  &    yrschool==0]    |
[inlist(serial3_21,5)  & inrange(yrschool,1,6)] | [inlist(serial3_21,5)    &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace serial3_z_21= 0.0  if  [[inlist(serial3_21,3,4)  &  yrschool==0]  |
[inlist(serial3_21,4)  & inrange(yrschool,1,6)] | [inlist(serial3_21,4)    &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace serial3_z_21=-1.0  if  [[inlist(serial3_21,2)     &  yrschool==0]  |
[inlist(serial3_21,3)   & inrange(yrschool,1,6)]
                                          ] & inrange(age_21,60,69)
replace serial3_z_21=-1.5  if [
      [inlist(serial3_21,2)   & inrange(yrschool,1,6)]
                                                  ]                        &
inrange(age_21,60,69)
replace serial3_z_21=-2.0  if [[inlist(serial3_21,0,1) & yrschool==0] |

[inlist(serial3_21,3)     & inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace serial3_z_21=-3.0  if [
      [inlist(serial3_21,0,1)       &       inrange(yrschool,1,6)]       |
[inrange(serial3_21,0,2)  & inrange(yrschool,7,25)]] & inrange(age_21,60,69)

//replace serial3_z_21= 3.0 if [

                                                                                ]      &
inrange(age_21,70,79)
//replace serial3_z_21= 2.0 if [

                                                                                ]      &
inrange(age_21,70,79)
//replace serial3_z_21= 1.5 if [

                                                                              ]        &
inrange(age_21,70,79)
replace serial3_z_21= 1.0  if [[inlist(serial3_21,5)   & yrschool==0]

                                                              ]                 &
inrange(age_21,70,79)
replace serial3_z_21= 0.0  if  [[inlist(serial3_21,3,4)  &  yrschool==0]  |
[inrange(serial3_21,3,5) & inrange(yrschool,1,6)] | [inlist(serial3_21,4,5)
& inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace serial3_z_21=-1.0  if [[inlist(serial3_21,2)  & yrschool==0]

                                                              ]                 &
inrange(age_21,70,79)
replace serial3_z_21=-1.5  if [[inlist(serial3_21,1)  &    yrschool==0]    |
[inlist(serial3_21,2)   & inrange(yrschool,1,6)] | [inlist(serial3_21,3)    &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
```

```
replace serial3_z_21=-2.0  if [[inlist(serial3_21,0)  &    yrschool==0]    |
[inlist(serial3_21,1)    & inrange(yrschool,1,6)]
                                        ] & inrange(age_21,70,79)
replace serial3_z_21=-3.0  if [
      [inlist(serial3_21,0)                &    inrange(yrschool,1,6)]    |
[inrange(serial3_21,0,2) & inrange(yrschool,7,25)]] & inrange(age_21,70,79)

//replace serial3_z_21= 3.0 if [

                                                                      ]       &
inrange(age_21,80,120)
//replace serial3_z_21= 2.0 if [

                                                                      ]       &
inrange(age_21,80,120)
//replace serial3_z_21= 1.5 if [

                                                                        ]      &
inrange(age_21,80,120)
replace serial3_z_21= 1.0  if  [[inlist(serial3_21,4,5)  &  yrschool==0]  |
[inlist(serial3_21,5)  & inrange(yrschool,1,6)] | [inlist(serial3_21,4,5)  &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace serial3_z_21= 0.0  if  [[inlist(serial3_21,3)     &  yrschool==0]  |
[inlist(serial3_21,3,4) & inrange(yrschool,1,6)]
                                        ] & inrange(age_21,80,120)
replace serial3_z_21=-1.0  if  [[inlist(serial3_21,2)     &  yrschool==0]  |
[inlist(serial3_21,2)   & inrange(yrschool,1,6)] | [inlist(serial3_21,2)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace serial3_z_21=-1.5  if [[inlist(serial3_21,1)   & yrschool==0] |

     [inlist(serial3_21,1)             &     inrange(yrschool,7,25)]]     &
inrange(age_21,80,120)
replace serial3_z_21=-2.0  if  [[inlist(serial3_21,0)     &  yrschool==0]  |
[inlist(serial3_21,1)   & inrange(yrschool,1,6)] | [inlist(serial3_21,0)   &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace serial3_z_21=-3.0  if [
      [inlist(serial3_21,0)               &     inrange(yrschool,1,6)]     |
[inlist(serial3_21,0)   & inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace serial3_z_21= serial3_21 if mi(serial3_21) & serial3_z_21==.
replace     serial3_z_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & serial3_z_21==.
replace serial3_z_21= .p if inlist(tipent_21,3,4)
label variable serial3_z_21 "MHAS Executive Function - Series of subtractions
Z-Scores"

** Impairment Status **
gen serial3_imp_21 = .
label variable serial3_imp_21 "MHAS 2021 Executive function - Verbal Fluency
Impairment"

* age_21 <59 & Years of Education=0
```

```
replace  serial3_imp_21=1    if  inrange(age_21,17,59)  &  yrschool==0  &
inlist(serial3_21,0)
replace  serial3_imp_21=0    if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(serial3_21,1,5)
* age_21 <59 & Years of Education 1-6
replace serial3_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,1,6) &
inrange(serial3_21,0,1)
replace serial3_imp_21=0  if inrange(age_21,17,59) & inrange(yrschool,1,6) &
inrange(serial3_21,2,5)
* age_21 <59 & Years of Education 7+
replace serial3_imp_21=1  if inrange(age_21,17,59) & inrange(yrschool,7,25) &
inrange(serial3_21,0,2)
replace serial3_imp_21=0  if inrange(age_21,17,59) & inrange(yrschool,7,25) &
inrange(serial3_21,3,5)

* age_21 60-69 & Years of Education=0
replace  serial3_imp_21=1    if  inrange(age_21,60,69)  &  yrschool==0  &
inlist(serial3_21,0)
replace  serial3_imp_21=0    if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(serial3_21,1,5)
* age_21 60-69 Years of Education 1-6
replace serial3_imp_21=1  if inrange(age_21,60,69) & inrange(yrschool,1,6) &
inrange(serial3_21,0,1)
replace serial3_imp_21=0  if inrange(age_21,60,69) & inrange(yrschool,1,6) &
inrange(serial3_21,2,5)
* age_21 60-69 & Years of Education 7+
replace serial3_imp_21=1  if inrange(age_21,60,69) & inrange(yrschool,7,25) &
inrange(serial3_21,0,2)
replace serial3_imp_21=0  if inrange(age_21,60,69) & inrange(yrschool,7,25) &
inrange(serial3_21,3,5)

* age_21 70-79 & Years of Education=0
replace  serial3_imp_21=1    if  inrange(age_21,70,79)  &  yrschool==0  &
inlist(serial3_21,0)
replace  serial3_imp_21=0    if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(serial3_21,1,5)
* age_21 70-79 & Years of Education 1-6
replace serial3_imp_21=1  if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inlist(serial3_21,0)
replace serial3_imp_21=0  if inrange(age_21,70,79) & inrange(yrschool,1,6) &
inrange(serial3_21,1,5)
* age_21 70-79 & Years of Education 7+
replace serial3_imp_21=1  if inrange(age_21,70,79) & inrange(yrschool,7,25) &
inrange(serial3_21,0,1)
replace serial3_imp_21=0  if inrange(age_21,70,79) & inrange(yrschool,7,25) &
inrange(serial3_21,2,5)

* age_21 80+ & Years of Education=0
//replace  serial3_imp_21=1    if  inrange(age_21,80,120)  &  yrschool==0  &
inlist(serial3_21,)
replace  serial3_imp_21=0    if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(serial3_21,0,5)
```

```
* age_21 80+ & Years of Education 1-6
replace serial3_imp_21=1  if inrange(age_21,80,120) & inrange(yrschool,1,6) &
inlist(serial3_21,0)
replace serial3_imp_21=0  if inrange(age_21,80,120) & inrange(yrschool,1,6) &
inrange(serial3_21,1,5)
* age_21 80+ & Years of Education 7+
replace serial3_imp_21=1  if inrange(age_21,80,120) & inrange(yrschool,7,25)
& inrange(serial3_21,0,1)
replace serial3_imp_21=0  if inrange(age_21,80,120) & inrange(yrschool,7,25)
& inrange(serial3_21,2,5)

replace serial3_imp_21= serial3_21 if mi(serial3_21) & serial3_imp_21==.
replace     serial3_imp_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & serial3_imp_21==.
replace serial3_imp_21= .p if inlist(tipent_21,3,4)
label values serial3_imp_21 cognitive_21


***CONSTRUCTIONAL PRAXIS Z-SCORES ***
** Copy one figure z-scores **
gen construction_z_21=.
//replace construction_z_21= 3.0 if [


            ] & inrange(age_21,17,59)
//replace construction_z_21= 2.0 if [


            ] & inrange(age_21,17,59)
//replace construction_z_21= 1.5 if [


            ] & inrange(age_21,17,59)
replace construction_z_21= 1.0   if   [[inlist(construction_21,6)          &
yrschool==0]

            ] & inrange(age_21,17,59)
replace construction_z_21= 0.0   if   [[inlist(construction_21,5)          &
yrschool==0]  |  [  inlist(construction_21,6)     &  inrange(yrschool,1,6)]  |
[inlist(construction_21,6)               &    inrange(yrschool,7,25)]]    &
inrange(age_21,17,59)
replace construction_z_21=-1.0   if   [[inlist(construction_21,4)          &
yrschool==0] | [ inlist(construction_21,5)   & inrange(yrschool,1,6)]
                                                                    ]       &
inrange(age_21,17,59)
replace construction_z_21=-1.5   if   [[inlist(construction_21,3)          &
yrschool==0]

            ] & inrange(age_21,17,59)
replace construction_z_21=-2.0   if   [[inlist(construction_21,2)          &
yrschool==0]  |  [  inlist(construction_21,4)     &  inrange(yrschool,1,6)]  |
```

```
[inlist(construction_21,5)                &    inrange(yrschool,7,25)]]    &
inrange(age_21,17,59)
replace construction_z_21=-3.0   if      [[inlist(construction_21,0,1)      &
yrschool==0]  |  [inrange(construction_21,0,3)  &  inrange(yrschool,1,6)]  |
[inrange(construction_21,0,4)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)

//replace construction_z_21= 3.0 if [


            ] & inrange(age_21,60,69)
//replace construction_z_21= 2.0 if [


            ] & inrange(age_21,60,69)
//replace construction_z_21= 1.5 if [


            ] & inrange(age_21,60,69)
replace construction_z_21= 1.0   if   [[inlist(construction_21,6)          &
yrschool==0]

            ] & inrange(age_21,60,69)
replace construction_z_21= 0.0   if      [[inlist(construction_21,4,5)      &
yrschool==0] | [ inlist(construction_21,6)    & inrange(yrschool,1,6)] | [
inlist(construction_21,6)               &     inrange(yrschool,7,25)]]     &
inrange(age_21,60,69)
replace construction_z_21=-1.0   if   [[inlist(construction_21,3)          &
yrschool==0] | [ inlist(construction_21,5)   & inrange(yrschool,1,6)]
                                                                    ]       &
inrange(age_21,60,69)
replace construction_z_21=-1.5   if [

                  [ inlist(construction_21,5)    & inrange(yrschool,7,25)]]
& inrange(age_21,60,69)
replace construction_z_21=-2.0   if   [[inlist(construction_21,2)          &
yrschool==0] | [ inlist(construction_21,4)   & inrange(yrschool,1,6)]
                                                                    ]       &
inrange(age_21,60,69)
replace construction_z_21=-3.0   if      [[inlist(construction_21,0,1)      &
yrschool==0]  |  [inrange(construction_21,0,3)  &  inrange(yrschool,1,6)]  |
[inrange(construction_21,0,4)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,60,69)

//replace construction_z_21= 3.0 if [


            ] & inrange(age_21,70,79)
//replace construction_z_21= 2.0 if [


            ] & inrange(age_21,70,79)
```

```
//replace construction_z_21= 1.5 if [


            ] & inrange(age_21,70,79)
replace construction_z_21= 1.0   if    [[inlist(construction_21,6)          &
yrschool==0]

            ] & inrange(age_21,70,79)
replace construction_z_21= 0.0   if      [[inlist(construction_21,4,5)      &
yrschool==0]  |  [ inlist(construction_21,5,6)  &  inrange(yrschool,1,6)]  |  [
inlist(construction_21,6)               &     inrange(yrschool,7,25)]]     &
inrange(age_21,70,79)
replace construction_z_21=-1.0   if    [[inlist(construction_21,3)          &
yrschool==0] |
             [  inlist(construction_21,5)     &  inrange(yrschool,7,25)]]  &
inrange(age_21,70,79)
replace construction_z_21=-1.5   if [
                 [ inlist(construction_21,4)   & inrange(yrschool,1,6)]
                                                                      ]  &
inrange(age_21,70,79)
replace construction_z_21=-2.0   if    [[inlist(construction_21,2)          &
yrschool==0] | [ inlist(construction_21,3)    & inrange(yrschool,1,6)] | [
inlist(construction_21,4)               &     inrange(yrschool,7,25)]]     &
inrange(age_21,70,79)
replace construction_z_21=-3.0   if      [[inlist(construction_21,0,1)      &
yrschool==0]  |  [inrange(construction_21,0,2)  &  inrange(yrschool,1,6)]  |
[inrange(construction_21,0,3)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,70,79)

//replace construction_z_21= 3.0 if [


            ] & inrange(age_21,80,120)
//replace construction_z_21= 2.0 if [


            ] & inrange(age_21,80,120)
//replace construction_z_21= 1.5 if [


            ] & inrange(age_21,80,120)
replace construction_z_21= 1.0   if    [[inlist(construction_21,6)          &
yrschool==0]

            ] & inrange(age_21,80,120)
replace construction_z_21= 0.0   if      [[inlist(construction_21,4,5)      &
yrschool==0]  |  [inlist(construction_21,5,6)  &  inrange(yrschool,1,6)]  |  [
inlist(construction_21,5,6)        &        inrange(yrschool,7,25)]]        &
inrange(age_21,80,120)
replace construction_z_21=-1.0   if      [[inlist(construction_21,2,3)      &
yrschool==0] | [inlist(construction_21,4)   & inrange(yrschool,1,6)]
```

```
                                                                    ]        &
inrange(age_21,80,120)
replace construction_z_21=-1.5   if [
                  [inlist(construction_21,3)    & inrange(yrschool,1,6)] | [
inlist(construction_21,4)               &     inrange(yrschool,7,25)]]     &
inrange(age_21,80,120)
replace construction_z_21=-2.0   if [
                  [inlist(construction_21,2)   & inrange(yrschool,1,6)]
                                                                        ]   &
inrange(age_21,80,120)
replace construction_z_21=-3.0   if      [[inlist(construction_21,0,1)      &
yrschool==0]   |   [inlist(construction_21,0,1)   &   inrange(yrschool,1,6)]   |
[inrange(construction_21,0,3)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)

replace   construction_z_21=   construction_21   if   mi(construction_21)   &
construction_z_21==.
replace     construction_z_21=     .m     if     [inlist(age_21,888,999)     |
inlist(yrschool,.d,.r,.m)] & construction_z_21==.
replace construction_z_21= .p if inlist(tipent_21,3,4)
label variable construction_z_21 "MHAS 2021 Constructional Praxis Z-Scores"

** Impairment Status (using 0-6 score) **
gen construction_imp_21 = .
label   variable   construction_imp_21   "MHAS   2021   Constructional   Praxis
Impairment"

* age_21 <59 & Years of Education=0
replace  construction_imp_21=1  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(construction_21,0,3)
replace  construction_imp_21=0  if  inrange(age_21,17,59)  &  yrschool==0  &
inrange(construction_21,4,6)
* age_21 <59 & Years of Education 1-6
replace     construction_imp_21=1          if     inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(construction_21,0,4)
replace     construction_imp_21=0          if     inrange(age_21,17,59)     &
inrange(yrschool,1,6) & inrange(construction_21,5,6)
* age_21 <59 & Years of Education 7+
replace     construction_imp_21=1          if     inrange(age_21,17,59)     &
inrange(yrschool,7,25) & inrange(construction_21,0,5)
replace     construction_imp_21=0          if     inrange(age_21,17,59)     &
inrange(yrschool,7,25) & inlist(construction_21,6)

* age_21 60-69 & Years of Education=0
replace  construction_imp_21=1  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(construction_21,0,2)
replace  construction_imp_21=0  if  inrange(age_21,60,69)  &  yrschool==0  &
inrange(construction_21,3,6)
* age_21 60-69 & Years of Education 1-6
replace     construction_imp_21=1          if     inrange(age_21,60,69)     &
inrange(yrschool,1,6) & inrange(construction_21,0,4)
```

```
replace     construction_imp_21=0          if     inrange(age_21,60,69)     &
inrange(yrschool,1,6) & inrange(construction_21,5,6)
* age_21 60-69 & Years of Education 7+
replace     construction_imp_21=1          if     inrange(age_21,60,69)     &
inrange(yrschool,7,25) & inrange(construction_21,0,5)
replace     construction_imp_21=0          if     inrange(age_21,60,69)     &
inrange(yrschool,7,25) & inrlist(construction_21,6)

* age_21 70-79 & Years of Education=0
replace  construction_imp_21=1  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(construction_21,0,2)
replace  construction_imp_21=0  if  inrange(age_21,70,79)  &  yrschool==0  &
inrange(construction_21,3,6)
* age_21 70-79 & Years of Education 1-6
replace      construction_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,1,6) & inrange(construction_21,0,4)
replace      construction_imp_21=0      if      inrange(age_21,70,79)      &
inrange(yrschool,1,6) & inrange(construction_21,5,6)
* age_21 70-79 & Years of Education 7+
replace      construction_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(construction_21,0,4)
replace      construction_imp_21=0      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(construction_21,5,6)

* age_21 80+ & Years of Education=0
replace  construction_imp_21=1  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(construction_21,0,1)
replace  construction_imp_21=0  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(construction_21,2,6)
* age_21 80+ & Years of Education 1-6
replace      construction_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(construction_21,0,3)
replace      construction_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(construction_21,4,6)
* age_21 80+ & Years of Education 7+
replace      construction_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(construction_21,0,4)
replace      construction_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(construction_21,5,6)

replace  construction_imp_21=  construction_21  if  mi(construction_21)  &
construction_imp_21==.
replace    construction_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & construction_imp_21==.
replace construction_imp_21= .p if inlist(tipent_21,3,4)
label values construction_imp_21 cognitive_21


** Copy one figure z-scores comparable with 2001 & 2003 **
gen construction_z_v01_21=.
replace construction_z_v01_21= 0.0    if  [[inlist(construction_v01_21,2)  &
yrschool==0] | [inlist(construction_v01_21,2)    & inrange(yrschool,1,6)] |
```

```
[inlist(construction_v01_21,2)            &    inrange(yrschool,7,25)]]    &
inrange(age_21,17,59)
replace construction_z_v01_21=-1.0    if  [[inlist(construction_v01_21,1)  &
yrschool==0]

                        ] & inrange(age_21,17,59)
//replace construction_z_v01_21=-1.5  if [


                            ] & inrange(age_21,17,59)
//replace construction_z_v01_21=-2.0  if [


                            ] & inrange(age_21,17,59)
replace construction_z_v01_21=-3.0    if  [[inlist(construction_v01_21,0)  &
yrschool==0]  |  [inlist(construction_v01_21,0,1)  &  inrange(yrschool,1,6)]  |
[inlist(construction_v01_21,0,1)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,17,59)

replace construction_z_v01_21= 0.0    if  [[inlist(construction_v01_21,2)  &
yrschool==0]  |  [inlist(construction_v01_21,2)  &  inrange(yrschool,1,6)]  |
[inlist(construction_v01_21,2)            &    inrange(yrschool,7,25)]]    &
inrange(age_21,60,69)
replace construction_z_v01_21=-1.0    if  [[inlist(construction_v01_21,1)  &
yrschool==0] | [inlist(construction_v01_21,1) & inrange(yrschool,1,6)]

     ] & inrange(age_21,60,69)
//replace construction_z_v01_21=-1.5  if [


                           ] & inrange(age_21,60,69)
//replace construction_z_v01_21=-2.0  if [


                           ] & inrange(age_21,60,69)
replace construction_z_v01_21=-3.0    if  [[inlist(construction_v01_21,0)  &
yrschool==0]  |  [inlist(construction_v01_21,0)  &  inrange(yrschool,1,6)]  |
[inlist(construction_v01_21,0,1)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,60,69)

replace construction_z_v01_21= 1.0    if  [[inlist(construction_v01_21,2)  &
yrschool==0]

                      ] & inrange(age_21,70,79)
replace construction_z_v01_21= 0.0    if [
                           [inlist(construction_v01_21,2)                  &
inrange(yrschool,1,6)]     |     [inlist(construction_v01_21,2)            &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace construction_z_v01_21=-1.0    if  [[inlist(construction_v01_21,1)  &
yrschool==0]

                      ] & inrange(age_21,70,79)
```

```
//replace construction_z_v01_21=-1.5  if [


                          ] & inrange(age_21,70,79)
replace construction_z_v01_21=-2.0    if  [[inlist(construction_v01_21,0)  &
yrschool==0] | [inlist(construction_v01_21,1) & inrange(yrschool,1,6)]
                                                                          ]
& inrange(age_21,70,79)
replace construction_z_v01_21=-3.0    if [
                         [inlist(construction_v01_21,0)                   &
inrange(yrschool,1,6)]       |       [inlist(construction_v01_21,0,1)      &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)

replace construction_z_v01_21= 1.0    if  [[inlist(construction_v01_21,2)  &
yrschool==0]

                 ] & inrange(age_21,80,120)
replace construction_z_v01_21= 0.0    if  [[inlist(construction_v01_21,1)  &
yrschool==0]  |  [inlist(construction_v01_21,2)  &  inrange(yrschool,1,6)]  |
[inlist(construction_v01_21,2)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace construction_z_v01_21=-1.0    if [
                           [inlist(construction_v01_21,1)                   &
inrange(yrschool,1,6)]
                       ] & inrange(age_21,80,120)
//replace construction_z_v01_21=-1.5  if [


                    ] & inrange(age_21,80,120)
replace construction_z_v01_21=-2.0    if  [[inlist(construction_v01_21,0)  &
yrschool==0] |
                  [inlist(construction_v01_21,1)  &  inrange(yrschool,7,25)]]
& inrange(age_21,80,120)
replace construction_z_v01_21=-3.0    if [
                           [inlist(construction_v01_21,0)                   &
inrange(yrschool,1,6)]       |       [inlist(construction_v01_21,0)        &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace construction_z_v01_21= construction_v01_21 if mi(construction_v01_21)
& construction_z_v01_21==.
replace    construction_z_v01_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & construction_z_v01_21==.
replace construction_z_v01_21= .p if inlist(tipent_21,3,4)
label variable construction_z_v01_21 "MHAS 2021 Constructional Praxis Z-Scores
v2001-2003"

** Impairment Status (using 0-2 score comparable to 2001 and 2003) **
gen construction_imp_v01_21 = .
label  variable  construction_imp_v01_21  "MHAS  2021  Constructional  Praxis
Impairment v2001-2003"

* age_21 69< & Years of Education=0
```

```
replace construction_imp_v01_21=1 if inrange(age_21,17,69) & yrschool==0 &
inlist(construction_v01_21,0)
replace construction_imp_v01_21=0 if inrange(age_21,17,69) & yrschool==0 &
inlist(construction_v01_21,1,2)
* age_21 69< & Years of Education 1-6
replace    construction_imp_v01_21=1        if    inrange(age_21,17,69)    &
inrange(yrschool,1,6) & inlist(construction_v01_21,0,1)
replace    construction_imp_v01_21=0        if    inrange(age_21,17,69)    &
inrange(yrschool,1,6) & inlist(construction_v01_21,2)
* age_21 69< & Years of Education 7+
replace    construction_imp_v01_21=1        if    inrange(age_21,17,69)    &
inrange(yrschool,7,25) & inlist(construction_v01_21,0,1)
replace    construction_imp_v01_21=0        if    inrange(age_21,17,69)    &
inrange(yrschool,7,25) & inlist(construction_v01_21,2)

* age_21 70-79 & Years of Education=0
replace construction_imp_v01_21=1 if inrange(age_21,70,79) & yrschool==0 &
inlist(construction_v01_21,0)
replace construction_imp_v01_21=0 if inrange(age_21,70,79) & yrschool==0 &
inlist(construction_v01_21,1,2)
* age_21 70-79 & Years of Education 1-6
replace     construction_imp_v01_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inlist(construction_v01_21,0,1)
replace     construction_imp_v01_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,1,6) & inlist(construction_v01_21,2)
* age_21 70-79 & Years of Education 7+
replace     construction_imp_v01_21=1     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inlist(construction_v01_21,0,1)
replace     construction_imp_v01_21=0     if     inrange(age_21,70,79)     &
inrange(yrschool,7,25) & inlist(construction_v01_21,2)

* age_21 80+ & Years of Education=0
replace construction_imp_v01_21=1 if inrange(age_21,80,120) & yrschool==0 &
inlist(construction_v01_21,0)
replace construction_imp_v01_21=0 if inrange(age_21,80,120) & yrschool==0 &
inlist(construction_v01_21,1,2)
* age_21 80+ & Years of Education 1-6
replace     construction_imp_v01_21=1     if     inrange(age_21,80,120)     &
inrange(yrschool,1,6) & inlist(construction_v01_21,0)
replace     construction_imp_v01_21=0     if     inrange(age_21,80,120)     &
inrange(yrschool,1,6) & inlist(construction_v01_21,1,2)
* age_21 80+ & Years of Education 7+
replace     construction_imp_v01_21=1     if     inrange(age_21,80,120)     &
inrange(yrschool,7,25) & inlist(construction_v01_21,0,1)
replace     construction_imp_v01_21=0     if     inrange(age_21,80,120)     &
inrange(yrschool,7,25) & inlist(construction_v01_21,2)

replace        construction_imp_v01_21=        construction_v01_21        if
mi(construction_v01_21) & construction_imp_v01_21==.
replace   construction_imp_v01_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & construction_imp_v01_21==.
replace construction_imp_v01_21= .p if inlist(tipent_21,3,4)
```

```
label values construction_imp_v01_21 cognitive_21


*** Delayed recall/copy of figure ***
** Z-SCORES **
gen construction_m_z_21=.
//replace construction_m_z_21= 3.0    if [


                                ] & inrange(age_21,17,59)
//replace construction_m_z_21= 2.0    if [


                               ] & inrange(age_21,17,59)
//replace construction_m_z_21= 1.5    if [


                                ] & inrange(age_21,17,59)
replace construction_m_z_21= 1.0 if   [[inlist(construction_m_21,6)         &
yrschool==0]

                    ] & inrange(age_21,17,59)
replace construction_m_z_21= 0.0 if     [[inlist(construction_m_21,4,5)     &
yrschool==0]  |  [inlist(construction_m_21,5,6)  &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,5,6)           &     inrange(yrschool,7,25)]]     &
inrange(age_21,17,59)
replace construction_m_z_21=-1.0 if   [[inlist(construction_m_21,3)         &
yrschool==0] | [inlist(construction_m_21,4)   & inrange(yrschool,1,6)] |
[inlist(construction_m_21,4)                &    inrange(yrschool,7,25)]]    &
inrange(age_21,17,59)
replace construction_m_z_21=-1.5 if   [[inlist(construction_m_21,2)         &
yrschool==0] | [inlist(construction_m_21,3)   & inrange(yrschool,1,6)]
                                                                          ]
& inrange(age_21,17,59)
replace construction_m_z_21=-2.0 if   [[inlist(construction_m_21,1)         &
yrschool==0]  |  [inlist(construction_m_21,2)     &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,3)                &    inrange(yrschool,7,25)]]    &
inrange(age_21,17,59)
replace construction_m_z_21=-3.0 if   [[inlist(construction_m_21,0)         &
yrschool==0]  |  [inlist(construction_m_21,0,1)  &  inrange(yrschool,1,6)]  |
[inrange(construction_m_21,0,2)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,17,59)

//replace construction_m_z_21= 3.0    if [


                                 ] & inrange(age_21,60,69)
//replace construction_m_z_21= 2.0    if [


                                 ] & inrange(age_21,60,69)
```

```
//replace construction_m_z_21= 1.5    if [


                                ] & inrange(age_21,60,69)
//replace construction_m_z_21= 1.0    if [


                                ] & inrange(age_21,60,69)
replace construction_m_z_21= 0.0 if    [[inlist(construction_m_21,4,5,6)    &
yrschool==0]  |  [inlist(construction_m_21,5,6)  &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,5,6)          &     inrange(yrschool,7,25)]]     &
inrange(age_21,60,69)
replace construction_m_z_21=-1.0 if   [[inlist(construction_m_21,2,3)       &
yrschool==0]  |  [inlist(construction_m_21,4)     &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,4)               &    inrange(yrschool,7,25)]]    &
inrange(age_21,60,69)
replace construction_m_z_21=-1.5 if [
                      [inlist(construction_m_21,3)                          &
inrange(yrschool,1,6)]
                           ] & inrange(age_21,60,69)
replace construction_m_z_21=-2.0 if   [[inlist(construction_m_21,0,1)       &
yrschool==0]  |  [inlist(construction_m_21,2)     &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,3)               &    inrange(yrschool,7,25)]]    &
inrange(age_21,60,69)
replace construction_m_z_21=-3.0 if [
                        [inlist(construction_m_21,0,1)                      &
inrange(yrschool,1,6)]       |       [inrange(construction_m_21,0,2)       &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)

//replace construction_m_z_21= 3.0    if [


                           ] & inrange(age_21,70,79)
//replace construction_m_z_21= 2.0    if [


                           ] & inrange(age_21,70,79)
//replace construction_m_z_21= 1.5    if [


                           ] & inrange(age_21,70,79)
replace construction_m_z_21= 1.0 if     [[inlist(construction_m_21,5,6)     &
yrschool==0]

                     ] & inrange(age_21,70,79)
replace construction_m_z_21= 0.0 if     [[inlist(construction_m_21,3,4)     &
yrschool==0]  |  [inlist(construction_m_21,4,5,6)  &  inrange(yrschool,1,6)]  |
[inlist(construction_m_21,4,5,6)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,70,79)
replace construction_m_z_21=-1.0 if   [[inlist(construction_m_21,2)         &
yrschool==0] | [inlist(construction_m_21,3)     & inrange(yrschool,1,6)]
```

```
] & inrange(age_21,70,79)
replace construction_m_z_21=-1.5 if   [[inlist(construction_m_21,1)         &
yrschool==0]
                 | [inlist(construction_m_21,3)    & inrange(yrschool,7,25)]]
& inrange(age_21,70,79)
replace construction_m_z_21=-2.0 if   [[inlist(construction_m_21,0)         &
yrschool==0] | [inlist(construction_m_21,1,2)    & inrange(yrschool,1,6)] |
[inlist(construction_m_21,2)                &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)
replace construction_m_z_21=-3.0 if [
                     [inlist(construction_m_21,0)    & inrange(yrschool,1,6)]
|   [inlist(construction_m_21,0,1)          &   inrange(yrschool,7,25)]]   &
inrange(age_21,70,79)

//replace construction_m_z_21= 3.0    if [


                            ] & inrange(age_21,80,120)
//replace construction_m_z_21= 2.0    if [


                            ] & inrange(age_21,80,120)
replace construction_m_z_21= 1.5 if  [[inlist(construction_m_21,6)          &
yrschool==0]

                 ] & inrange(age_21,80,120)
replace construction_m_z_21= 1.0 if  [[inlist(construction_m_21,5)          &
yrschool==0] | [inlist(construction_m_21,6)       & inrange(yrschool,1,6)] |
[inlist(construction_m_21,6)              &     inrange(yrschool,7,25)]]    &
inrange(age_21,80,120)
replace construction_m_z_21= 0.0 if    [[inlist(construction_m_21,2,3,4)    &
yrschool==0] | [inlist(construction_m_21,3,4,5)  & inrange(yrschool,1,6)] |
[inlist(construction_m_21,4,5)       &       inrange(yrschool,7,25)]]       &
inrange(age_21,80,120)
replace construction_m_z_21=-1.0 if  [[inlist(construction_m_21,1)          &
yrschool==0] | [inlist(construction_m_21,2)       & inrange(yrschool,1,6)] |
[inlist(construction_m_21,3)              &     inrange(yrschool,7,25)]]    &
inrange(age_21,80,120)
replace construction_m_z_21=-1.5 if [

                           [inlist(construction_m_21,2)                     &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace construction_m_z_21=-2.0 if  [[inlist(construction_m_21,0)          &
yrschool==0] | [inlist(construction_m_21,0,1)    & inrange(yrschool,1,6)] |
[inlist(construction_m_21,1)              &     inrange(yrschool,7,25)]]    &
inrange(age_21,80,120)
replace construction_m_z_21=-3.0 if [

                          [inlist(construction_m_21,0)                      &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
```

```
replace construction_m_z_21= construction_m_21 if mi(construction_m_21) &
construction_m_z_21==.
replace    construction_m_z_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & construction_m_z_21==.
replace construction_m_z_21= .p if inlist(tipent_21,3,4)
label variable construction_m_z_21 "MHAS 2021 Delayed Memory - Delayed recall
of figure Z-Scores"

** Impairment Status **
gen construction_m_imp_21 = .
label variable construction_m_imp_21 "MHAS 2021 Delayed Memory - Delayed
recall of figure Impairment"

* age_21 <59 & Years of Education=0
replace construction_m_imp_21=1 if inrange(age_21,17,59) & yrschool==0 &
inrange(construction_m_21,0,2)
replace construction_m_imp_21=0 if inrange(age_21,17,69) & yrschool==0 &
inrange(construction_m_21,3,6)
* age_21 <59 & Years of Education 1-6
replace    construction_m_imp_21=1         if    inrange(age_21,17,59)    &
inrange(yrschool,1,6) & inrange(construction_m_21,0,3)
replace    construction_m_imp_21=0         if    inrange(age_21,17,59)    &
inrange(yrschool,1,6) & inrange(construction_m_21,4,6)
* age_21 <59 & Years of Education 7+
replace    construction_m_imp_21=1         if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(construction_m_21,0,3)
replace    construction_m_imp_21=0         if    inrange(age_21,17,59)    &
inrange(yrschool,7,25) & inrange(construction_m_21,4,6)

* age_21 60-69 & Years of Education=0
replace construction_m_imp_21=1 if inrange(age_21,60,69) & yrschool==0 &
inrange(construction_m_21,0,1)
replace construction_m_imp_21=0 if inrange(age_21,60,69) & yrschool==0 &
inrange(construction_m_21,2,6)
* age_21 60-69 & Years of Education 1-6
replace     construction_m_imp_21=1     if     inrange(age_21,60,69)     &
inrange(yrschool,1,6) & inrange(construction_m_21,0,3)
replace     construction_m_imp_21=0     if     inrange(age_21,60,69)     &
inrange(yrschool,1,6) & inrange(construction_m_21,4,6)
* age_21 60-69 & Years of Education 7+
replace     construction_m_imp_21=1     if     inrange(age_21,60,69)     &
inrange(yrschool,7,25) & inrange(construction_m_21,0,3)
replace     construction_m_imp_21=0     if     inrange(age_21,60,69)     &
inrange(yrschool,7,25) & inrange(construction_m_21,4,6)

* age_21 70-79 & Years of Education=0
replace construction_m_imp_21=1 if inrange(age_21,70,79) & yrschool==0 &
inrange(construction_m_21,0,1)
replace construction_m_imp_21=0 if inrange(age_21,70,79) & yrschool==0 &
inrange(construction_m_21,2,6)
* age_21 70-79 & Years of Education 1-6
```

```
replace      construction_m_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,1,6) & inrange(construction_m_21,0,2)
replace      construction_m_imp_21=0      if      inrange(age_21,70,79)      &
inrange(yrschool,1,6) & inrange(construction_m_21,3,6)
* age_21 70-79 & Years of Education 7+
replace      construction_m_imp_21=1      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(construction_m_21,0,3)
replace      construction_m_imp_21=0      if      inrange(age_21,70,79)      &
inrange(yrschool,7,25) & inrange(construction_m_21,4,6)

* age_21 80+ & Years of Education=0
replace  construction_m_imp_21=1  if  inrange(age_21,80,120)  &  yrschool==0  &
inlist(construction_m_21,0)
replace  construction_m_imp_21=0  if  inrange(age_21,80,120)  &  yrschool==0  &
inrange(construction_m_21,1,6)
* age_21 80+ & Years of Education 1-6
replace      construction_m_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(construction_m_21,0,1)
replace      construction_m_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,1,6) & inrange(construction_m_21,2,6)
* age_21 80+ & Years of Education 7+
replace      construction_m_imp_21=1      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(construction_m_21,0,2)
replace      construction_m_imp_21=0      if      inrange(age_21,80,120)      &
inrange(yrschool,7,25) & inrange(construction_m_21,3,6)

replace construction_m_imp_21= construction_m_21 if mi(construction_m_21) &
construction_m_imp_21==.
replace    construction_m_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & construction_m_imp_21==.
replace construction_m_imp_21= .p if inlist(tipent_21,3,4)
label values construction_m_imp_21 cognitive_21


** Delayed recall/copy of figure z-scores comparable with 2001 & 2003 **
gen construction_m_z_v01_21=.
replace construction_m_z_v01_21= 1.0  if [[inlist(construction_m_v01_21,2) &
yrschool==0]

                     ] & inrange(age_21,17,59)
replace construction_m_z_v01_21= 0.0  if [[inlist(construction_m_v01_21,1) &
yrschool==0] | [inlist(construction_m_v01_21,2) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_21,2)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,17,59)
replace construction_m_z_v01_21=-1.0  if [
                            [inlist(construction_m_v01_21,1)              &
inrange(yrschool,1,6)]
                                ] & inrange(age_21,17,59)
replace construction_m_z_v01_21=-2.0  if [[inlist(construction_m_v01_21,0) &
yrschool==0]    |
                        [inlist(construction_m_v01_21,1)                  &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)
```

```
replace construction_m_z_v01_21=-3.0  if [
                              [inlist(construction_m_v01_21,0)              &
inrange(yrschool,1,6)]       |       [inlist(construction_m_v01_21,0)       &
inrange(yrschool,7,25)]] & inrange(age_21,17,59)

replace construction_m_z_v01_21= 1.0  if [[inlist(construction_m_v01_21,2) &
yrschool==0]

                           ] & inrange(age_21,60,69)
replace construction_m_z_v01_21= 0.0  if [[inlist(construction_m_v01_21,1) &
yrschool==0] | [inlist(construction_m_v01_21,2) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_21,2)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,60,69)
replace construction_m_z_v01_21=-1.0  if [
                              [inlist(construction_m_v01_21,1)              &
inrange(yrschool,1,6)]
                           ] & inrange(age_21,60,69)
replace construction_m_z_v01_21=-1.5  if [

                                    [inlist(construction_m_v01_21,1)        &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)
replace construction_m_z_v01_21=-2.0  if [[inlist(construction_m_v01_21,0) &
yrschool==0]

                       ] & inrange(age_21,60,69)
replace construction_m_z_v01_21=-3.0  if [
                              [inlist(construction_m_v01_21,0)              &
inrange(yrschool,1,6)]       |       [inlist(construction_m_v01_21,0)       &
inrange(yrschool,7,25)]] & inrange(age_21,60,69)

replace construction_m_z_v01_21= 1.0  if [[inlist(construction_m_v01_21,2) &
yrschool==0]

                           ] & inrange(age_21,70,79)
replace construction_m_z_v01_21= 0.0  if [[inlist(construction_m_v01_21,1) &
yrschool==0] | [inlist(construction_m_v01_21,2) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_21,2)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,70,79)
replace construction_m_z_v01_21=-1.0  if [
                              [inlist(construction_m_v01_21,1)              &
inrange(yrschool,1,6)]       |       [inlist(construction_m_v01_21,1)       &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)
replace construction_m_z_v01_21=-1.5  if [[inlist(construction_m_v01_21,0) &
yrschool==0]

                       ] & inrange(age_21,70,79)
replace construction_m_z_v01_21=-2.0  if [
                              [inlist(construction_m_v01_21,0)              &
inrange(yrschool,1,6)]
                           ] & inrange(age_21,70,79)
replace construction_m_z_v01_21=-3.0  if [
```

```
                                    [inlist(construction_m_v01_21,0)        &
inrange(yrschool,7,25)]] & inrange(age_21,70,79)

replace construction_m_z_v01_21= 1.5  if [[inlist(construction_m_v01_21,2) &
yrschool==0]

                      ] & inrange(age_21,80,120)
replace construction_m_z_v01_21= 1.0  if [
                                [inlist(construction_m_v01_21,2)             &
inrange(yrschool,1,6)]       |       [inlist(construction_m_v01_21,2)       &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)
replace construction_m_z_v01_21= 0.0  if [[inlist(construction_m_v01_21,1) &
yrschool==0] | [inlist(construction_m_v01_21,1) & inrange(yrschool,1,6)] |
[inlist(construction_m_v01_21,1)      &      inrange(yrschool,7,25)]]      &
inrange(age_21,80,120)
replace construction_m_z_v01_21=-1.0  if [[inlist(construction_m_v01_21,0) &
yrschool==0]

                             ] & inrange(age_21,80,120)
replace construction_m_z_v01_21=-2.0  if [
                                [inlist(construction_m_v01_21,0)             &
inrange(yrschool,1,6)]       |       [inlist(construction_m_v01_21,0)       &
inrange(yrschool,7,25)]] & inrange(age_21,80,120)

replace       construction_m_z_v01_21=       construction_m_v01_21       if
mi(construction_m_v01_21) & construction_m_z_v01_21==.
replace   construction_m_z_v01_21=   .m   if   [inlist(age_21,888,999)   |
inlist(yrschool,.d,.r,.m)] & construction_m_z_v01_21==.
replace construction_m_z_v01_21= .p if inlist(tipent_21,3,4)
label variable construction_m_z_v01_21 "MHAS 2021 Delayed Memory - Delayed
recall of figure Z-Scores v2001-2003"

** Impairment Status **
gen construction_m_imp_v01_21 = .
label variable construction_m_imp_v01_21 "MHAS 2021 Delayed Memory - Delayed
recall of figure Impairment v2001-2003"

* age_21 <59 & Years of Education=0
replace construction_m_imp_v01_21=1 if inrange(age_21,17,59) & yrschool==0 &
inlist(construction_m_v01_21,0)
replace construction_m_imp_v01_21=0 if inrange(age_21,17,59) & yrschool==0 &
inlist(construction_m_v01_21,1,2)
* age_21 <59 & Years of Education 1-6
replace   construction_m_imp_v01_21=1       if   inrange(age_21,17,59)   &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,0)
replace   construction_m_imp_v01_21=0       if   inrange(age_21,17,59)   &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,1,2)
* age_21 <59 & Years of Education 7+
replace   construction_m_imp_v01_21=1       if   inrange(age_21,17,59)   &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,0,1)
replace   construction_m_imp_v01_21=0       if   inrange(age_21,17,59)   &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,2)
```

```
* age_21 60-69 & Years of Education=0
replace construction_m_imp_v01_21=1 if inrange(age_21,60,69) & yrschool==0 &
inlist(construction_m_v01_21,0)
replace construction_m_imp_v01_21=0 if inrange(age_21,60,69) & yrschool==0 &
inlist(construction_m_v01_21,1,2)
* age_21 60-69 & Years of Education 1-6
replace    construction_m_imp_v01_21=1       if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,0)
replace    construction_m_imp_v01_21=0       if    inrange(age_21,60,69)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,1,2)
* age_21 60-69 & Years of Education 7+
replace    construction_m_imp_v01_21=1       if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,0,1)
replace    construction_m_imp_v01_21=0       if    inrange(age_21,60,69)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,2)

* age_21 70-79 & Years of Education=0
replace construction_m_imp_v01_21=1 if inrange(age_21,70,79) & yrschool==0 &
inlist(construction_m_v01_21,0)
replace construction_m_imp_v01_21=0 if inrange(age_21,70,79) & yrschool==0 &
inlist(construction_m_v01_21,1,2)
* age_21 70-79 & Years of Education 1-6
replace    construction_m_imp_v01_21=1    if    inrange(age_21,70,79)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,0)
replace    construction_m_imp_v01_21=0    if    inrange(age_21,70,79)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,1,2)
* age_21 70-79 & Years of Education 7+
replace    construction_m_imp_v01_21=1    if    inrange(age_21,70,79)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,0)
replace    construction_m_imp_v01_21=0    if    inrange(age_21,70,79)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,1,2)

* age_21 80+ & Years of Education=0
//replace construction_m_imp_v01_21=1 if inrange(age_21,80,120) & yrschool==0
& inlist(construction_m_v01_21,)
replace construction_m_imp_v01_21=0 if inrange(age_21,80,120) & yrschool==0 &
inlist(construction_m_v01_21,0,1,2)
* age_21 80+ & Years of Education 1-6
replace    construction_m_imp_v01_21=1    if    inrange(age_21,80,120)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,0)
replace    construction_m_imp_v01_21=0    if    inrange(age_21,80,120)    &
inrange(yrschool,1,6) & inlist(construction_m_v01_21,1,2)
* age_21 80+ & Years of Education 7+
replace    construction_m_imp_v01_21=1    if    inrange(age_21,80,120)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,0)
replace    construction_m_imp_v01_21=0    if    inrange(age_21,80,120)    &
inrange(yrschool,7,25) & inlist(construction_m_v01_21,1,2)

replace      construction_m_imp_v01_21=      construction_m_v01_21      if
mi(construction_m_v01_21) & construction_m_imp_v01_21==.
```

```
replace   construction_m_imp_v01_21=   .m   if   [inlist(age_21,888,999)   |
inlist(yrschool,.d,.r,.m)] & construction_m_imp_v01_21==.
replace construction_m_imp_v01_21= .p if inlist(tipent_21,3,4)
label values construction_m_imp_v01_21 cognitive_21


***  TOTAL NUMBER OF TASKS COMPLETED (max. 5) ***
egen   numb_tasks_miss_v01_21=   rowmiss(iwr_21   dwr_21   construction_m_21
visual_scan_21 construction_21) if inlist(tipent_21,1,2)

gen numb_tasks_comp_v01_21= 5-numb_tasks_miss_v01_21 if inlist(tipent_21,1,2)
replace numb_tasks_comp_v01_21= .p if inlist(tipent_21,3,4)
label variable numb_tasks_comp_v01_21 "MHAS 2021 Number of Tasks Completed
(0-5)"

drop numb_tasks_miss_v01_21

***  TOTAL NUMBER OF TASKS WITH IMPAIRMENT IF COMPLETED 2 OR MORE TASKS(max.
5) ***
egen       numb_tasks_imp_v01_21=       rowtotal(iwr_imp_21       dwr_imp_21
construction_m_imp_v01_21   visual_scan_imp_21   construction_imp_v01_21)   if
inrange(numb_tasks_comp_v01_21,2,5)
replace numb_tasks_imp_v01_21= .i if inrange(numb_tasks_comp_v01_21,0,1)
replace    numb_tasks_imp_v01_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & mi(numb_tasks_imp_v01_21)
replace numb_tasks_imp_v01_21= .s if [iwr_imp_21==.s ///
                                                  & dwr_imp_21==.s ///
                                                  &
construction_m_imp_v01_21==.s ///
                                                  &     visual_scan_imp_21==.s
///
                                                  &
construction_imp_v01_21==.s] ///
                                                  & mi(numb_tasks_imp_v01_21)
replace numb_tasks_imp_v01_21= .p if inlist(tipent_21,3,4)
label  variable  numb_tasks_imp_v01_21  "MHAS  2021  Number  of  Tasks  with
Impairment if completed 2+ (0-5)"

gen two_more_tasks_imp_v01_21=.
replace two_more_tasks_imp_v01_21=0 if inrange(numb_tasks_imp_v01_21,0,1)
replace two_more_tasks_imp_v01_21=1 if inrange(numb_tasks_imp_v01_21,2,5)
replace two_more_tasks_imp_v01_21= .i if inrange(numb_tasks_comp_v01_21,0,1)
replace   two_more_tasks_imp_v01_21=   .m   if   [inlist(age_21,888,999)   |
inlist(yrschool,.d,.r,.m)] & mi(two_more_tasks_imp_v01_21)
replace two_more_tasks_imp_v01_21= .s if [iwr_imp_21==.s ///
                                                        & dwr_imp_21==.s ///
                                                        &
construction_m_imp_v01_21==.s ///
                                                        &
visual_scan_imp_21==.s ///
                                                        &
construction_imp_v01_21==.s] ///
```

```
                                                    &
mi(two_more_tasks_imp_v01_21)
replace two_more_tasks_imp_v01_21= .p if inlist(tipent_21,3,4)
label variable two_more_tasks_imp_v01_21 "MHAS 2021 Dummy of 2+/5 Tasks with
Impairment if completed 2+ (0-1)"


***  TOTAL NUMER OF TASKS COMPLETED (max. 8) ***
egen    numb_tasks_miss_21=    rowmiss(iwr_21    dwr_21    construction_m_21
visual_scan_21 construction_21 orientation_21 verbal_fluency_21 serial3_21)
if inlist(tipent_21,1,2)

gen numb_tasks_comp_21= 8-numb_tasks_miss_21 if inlist(tipent_21,1,2)
replace numb_tasks_comp_21= .p if inlist(tipent_21,3,4)
label variable numb_tasks_comp_21 "MHAS 2021 Number of Tasks Completed (0-7)"

drop numb_tasks_miss_21

***  TOTAL NUMBER OF TASKS WITH IMPAIRMENT IF COMPLETED 2 OR MORE TASKS(max.
8) ***
egen numb_tasks_imp_21= rowtotal(iwr_imp_21 dwr_imp_21 construction_m_imp_21
visual_scan_imp_21          construction_imp_21          orientation_imp_21
verbal_fluency_imp_21 serial3_imp_21) ///
                                          if
inrange(numb_tasks_comp_21,2,8)
replace numb_tasks_imp_21= .i if inrange(numb_tasks_comp_21,0,1)
replace    numb_tasks_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)]
replace numb_tasks_imp_21= .s if [iwr_imp_21==.s ///
                                          & dwr_imp_21==.s ///
                                          & construction_m_imp_21==.s ///
                                          & visual_scan_imp_21==.s ///
                                          & construction_imp_21==.s ///
                                          & orientation_imp_21==.s ///
                                          & verbal_fluency_imp_21==.s] ///
                                          & mi(numb_tasks_imp_21)
replace numb_tasks_imp_21= .p if inlist(tipent_21,3,4)
label variable numb_tasks_imp_21 "MHAS 2012 Number of Tasks with Impairment
if completed 2+ (0-8)"

gen two_more_tasks_imp_21=.
replace two_more_tasks_imp_21=0 if inrange(numb_tasks_imp_21,0,1)
replace two_more_tasks_imp_21=1 if inrange(numb_tasks_imp_21,2,8)
replace two_more_tasks_imp_21= .i if inrange(numb_tasks_comp_21,0,1)
replace   two_more_tasks_imp_21=    .m    if    [inlist(age_21,888,999)    |
inlist(yrschool,.d,.r,.m)] & mi(two_more_tasks_imp_21)
replace two_more_tasks_imp_21= .s if [iwr_imp_21==.s ///
                                                & dwr_imp_21==.s ///
                                                & construction_m_imp_21==.s
///
                                                &    visual_scan_imp_21==.s
///
```

```
                                                    &    construction_imp_21==.s
///
                                                    &     orientation_imp_21==.s
///
                                                    &
verbal_fluency_imp_21==.s] ///
                                                    & mi(two_more_tasks_imp_21)
replace two_more_tasks_imp_21= .p if inlist(tipent_21,3,4)
label variable two_more_tasks_imp_21 "MHAS 2021 Dummy of 2+/8 Tasks with
Impairment if completed 2+ (0-1)"


*********************************************************************
********* MHAS 2021 INSTRUMENTAL ACTIVITIES OF DAILY LIVING *********
*********************************************************************

*** IADL ***
** Managing money **
gen iadl_money_21 =.
replace iadl_money_21 = .d if h29a_21 == 9
replace iadl_money_21 = .r if h29a_21 == 8
replace iadl_money_21 = .p if inlist(tipent_21,3,4)
replace iadl_money_21 = .x if h29a_21 == 7 & inlist(h29b_21,8,9)
replace iadl_money_21 = 0 if h29a_21 == 2 | (h29a_21 == 7 & h29b_21 == 2)
replace iadl_money_21 = 1 if inlist(h29a_21,1,6) | (h29a_21 == 7 & h29b_21 ==
1)
label variable iadl_money_21 "IADLs: Difficulty Managing Money 2021"

** Taking medicines **
gen iadl_medicines_21 =.
replace iadl_medicines_21 = .d if h28a_21 == 9
replace iadl_medicines_21 = .r if h28a_21 == 8
replace iadl_medicines_21 = .p if inlist(tipent_21,3,4)
replace iadl_medicines_21 = .x if h28a_21 == 7 & inlist(h28b_21,8,9)
replace iadl_medicines_21 = 0 if h28a_21 == 2 | (h28a_21 == 7 & h28b_21 == 2)
replace iadl_medicines_21 = 1 if inlist(h28a_21,1,6) | (h28a_21 == 7 & h28b_21
== 1)
label variable iadl_medicines_21 "IADLs: Difficulty Taking Medicines 2021"

** Shopping for groceries **
gen iadl_shopping_21 =.
replace iadl_shopping_21 = .d if h27a_21 == 9
replace iadl_shopping_21 = .r if h27a_21 == 8
replace iadl_shopping_21 = .p if inlist(tipent_21,3,4)
replace iadl_shopping_21 = .x if h27a_21 == 7 & inlist(h27b_21,8,9)
replace iadl_shopping_21 = 0 if h27a_21 == 2 | (h27a_21 == 7 & h27b_21 == 2)
replace iadl_shopping_21 = 1 if inlist(h27a_21,1,6) | (h27a_21 == 7 & h27b_21
== 1)
label variable iadl_shopping_21 "IADLs: Difficulty Shopping for Groceries
2021"

** Cooking a hot meal **
```

```
gen iadl_cooking_21 =.
replace iadl_cooking_21 = .d if h26a_21 == 9
replace iadl_cooking_21 = .r if h26a_21 == 8
replace iadl_cooking_21 = .p if inlist(tipent_21,3,4)
replace iadl_cooking_21 = .x if h26a_21 == 7 & inlist(h26b_21,8,9)
replace iadl_cooking_21 = 0 if h26a_21 == 2 | (h26a_21 == 7 & h26b_21 == 2)
replace iadl_cooking_21 = 1 if inlist(h26a_21,1,6) | (h26a_21 == 7 & h26b_21
== 1)
label variable iadl_cooking_21 "IADLs: Difficulty Cooking a Meal 2021"
label values iadl_cooking_21 adls

** Number of Limitations with IADLs (0-4) **
egen iadls_21 = rowtotal(iadl_money_21 iadl_medicines_21 iadl_shopping_21
iadl_cooking_21) if inlist(tipent_21,1,2)
replace iadls_21 = .d if iadl_money_21 == .d & iadl_medicines_21 == .d &
iadl_shopping_21 == .d & iadl_cooking_21 == .d
replace iadls_21 = .r if iadl_money_21 == .r & iadl_medicines_21 == .r &
iadl_shopping_21 == .r & iadl_cooking_21 == .r
replace iadls_21 = .x if iadl_money_21 == .x & iadl_medicines_21 == .x &
iadl_shopping_21 == .x & iadl_cooking_21 == .x
replace iadls_21 = .x if iadl_money_21 == .x & iadl_medicines_21 == .x &
iadl_shopping_21 == .x & iadl_cooking_21 == .x
replace iadls_21 = .p if inlist(tipent_21,3,4)
label variable iadls_21 "Number of Limitations with IADLs 2021 (0-4)"



************************************************************
********* MHAS 2021 COGNITIVE STATUS ASSESSMENT  *********
************************************************************

* COGNITIVE STATUS CLASIFICATION: 1) CIND: impairment in 2+ cognitive tasks
and NO IADLs difficulties
                                /// 2) Dementia: 2a) Proxy subjects who had a
score equal or above 3.4 in the IQCODE
                                        ///               2b) Direct interview
subjects impaired in 2+ cognitive tasks and
                                       ///                  had difficulty in
one or more IADLs
                                         /// 3) Normal with IADLS:
                                         ///              3a) Direct interview
subjects with no impairment
                                         ///                or with impairment
in only one cognitive task and who had difficulty in one or more IADLs
                                         /// 4) Normal NO IADLS:
                                         ///             4a) Proxy respondents
with a score below 3.4 in the IQCODE
                                         ///              4b) Direct interview
subjects with no impairment
                                         ///                or with impairment
in only one cognitive task and who did not have difficulty with IADLs
```

```
** Cognitive Status Classification using 8 tasks (if 2 or more completed) **
gen cognitive_status_21=.
replace cognitive_status_21=3 if cog_imp_iqcode_21==1 |
[inrange(numb_tasks_imp_21,2,8) & inrange(iadls_21,1,4)]
replace cognitive_status_21=2 if inrange(numb_tasks_imp_21,2,8) & iadls_21==0
replace cognitive_status_21=1 if inrange(numb_tasks_imp_21,0,1) &
inrange(iadls_21,1,4)
replace cognitive_status_21=0 if cog_imp_iqcode_21==0 |
[inrange(numb_tasks_imp_21,0,1) & iadls_21==0]
replace cognitive_status_21=.d if [iadls_21==.d | cog_imp_iqcode_21==.d] &
cognitive_status_21==.
replace cognitive_status_21=.r if [iadls_21==.r | cog_imp_iqcode_21==.r] &
cognitive_status_21==.
replace cognitive_status_21=.m if [inlist(age_21,888,999) |
inlist(yrschool,.d,.r,.m) | inlist(iadls_21,.m,.)] & cognitive_status_21==.
replace cognitive_status_21=.i if numb_tasks_imp_21==.i &
cognitive_status_21==.
label variable cognitive_status_21 "MHAS 2021 Cognitive Status (using 8
tasks)"
label define cognitive_status_21 0 "0.Normal" 1 "1.Normal with Instrumental
Impairment" 2 "2.CIND" 3 "3.Dementia", replace
label values cognitive_status_21 cognitive_status_21

** Cognitive Status Classification using 5 tasks (if 2 or more completed) **
gen cognitive_status_v01_21=.
replace cognitive_status_v01_21=3 if cog_imp_iqcode_21==1 |
[inrange(numb_tasks_imp_v01_21,2,5) & inrange(iadls_21,1,4)]
replace cognitive_status_v01_21=2 if inrange(numb_tasks_imp_v01_21,2,5) &
iadls_21==0
replace cognitive_status_v01_21=1 if inrange(numb_tasks_imp_v01_21,0,1) &
inrange(iadls_21,1,4)
replace cognitive_status_v01_21=0 if cog_imp_iqcode_21==0 |
[inrange(numb_tasks_imp_v01_21,0,1) & iadls_21==0]
replace cognitive_status_v01_21=.d if [iadls_21==.d | cog_imp_iqcode_21==.d]
& cognitive_status_v01_21==.
replace cognitive_status_v01_21=.r if [iadls_21==.r | cog_imp_iqcode_21==.r]
& cognitive_status_v01_21==.
replace cognitive_status_v01_21=.m if [inlist(age_21,888,999) |
inlist(yrschool,.d,.r,.m) | inlist(iadls_21,.m,.)] &
cognitive_status_v01_21==.
replace cognitive_status_v01_21=.i if numb_tasks_imp_v01_21==.i &
cognitive_status_v01_21==.
label variable cognitive_status_v01_21 "MHAS 2021 Cognitive Status v2001-2003"
label values cognitive_status_v01_21 cognitive_status_21
```